

Ice sheets form from the accumulation of snowfall over many years, followed by compression of the snow as the weight of new snow presses down on previous years' fall. Ice sheets are composed of 90% fresh water that would otherwise be added to the oceans and raise global sea level by 58 metres. The Antarctic ice sheet is the largest single mass of ice on Earth, with a size of about 14 million km² and contains 30 million km³ of ice.

ICE SHELF: A thick mass of floating ice sheet that is attached to a coast but projects into and floats on the sea.

ICE VOLCANISM: The eruption of molten ice or gas-driven solid fragments from the surface of a planet or moon.

ICE-MINUS: Bacteria genetically altered that are applied to crop plants to retard frost formation on plant leaves.

ICE-NUCLEATING AGENT: Any agent that promotes the formation of ice crystals. They include minute solid particles such as dust or food particles and large molecules.

ICEBERG: A large, floating mass of ice, with only about 10% of its mass above the surface of the water.

ICELAND SPAR: Very pure transparent form of crystallised calcite, which is noted for its property of double refraction. It is used in the manufacture of Nicol prisms, which are essential parts of polarising microscopes and other optical instruments.

ICHTHYOLOGY: A specialised branch of biology that concerns the study of fish behaviour, anatomy, physiology and evolution. **Ichthyologist** is a biologist who specialises in the study of fish behaviour, anatomy, physiology and evolution.

ICON: In computer terminology, it refers to a visual representation of an object or program such as a specific file, directory, window, option or program on a computer screen. Examples are the folders on a desktop or in open windows are icons, files in folders and the recycle bin on Windows.

ICONOSCOPE: An early type of television camera tube whose light beam is focused on to a thin mica plate of which one side of the plate is faced with a thin metallic electrode and the other side coated with a mosaic of small globules of a photo-emissive material.

ICOSA-: A prefix denoting the number twenty. An example is **icosahedron**, a regular polyhedron that has twenty identical equilateral triangular faces, 30 edges and 12 vertices. Icosahedron symmetry also occurs in certain quasi-crystals such as alloys of aluminium and manganese and also in biology as icosahedron head of the bacteriophage.

ICOSAHEDRO-: An affix used in names to denote twelve atoms bound into a triangular icosahedron.

ICOSANOIDS (EICOSANOIDS): Any of a group of unsaturated 20-carbon (C₂₀) fatty acids or their related compounds having one or more double bonds. They include signalling molecules and various biologically important molecules made by oxidation of 20-carbon essential fatty acids. Unlike hormones, they are not produced in a central location and then transported to the target organ. They act locally in the cells in which they are created and disappear quickly due to their rapid inactivation and extremely short half-lives. In general, eicosanoids bind to receptors on target cell plasma membranes in various tissues and stimulate or inhibit the synthesis of other messengers such as cyclic AMP. There are many different types of icosanoids and they have a wide variety of functions. Three types: the leukotrienes, the thromboxanes and particularly the prostaglandins have been the focus of the majority of eicosanoid research.

ICT (INFORMATION AND COMMUNICATION TECHNOLOGIES): Technologies that provide access to information through telecommunications. It is similar to Information Technology (IT), but focuses primarily on communication technologies. This includes the Internet, wireless networks, cell phones and other communication mediums.

ID SOFTWARE: Computer software company that publishes popular games such as Doom and Quake, based in Texas, USA. An entire sub-culture has built up around id's games because of its habit of

releasing source code to enable fans to write their own additional game levels using settings of their own choice.

IDEA (INTERNATIONAL DATA ENCRYPTION ALGORITHM): In computing, it is an encryption algorithm that operates on 64-bit blocks using a 128-bit key. It is used in the encryption program Pretty Good Privacy (PGP), a very secure public key encryption application for MS-DOS, UNIX and VAX/VMS. It was developed in 1990 by Xuejia Lai and James Massey in Zurich, Switzerland.

IDEAL: In mathematics, a subset I of elements in a ring R that forms an additive group and has the property that, whenever x belongs to R and y belongs to I , then xy and yx belong to I . For example, the set of even integers is an ideal in the ring of integers Z .

IDEAL BURNING: A term applied to solid propellants. It occurs when the propellant burns in such a way that the thrust produced by the rocket motor and the pressure of combustion remain constant throughout the burn period.

IDEAL CHROMATOGRAPHY: Chromatographic process, in which no peak-broadening effects (such as diffusion, slow mass transfer, etc.) operate. This is a hypothetical case, implying that the plate number is infinite.

IDEAL CRYSTAL: A single crystal with a perfectly regular lattice that contains no impurities, imperfections and other defects.

IDEAL ELEMENT: Any element added to a mathematical theory in order to eliminate special cases. For example, adding the ideal element $i = \sqrt{-1}$ to the reals allows all algebraic equations to be solved and the improper points of the projective plane permit the assertion, without excluding parallel lines that every pair of lines intersect.

IDEAL FLUID: An imaginary fluid that is neither viscous nor compressible.

IDEAL GAS: A hypothetical gas that obeys the gas laws ($PV = nRT$) perfectly at all temperatures and pressures.

IDEAL GAS LAW: A law relating the pressure (P), absolute temperature (T) and volume (V) of an ideal gas. Mathematically, the ideal gas law can be written as: $pV = nRT$, where n is the number of moles of the gas and R is the gas constant. The ideal gas law implies that at constant temperature (T), the product of pressure and volume (pV) is constant (Boyle's law) and at a constant pressure, the volume is proportional to the temperature (Charles' law).

IDEAL AMMETER: An ammeter with zero internal resistance which must be placed in series with a load in a circuit in order to drop as little voltage as possible as electrons flow through it.

IDEAL POINT: An ideal element, especially a point at infinity.

IDEAL SOLUTION (IDEAL MIXTURE): Hypothetical solution that obeys Raoult's law in which the enthalpy of solution is zero; or a mixture in which the activity coefficients are equal to one.

IDEAL VOLTMETER: A voltmeter with infinite internal resistance which must be placed in parallel with a load in a circuit.

IDEMPOTENT: 1. Of a matrix, function or ring element, having the property that it is equal to its own square; the square of a symmetric matrix gives the same matrix. For example, if $A = A^2$ in matrix A , it is idempotent. An example is the unit matrix, given as follows:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and also } \begin{pmatrix} 3 & 3 \\ -2 & -2 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} 3 & 3 \\ -2 & -2 \end{pmatrix}.$$

2. Of an operation, having the property that every element of its domain is idempotent with respect to it; for example, set-theoretic intersection and union are idempotent, since $S \cup S = S = S \cap S$. A ring of which every member is idempotent is called a Boolean ring.

IDENTICAL: 1. Also known as **numerically identical**, of an entity, being referred to in two different ways, being one and the same individual or set of individuals. For example, the classes $\{x : x \text{ is even}\}$ and $\{x : x + 1 \text{ is odd}\}$ are identical. Congruent triangles have numerically identical properties but are not themselves numerically identical. If A is identical with B , written $A = B$, then the set $\{A, B\}$

has only one member. 2. Also called **qualitatively identical**, usually of two numerically distinct individuals, exactly alike, having the same properties. For example, congruent triangles are qualitatively identical, but since they are distinct triangles, they are not numerically identical.

IDENTICAL EQUATION (IDENTITY): An equation indicating the equality of two expressions for all values of any variables that occur. In other words, a universally true equation that is not to be solved for the value of its variables that makes it true, but that is true for all values of its variables. Examples are $(x^2 - a^2) = (x + a)(x - a)$, $(x^2 + 6x - 2 = ax^2 + bc - c)$ and the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$.

IDENTICAL PROPOSITION: Any necessary truth, but especially a categorial identity, such as whatever is quadrilateral has four sides.

IDENTICAL TWINS (MONOZYGOTIC TWINS): Two individuals that develop from a single fertilised ovum or egg cell. At an early stage of development, the single fertilised ovum becomes separated, giving rise to two individuals of the same sex, with identical genetic makeup and closely similar appearance. **Fraternal twins**, also known as **dizygotic twins** develop from two separate eggs fertilised at different but close times. Fraternal twins have different genetic makeup and may be of the same or opposite sex.

IDENTICALLY DISTRIBUTED: Of a set of random variables, having the same cumulative distribution function.

IDENTIFICATION: In animal husbandry, it refers to a sign or mark made on the body of an animal for easy recognition.

IDENTITY: 1. Also known as **identical equation**, an equation indicating the equality of two expressions for all values of any variables that occur. In other words, a universally true equation that is not to be solved for the value of its variables that makes it true, but that is true for all values of its variables. Examples are

$(x^2 - a^2) \equiv (x + a)(x - a)$ and $(x^2 + 6x - 2 \equiv ax^2 + bx - c)$. Another example is the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$. 2. Also known as **numerical identity**, the property or fact of being the same individual. An example is the identity of $a^{1/2}$ and $\sqrt{a}$. 3. Also known as **qualitative identity**, the property of being exactly alike or the relation that holds between such entities. For example, the identity of congruent triangles can be shown by superimposition. 4. In logic, also known as **numerical identity**, it is the relation that trivially holds between every entity and itself. It is formally defined as the set of ordered pairs (x, x) for all x in the underlying domain. 5. In logic, an assertion that the relation of identity holds, such as ' e^x is $\exp x$ '; any statement of which the operator of widest scope is '='.

IDENTITY ELEMENT (IDENTITY; NEUTRAL ELEMENT): For a set endowed with a binary operation, a member of the set such that the result of applying the operation to that element and any member of the set is equal to the latter member. A multiplicative identity is a unity; an additive identity is often called a zero. For example, the identity for multiplication in ordinary arithmetic is 1, since $x \times 1 = 1 \times x = x$, and identity for addition in ordinary arithmetic is 0 since $0 + x = x + 0 = x$.

IDENTITY FUNCTION: A function is said to be identity, when it maps every element to itself, for example, $f(a) = a$ for every a .

IDENTITY MATRIX: The $m \times m$ identity matrix I or I_m is the $m \times m$ matrix, which is given by: $I_m = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$. Each entry on the leading

diagonal is 1 and all other entries are 0. Consider the $m \times m$ matrix A , $A.I = I.A = A$, if A^{-1} is the inverse matrix of A , then $A^{-1}A = A^{-1}A = I$.

IDENTITY REACTION (DEGENERATE CHEMICAL REACTION): A chemical reaction whose products are chemically identical with the reactants, for example, the bimolecular self exchange reaction of CH_3I with I^- .

IDENTITY PRESERVED (IP): Identification and maintenance through marketing channels (usually through contract growing for a higher or premium price) of seed with specific traits or characteristics that are sought or preferred by users. It is often used to market

commodities like food-grade corn and soybeans, but there are IP contracts for conventional, organic and biotech seeds.

IDENTITY PROPERTY OF ADDITION: It states that the sum of any number, n and zero equals the original number: $n + 0 = n$.

IDENTITY PROPERTY OF MULTIPLICATION: It states that the product of any number, n and one equals the original number: $n \times 1 = n$.

IDENTITY THEOREM (UNIQUENESS THEOREM): The theorem asserting that if f and g are complex functions analytic on a region G and $f(z) = g(z)$ for all z in any set containing a cluster point in G , then $f(z) = g(z)$ for all z in G .

IDIOBLAST: Any cell markedly differing from the cells of the neighbouring tissue in plants. They have various functions such as storage of reserves, excretory materials, pigments and minerals. The three main categories of plant idioblasts are excretory, tracheoid and sclerenchymatous.

IDIOSYNCRASY: An abnormal reaction to a drug, food or other foreign substances shown by an individual that is not caused by an allergy but is usually genetically determined. An individual that shows immunological idiosyncrasy is said to be **hypersensitive** to a particular substance, agent, etc, without connection to pharmacological toxicity.

IF: 1. In mathematics, the usual conjunction in a conditional sentence. 2. In logic, the usual rendition of the sentential connective in material conditionals as a truth function in the sentential calculus. Since the truth of the compound sentence depends only on the truth-values of the components, 'if $2 + 2 = 4$, then gold is a metal' must be assigned the same truth-value as 'if $2 + 2 = 4$, then $2 + 3 = 5$ ', namely true.

IFF: Abbreviation for "if and only if". It is biconditional, defined to be true whenever both parts are the same. For example, a triangle is isosceles if and only if the base angles are equal. Its symbol is $\leftrightarrow$ or $\rightleftarrows$.

IGNEOUS AGE: The period of time since a rock, such as some types of meteorite, solidified and crystallised from a molten state. This can be derived from the ratios of the radioactive elements of certain elements, such as samarium-neodymium (Sm-Nd) and rubidium-strontium (Rb-Sr).

IGNEOUS ROCK: A hard volcanic rock formed from cooling and solidification of magma or lava from a molten state. Solidification into rock occurs either below the surface, called intrusive rocks or on the surface, called extrusive rocks. They mostly consist of quartz, feldspars, pyroxenes, amphiboles, micas, olivines, nepheline, leucite and apatite. Igneous rocks are one of the three principal classes of rocks, the others being metamorphic and sedimentary rocks.

IGNIS FATUUS (WILL-O'-THE-WISP; FRIAR'S LANTERN; JACK-O'-LANTERN): A phosphorescent light seen over marshy ground caused by the ignition of methane by traces of phosphorus compounds such as phosgene.

IGNITION: Initial combustion of a substance particularly of an explosive mixture in an internal combustion engine.

IGNITION COIL (SPARK COIL): A transformer or induction coil that transforms a battery's 12 volts to the thousands of volts needed to spark the plugs in a petrol engine's ignition system. Some coils have an internal resistor to reduce the voltage and some rely on a resistor wire or an external resistor to reduce the voltage from the vehicle's 12 volt wiring flowing into the coil. The wires which go from the ignition coil to the distributor and from the distributor to each of the spark plugs are called **spark plug wires** or **high tension leads**.

IGNITION LAG: In the internal combustion engine, the time interval measured from the instant that electrical contact is made to the inflammation of the air-fuel mixture.

IGNITION TEMPERATURE: 1. The lowest temperature to which a combustion begins and continues in substance heated in air. 2. The lowest temperature to which plasma has to be raised in order that nuclear fusion will occur.

IGNITRON: A mercury-containing rectifier tube in which the arc is struck again at the beginning of each cycle by a special electrode separately energised by an auxiliary circuit.

ILEUM: The final region of the mammalian small intestine terminating at the caecum in the higher vertebrates (mammals, reptiles and birds). It is the third and lowest portion of the small intestine from the jejunum to the pouch-shaped caecum which lies at the beginning of the large intestine. The function of the ileum is mainly to absorb vitamin B₁₂ and bile salts and whatever products of digestion were not absorbed by the jejunum.

ILIAC ARTERIES: The arteries that supply most of the blood to the lower limbs and the pelvic region. They are right and left common iliac arteries that form the terminal branches of the abdominal aorta. Each branches into the external iliac artery and the smaller internal iliac artery.

ILIUM: The uppermost and largest of the three bones that make up each half of the pelvic girdle, appearing in most vertebrates including mammals and birds, all reptiles except snakes but not bony fish. It bears a flattened wing of bone that is attached by ligament to the sacrum.

ILKOVIC EQUATION: A relation used in polarography relating the diffusion current i_d and the concentration c . The equation has the form $i_d = kc$, where k is constant.

-ILLION: A suffix denoting a number of millions. The multiple is denoted by the prefix, for example, a **trillion** (10^{12}) or in British usage 10^{18} .

ILL-CONDITIONED: 1. In mathematics, of a problem, having a large condition number. For example, a matrix is ill-conditioned if the condition number is too large and singular if it is infinite. 2. Of a computation, numerically unstable. While it is possible to have an ill-conditioned implementation for a well-conditioned problem, the converse is implausible.

ILL-POSED PROBLEM (IMPROPERLY POSED PROBLEM): In mathematics, a problem that can be solved in one or more ways. Alternatively, a problem whose solutions depend discontinuously upon the initial data.

ILLITE: A clay mineral of hydrous mica group resembling micas, with a lattice structure which does not expand on absorption of water. It is found in shales and mudstones.

ILLUMINANCE (ILLUMINATION): The energy in the form of visible radiation reaching a surface per unit area in a unit time or the luminous flux per unit time. It is a measure of the intensity of the incident light, wavelength-weighted by the luminosity function to correlate with human brightness perception. It is measured in lux (lumens per square metre). Its symbol is E .

ILLUMINATION: A measure of the energy intensity or brightness of light reaching a unit area of a surface or the energy in the form of visible radiation reaching a surface per unit area in unit time. An **illuminant** is anything that provides or gives off light.

ILLUMINATED OBJECT: A visible object that is not a light source but capable of reflecting light to the eyes. An example of an illuminated object is the moon.

ILLUSION: An erroneous perception of reality, often the result of a misinterpretation by the brain of information received by the senses. Illusions may occur with more of the human senses than vision, but visual illusions or optical illusions are the most well known and understood. Optical illusions may have external causes, such as refraction, as in the observation of a stick held in water. Example of auditory illusion is the apparent change in pitch of a railroad train's whistle as it passes.

ILLUSTRATE: To explain or make something clear with pictures, diagrams, graphs, etc.

ILLUVIAL HORIZON: A soil horizon in which materials carried from an overlying layer have been precipitated from solution or deposited from suspension as a layer of accumulation. The most common illuvial horizon is the B horizon. Its opposite is eluvial horizon.

ILLUVIATION: The process of depositing soil materials that are removed from an upper to a lower soil horizon through leaching (percolation). Illuviated substances include silicate clay, hydrous oxides of iron and aluminium, and organic matter.

ILMENITE: A grey or black, weakly magnetic mineral found in basic igneous rocks, notably basalt and metamorphic rocks with hardness of 5-6 and relative density of 4.7. Ilmenite is nearly pure iron(II) titanium(IV) oxide (FeTiO_3). It occurs as tabular to fine, scaly crystals in the rhombohedral system and as compact masses or granules. The discovery of ilmenite on the Moon has opened up the possibility of mining this mineral to supply both building material and oxygen to a lunar base.

IM-: Abbreviation and symbol for the imaginary part of a number: that is, where a and b are real, $\text{im}(a + bi) = b$; for example, $\text{im}(2 + 3i) = 3$.

IMAGE: 1. In mathematics, of a point or number, the value of a function corresponding to the given value of the independent variable or the point into which the given point is mapped. For example, the image of $x = 2$ under the function $y = 3x + 2$ is 8. 2. Of a set, the set of values taken by a given function or set-valued function for arguments in that set. written $f(S)$, where S is the given set $f(S) = \{y : y = f(x), x \in S\}$. 3. Of a function, the set of all values of the function, its range as distinct from its codomain. 4. A figure that is produced by a transformation of another figure called the preimage. 5. An optical reproduction of a physical object formed by sound wave or electromagnetic radiation originating from that object and reflected from a mirror or refracted by a lens. An image of an object is also formed when this light passes through a very small opening like that of a pinhole camera. Examples are ultrasound scan and x-ray photograph. 6. An artefact or two-dimensional figure that has a similar appearance to some subject, usually a physical object or person. Images may be two-dimensional such as a photograph or maybe three-dimensional such as a statue. 7. A picture produced on an electronic display such as a television or computer screen.

IMAGE AMPLIFIER (IMAGE INTENSIFIER): A device for converting a low light level fluoroscopic image to one that can be seen by the eye in a lighted environment. It usually consists of an electronic light amplifier chained to a television tube.

IMAGE ANTENNA: In telecommunications and antenna design, it is an electrical mirror-image of an antenna element formed by the radio waves reflecting from a conductive surface called a ground plane, such as the surface of the earth. It is used as a geometrical technique in calculating the radiation pattern of the antenna.

IMAGE CAMERA TUBE: A photoemissive camera tube having its photosensitive electrode separate from the target and in which the electrons emitted are focused to generate a corresponding electron image on the target.

IMAGE CAPTURE: The use of a device, such as a scanner or digital camera, to create a digital representation of an image. This digital representation can then be stored and manipulated on a computer.

IMAGE CONVERTER: An electronic device in which an image formed by invisible radiation such as x-rays, gamma rays, infrared or ultraviolet ray is converted into a visible image. An image converter which produces an image with enhanced radiance is sometimes called an **image intensifier**.

IMAGE DISSECTION TUBE: A two-dimensional radiation detector in which the electron image produced by a photo-emitting surface, usually a photocathode, is focused in the plane of a defining aperture. Magnetic or electric fields scan this image across the defining aperture.

IMAGE INTENSIFIER: A device used in producing a brighter optical image than an initial image. It increases the intensity of available light in an optical system to allow use under low light conditions such as at night, to facilitate visual imaging of low-light processes such as fluorescence of materials to x-rays or gamma rays or for conversion of non-visible light sources such as near-infrared or short wave infrared to visible.

IMAGE PROCESSING: The processing of images using computer techniques. Data from the image are digitised and mathematical

operations are applied to the data, usually by a computer, to manipulate the imagery. The term covers a range of processes including enhancing images, emphasising particular features or storing the image digitally for later retrieval or transmission. Image processing is used routinely in many applications, for example to process image data received by remote sensors on board spacecraft and satellites.

IMAGE TUBE (IMAGE INTENSIFIER TUBE): A vacuum-filled tube with a photocathode at one end, used to amplify faint images. Electrons, released when light forms an image on the photo cathode, are accelerated by magnetic coils around the tube so that they form a second, brighter image when they strike a second phosphor screen.

IMAGINAL DISC: A group of undifferentiated cells in an insect larva that develops into a portion of the epithermal structures such as eye, antennae and wing under the control of hormone of the adult insect during the pupal transformation. They are derived from the blastoderm. They do not grow or undergo further development until the pupal stage when they start to differentiate.

IMAGINARY AXIS: It is the y -axis of an imaginary plane.

IMAGINARY NUMBER: A number that is a multiple of $\sqrt{-1}$, which is denoted by i . An example is $\sqrt{-3} = 3i$. The square of an imaginary number is less than or equal to zero.

IMAGINARY PART: In a complex number $x + iy$ where x and y are real, the imaginary part is the coefficient i . The symbol for imaginary part of a complex number is I_m .

IMAGING (PHOTOIMAGING): The use of a photosensitive system for the capture, recording and retrieval of information associated with an object using electromagnetic energy.

IMAGO: The last stage of development of an insect, when it is sexually mature and, if it is necessary has functional wings and become a fully grown adult form. Complete metamorphosis changes a pupa into an imago.

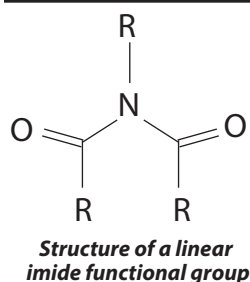
IMBALANCE: The situation in which reaction parameters that characterise different bond forming or bond breaking processes in the same reaction have developed to different extents as the transition state is approached along some arbitrarily defined reaction coordinate. For example, in the nitroalkane anomaly, the Brønsted exponent for proton removal is smaller than the Brønsted for the nitroalkane, because of imbalance between the amount of bond breaking and resonance delocalisation in the transition state. Imbalance is common in reactions such as elimination, addition and other complex reactions that involve proton (hydron) transfer.

IMBIBITION: The uptake of water by substances that do not dissolve in water, which may or may not be accompanied by swelling of the substance.

IMBRICATE: Arranged so that they overlap such as fish scales or roof tiles.

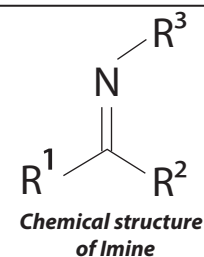
IMEI (INTERNATIONAL MOBILE EQUIPMENT IDENTITY): A unique code that identifies mobile phone devices. It is a 15- or 17-digit code that a Global System for Mobile communication (GSM) or Universal Mobile Telecommunications Service (UMTS) network uses to disable a misplaced or stolen phone from accessing a network. It is usually found printed inside the battery compartment, but can also be displayed on most phones by dialling *#06#. If the phone is lost, the owner can send the IMEI number to the network which can be used to disable it from accessing the network.

IMIDE: A functional group consisting of two acyl groups bound to nitrogen. These compounds are structurally related to acid anhydrides, although imides are less reactive. Imides are more acidic than amides (it is the $-NH$ group that loses the hydrogen) but less acidic than carboxylic acids. Sulfonamides are amides of sulfonic acids, for example, benzenesulfonamide acid and benzenesulfonic.



IMIDINES: Analogues of cyclic acid anhydrides in which $=O$ has been replaced by $=NR$ and $-O-$ by $-NR-$.

IMINES: Compounds containing the group $-NH-$ combined with a nonacid group and derived from ammonia, in which the nitrogen atom is part of a ring structure or the group $=NH$, in which the nitrogen atom is linked to a carbon atom by a double bond. In either case, the group is referred to as an imino group. Imine is used as a suffix in systematic nomenclature to denote the $C=NH$ group excluding the carbon atom.



IMINIUM COMPOUNDS: Salts in which the cation has the structure $R_2C = N^+R_2$.

IMINO ACID: A molecule containing a carboxylic acid group and an imino (NH) group or a molecule with both imine ($>C=NH$) and carboxylic ($-COOH$) functional groups attached to the same carbon atom. Examples are proline and hydroxyproline. The amino acid oxidase enzymes are able to convert amino acids into imino acids. Also the direct biosynthetic precursor to the amino acid proline is the imino acid (S)- Δ^1 -pyrroline-5-carboxylate (P5C). The structural formulae of two functional groups of imino acid that define it are shown in the diagram.

IMINYL RADICALS (ALKYLIDENEAMINYL RADICALS): Radicals having the structure $R_2C=N$. A contraction of alkaniminy radicals, for example, ethaniminyl ($CH_3-CH=N$).

IMMATURE SOIL: A soil profile whose physical features cannot be readily distinguished, usually because it has not had enough time to develop by normal soil-forming processes.

IMMEDIATE INFERENCE: A form of argument that derives a conclusion from a single premise. In the theory of the syllogism, the derivation from a categorical statement of its converse, obverse or contrapositive, when valid, are examples of immediate inferences.

IMMERSED: 1. To submerge completely in water. 2. A plant that grows beneath water.

IMMERSION OBJECTIVE (IMMERSION LENS): A high-power optical microscope objective with the front surface of the lens immersed in sugar solution or cedar-wood oil on the cover glass of the microscope specimen slide. Its refractive index is the same as that of the glass of the slide to enable the object to be immersed in it effectively. The liquid increases the effective aperture of the objective thus increasing the resolution.

IMMIGRATION: In biology, it is the movement of animals and people from their current place of habitation to a new habitat. It is a biological concept and is important in population ecology.

IMMISCIBLE: A term describing liquids that will not mix with each other to form a single homogeneous substance, but form two layers, one liquid on top of the other. Examples are oils and water. Immiscibility depends on the chemical structures, molar-mass distributions and molecular architectures of the components; and may be limited to certain ranges of temperature, pressure and composition.

IMMOBILE ADSORPTION: A situation in which the freedom of molecules of adsorbate to move about the surface is limited. Adsorption is immobile when kT is small compared to ΔE , the energy barrier separating adjacent sites. The adsorbate has little chance of migrating to neighbouring sites and such adsorption is necessarily localised.

IMMOBILISATION: In soil science, it is the conversion of inorganic compounds to organic compounds by micro-organisms or plants, so that the element is not readily available to other organisms or plants; the reverse of mineralisation.

IMMOBILISED ENZYME: A soluble enzyme bound to an insoluble organic or inorganic matrix or encapsulated within a membrane in order to increase its stability and make possible its repeated or continued use.

IMMUNE: A state of having efficient biological defence system to avoid infection, disease or other unwanted biological invasion or resistance acquired after catching the disease or resistance conferred by inoculation such as inoculation against smallpox. **Immune response** is the biological reaction of the body to foreign or potentially dangerous substances (antigens) particularly disease producing microbes. A system of structures and processes within an organism identifies and kills pathogens such as harmful viruses and bacteria and tumour cells. It is capable of distinguishing them from the organism's own healthy cells and tissues in order not to destroy them.

IMMUNE CLEARANCE: A small soluble cluster of bound antigen and antibody formed from the integral binding of an antibody to a soluble antigen. It contains small antigens, such as toxins or bacterial cell debris and commonly found following infections.

IMMUNE COMPLEX: A small soluble cluster of bound antigen and antibody formed from the integral binding of an antibody to a soluble antigen. It contains small antigens, such as toxins or bacterial cell debris and are commonly found following infections.

IMMUNE RESPONSE: A highly specific defensive reaction of the body in response to an antigen such as a virus, fungus, bacteria or transplanted organ. This may involve antibody production, induction of cell-mediated immunity, complement activation or development of immunological tolerance. It consists of a primary response in which the antigen is recognised as foreign and eliminated; and a secondary response to subsequent attacks by the same antigen. Antibody production is mediated by two types of lymphocytes B cells, which mature in the bone marrow. Cell-mediated immunity is by T cells, which mature in the thymus and are responsible. Complement activation is the sequential activation of serum components c1 through c9, initiated by an erythrocyte-antibody complex or by microbial polysaccharides and properdin and producing an inflammatory response.

IMMUNITY: The ability of an organism to resist disease against foreign micro-organisms, such as bacteria and viruses or cancerous cells. This is done either through the activities of specialised blood cells (white blood cells or leucocytes) or antibodies produced in response to natural exposure or inoculation. There are two main types of immunity: **acquired immunity**, also known as **adaptive immunity** and **innate immunity**. **Acquired immunity** is immunity to a particular disease that has been obtained by the development of antibodies after an attack of an infectious disease, for example, a pregnant mother passing antibodies through the placenta to a foetus or through breast milk, called **natural passive immunity**; or by vaccination or the injection of antiserum, called **artificial passive immunity**. **Active immunity** is a type of acquired immunity that results from the production of antibodies by the organism's own immune system in response to the presence of an antigen such as vaccine or exposure to an infectious disease. **Innate immunity** consists of cells and proteins that are already present in the body, which it mobilises to fight micro-organisms that invade the body. The innate immune system consists of physical epithelial barriers (such as the skin, tear fluid, acid in the stomach and mucus in the airways), phagocytic leukocytes, dendritic cells and a type of lymphocyte called a natural killer (NK) cell that destroys cells infected by viruses and cancerous cells and circulating plasma proteins.

IMMUNISATION (VACCINATION; INOCULATION): The prevention of infection by artificially introducing a pathogen into the body to stimulate resistance in the human body to specific diseases using microorganisms that have been modified or killed. This is called **passive immunity**. This triggers the body's immune system to build a defence mechanism against the disease.

IMMUNOASSAY: Any of various techniques that measure the amount of antigens and antibodies in tissue or the amount of a particular substance by virtue of its binding antigenically to a specific

antibody. It is also used to measure the composition or concentration of a substance in liquids such as serum or urine that frequently contain a complex mixture of substances.

IMMUNOBLOT: A laboratory test for analysing or identifying proteins through antigen-antibody specific reactions.

IMMUNOCHEMISTRY: Study of biochemical and molecular aspects of immunology, especially the nature of antibodies, antigens and their interactions.

IMMUNOELECTRON MICROSCOPY: A form of transmission electron microscopy in which the specimen is treated with gold-labelled antibodies specific for the protein component of interest.

IMMUNOELECTROPHORESIS (GAMMA GLOBULIN ELECTROPHORESIS; IMMUNOGLOBULIN ELECTROPHORESIS): An analytical technique used to separate antigens by electrophoresis to identify the blood levels of three major immunoglobulins: immunoglobulin M (IgM), immunoglobulin G (IgG) and immunoglobulin A (IgA). The gel medium such as agar used for the electrophoresis is irrigated by specific antibodies for the required antigens. When an antigen meets its corresponding antibody a precipitate forms, which can be seen and isolated.

IMMUNOFLOURESCENCE (FLUORESCENT ANTIBODY TECHNIQUE): A laboratory test used to detect the presence of specific proteins and enzymes in tissue samples by applying antibodies or disease-causing agents that have been labelled with fluorescent dye. The labelled antibody combines with the protein under investigation and the resulting complex can be identified under a microscope by its fluorescence.

IMMUNOGEN: A substance that elicits a cellular immune response and/or antibody production. **Immunogenicity** is the degree to which an antigen or epitope can produce an immune response.

IMMUNOGLOBULIN: One of a group of glycoproteins in the blood serum that acts as antibodies. They are synthesised by lymphocytes (white blood cell consisting of T cells and B cells) and plasma cells and are present in blood serum, the respiratory and digestive tracts and body secretions. They are grouped into five classes based on their structure and physiological activity. The various classes are: immunoglobulin A (IgA), immunoglobulin D (IgD), immunoglobulin E (IgE), immunoglobulin G (IgG) and immunoglobulin M (IgM).

IMMUNOLOGICAL MEMORY: The ability of the immune system to respond more rapidly and more effectively to subsequent contact with a specific antigen following an initial immune response to the same antigen, for example, as a result of an infection or vaccination. It depends on the formation of a small and persistent population of memory cells, induced by the primary immune response.

IMMUNOLOGICAL SYNAPSE (IMMUNE SYNAPSE): The specialised interface between antigen-presenting cells and a lymphocyte, where interaction of T cell antigen receptors with proteins of the major histocompatibility complex (antigen) takes place for the activation of T cell. It consists of a central cluster of T cell receptors surrounded by a ring of adhesion molecules.

IMMUNOLOGICALLY PRIVILEGED SITE (IMMUNE PRIVILEGE): Any of the various sites in the body such as the eye, brain, testis and the placenta and foetus where the introduction of antigen such as tissue grafts is tolerated without inducing any immune response. For example, a corneal graft placed in the eye of an unrelated recipient will generally not be rejected by the recipient's immune system.

IMMUNOLOGY: The branch of science that deals with the structure and anatomy functions of the immune system, malfunctions and immune responses in all organisms as well as immunologic techniques of analysis.

IMMUNORADIOMETRIC ASSAY: An assay based on the reversible and non-covalent binding of an antigen by a specific antibody labelled with a radioactive nuclide as a tracer.

IMMUNOSUPPRESSANT DRUG: Any of a group of drugs that reduces the activity of the body's immune system. Immunosuppressant drugs are prescribed after transplant surgery to prevent the rejection of

foreign tissues. They are also given to halt the progress of autoimmune diseases (in which the body's immune system attacks its own tissues) when other treatments are ineffective. Immunosuppressant drugs work by suppressing the production and activity of lymphocytes, a type of white blood cell that plays an important part in fighting infection and in eliminating abnormal cells that may form a malignant tumour. Commonly used immunosuppressants include corticosteroids such as prednisone and prednisolone; cytotoxic drugs such as azathioprine, chlorambucil, cyclophosphamide and methotrexate; and others such as antilymphocyte immunoglobulin and cyclosporin.

IMMUNOSUPPRESSION: The natural or induced active suppression of an immune response. This may be due to the following: (i) inhibition of the normal response of the immune system to an antigen; or (ii) prevention, by chemical or biological means, of the production of an antibody to an antigen by inhibition of the processes of transcription, translation or formation of tertiary structure.

IMMUNOTHERAPY: The concept of using the immune system to treat disease, for example, developing a vaccine against cancer. Immunotherapy may also refer to the therapy of diseases caused by the immune system, such as allergies.

IMMUNOTOXIN: A human-made protein consisting of a monodonal antibody linked to a toxin. When the protein binds to that cell, it folds inward to take in the toxin which kills the cell. It is used to target, for example, a particular tumour in a patient.

IMPACT ASSESSMENT (CLIMATE IMPACT ASSESSMENT): The practice of identifying and evaluating the detrimental and beneficial consequences of climate change on human systems, infrastructure, ecosystems, or valued natural resources.

IMPACT CRATER: An oval or near-circular depression on the surface of a planet, moon, asteroid, or other celestial body produced by the impact of a meteorite, volcanic activity, or an explosion.

IMPACT PARAMETER: In simple hard sphere collision theory the distance of closest approach that would result for two particles in a collision if the particle trajectories were undeflected by the collision.

IMPACT PRINTER: A computer-driven high-speed printer that has one or more character fonts, a ribbon or other inking device, a paper transport and some means of impacting desired characters or character elements on the paper. An impact printer automatically feeds continuous forms through as it prints, one line at a time, in contrast to a laser printer that prints the entire page at once. They are used to print cheshire labels, direct-mail advertising letters and forms, renewal notices, invoices, etc.

IMPACTS (CLIMATE IMPACTS): Consequences of climate change on natural and human systems. Depending on the consideration of adaptation, one can distinguish between potential impacts and residual impacts. Potential impacts are all impacts that may occur given a projected change in climate, without considering adaptation. Residual impacts are the impacts of climate change that would occur after adaptation.

IMPACTION: A forcible contact of particles of matter with a surface.

IMPARIPINNATE: A compound leaf that is divided up into many small leaflets (pinnate), arranged in pairs along an axis called rachis with a single leaflet at the apex. This makes the number of leaflets to be odd.

IMPATIENS (WILD BALSAM; BALSAM-WEED; IMPATIENS PALLIDA): Any of a large genus of herbs with stems enlarged at the joints, opposite leaves and irregular flowers, for example, 'touch-me-not' (*Mimosa pudica*). The flower colours are varied and include rose, rose-red, rose-purple, white, pink and salmon. The ripe elastic fruit wall when touched bursts open and scatter the seeds.

IMPEDANCE: A measure of the opposition of a circuit to the passage of alternating current, which determines the amplitude of the current. Its symbol is Z . In a d.c. circuit this is the resistance (R) alone. In an a.c. circuit, however, $Z^2 = R^2 + (X_L - X_C)^2$, where Z is impedance, R is the resistance, X_C is the capacitive reactance and X_L is the inductive reactance.

IMPEDANCE MATCHING: 1. An important function of the middle ear for transmitting, rather than reflecting, as much of the vibrations (sound energy) from the air to the cochlea as possible. It converts low pressure, high displacement vibrations into high pressure of the air into, low displacement vibrations suitable for driving fluids of the cochlea. Without this mechanism for pressure transformation between media of different densities (air and fluid), most sound would be reflected rather than transmitted into the cochlear fluid. 2. Designing electric circuits and components in such a way that the load impedance is equal to the internal impedance of the power source, in order to transfer the maximum amount of power from a source to a load.

IMPEDED DRAINAGE: A condition such as waterlogging that restricts the downward movement of water in soil by gravity.

IMPERFECT FLOWER: A flower lacking one or more of the following parts: stamen, pistil, petals or sepals. A dioecious plant has imperfect flowers on separate male and female plants. Examples are flowers of corn, marijuana, hops and persimmon. A monoecious plant has perfect flowers or has both male and female flowers on the same plant. Examples are apple, orange, carrot, pea, dandelion and rose.

IMPERIAL UNIT: The British system of units based on the pound and the yard and gallon, now replaced by SI units for most scientific purposes.

IMPERIAL SYSTEM (BRITISH IMPERIAL SYSTEM): A system of weights and measures originally developed in England that is largely based on the foot, pound and second (f.p.s.) system. It is similar to but not always the same as US standard units. In volume measurements, for example, one fluid ounce in the imperial system is equal to 28.413 ml and in the U.S. system it is equal to 29.57 ml.

IMPERMEABLE (IMPERVIOUS): Describing a solid that does not allow a fluid to pass through, which may be due to an absence of pores, joints or bedding planes or because pores are so small that the water within them is 'locked in' by surface tension.

IMPETIGO: A skin infection caused by bacteria. Usually the cause is staphylococcal but sometimes streptococcus can cause it, too. Impetigo is most common in children between the ages of two and six. It can be transmitted by contact with sores or nasal discharge from an infected person.

IMPINGER: A sampling instrument used to sample dust in the air that draws in a measured volume of dusty air and directs it through a jet to impact on a wetted glass plate; the dust particles adhering to the plate are counted.

IMPLANTATION: 1. In placental mammals, it is the attachment of the developing embryo to the uterine lining early in pregnancy enabling the foetus to receive oxygen and nutrients from the mother for growth. 2. The surgical procedure that places an organ such as a kidney in the human body.

IMPLEMENTATION: Of an algorithm, an explicit computer program that executes a given mathematical algorithm. Different implementations of the same algorithm may behave quite differently.

IMPLICATE: Describing plants, which intertwine or entangle.

IMPLICATION: A statement of the form; " $p \rightarrow q$ " (p implies q).

IMPLICIT: Of a function, not expressing the value of the dependent variable directly (explicitly) as a function of the independent variable, but rather stating a relationship that the variables must jointly satisfy. It is of the form $F(x, y) = 0$, $F(x, y, z) = 0$, etc. For example, $y^2 + xy + x^2 = 0$ cannot be universally expressed in the form $y = f(x)$.

IMPLICIT DIFFERENTIATION: A method for finding derivatives if the relationship between two variables is not expressed as an explicit function. For example, given $F(x, y) = 0$ with two real variables, the chain rule applied to $F(x, g(x)) = 0$ gives $F_x + F_y g'(x) = 0$ and hence $g'(x)$ must be $-F_x/F_y$. An example is given as follows: in differentiating $3x^2 + 2y^2 = xy$, applying the chain rule,

$$\frac{d}{dx}(3x^2 + 2y^2) = \frac{d}{dx}(xy)$$

$$3\frac{d}{dx}(x^2) + 2\frac{d}{dx}(y^2) = \frac{xd(y)}{dx} + y\frac{d}{dx}(x)$$

$$6x + 4y\frac{dy}{dx} = x\frac{dy}{dx} + y$$

$$\frac{dy}{dx}(4y - x) = y - 6x$$

$$\therefore \frac{dy}{dx} = \frac{y - 6x}{4y - x}$$

IMPLICIT FUNCTION THEOREM: A theorem that gives conditions under which an explicit equation equivalent to a given explicit function exists locally. For example, suppose $F(x, y) = 0$ represents equations in $m + n$ unknowns and has a solution $a(n$ variables) and $b(m$ variables), a unique function $y = g(x)$ may be found near a and b with $b = g(a)$, whenever F is continuously differentiable and the matrix of partial derivatives with respect to variables is full rank, then $F(x, g(x)) = 0$ and implicit differentiation allows the computation of the partial derivatives g at a .

I **IMPLOSION:** 1. The sudden reduction of pressure by chemical reaction or change of state which causes an intruding of the surrounding medium. 2. A bursting inward, as in the inward collapse of an evacuated container such as the glass envelope of a cathode-ray tube or the compression of fissionable material by ordinary explosives in a nuclear weapon. 3. The inward collapse of a building that is being demolished in a controlled fashion by the weakening and breaking of structural members by explosives.

IMPLY: It is the connective in propositional calculus which enables the valid conclusion to be inferred and has the meaning "if A is true, then B is also true", expressed as $A \rightarrow B$. Its symbol is $\rightarrow$.

IMPORT: In computer science, it refers to using data produced by another application. It is a command typically located within a program's File menu. Import is often used for importing parts of files, program settings, plug-ins or other unconventional file formats. The ability to import data is very important in software applications because it means that one application can complement another.

IMPORTATION: In logic, a rule for conjoining the antecedents of a set of repeated conditionals as in if P then Q , if Q then R , so if P and Q then R .

IMPOSSIBILITY THEOREM: Any theorem asserting the impossibility of some much sought after, often intuitively plausible, result. Examples are Arrow's impossibility theorem, Godel's incompleteness theorem or the impossibility of the solutions by radicals of the quintic.

IMPOSSIBLE FIGURE (ESCHER FIGURE): An image in two dimensions of an object because of spatial inconsistencies, is impossible to realise fully in three dimensions. Example is a picture of a staircase that re-enters itself while appearing to ascend continuously.

IMPOSSIBLE SET (LINEARLY ACCESSIBLE SET; NIKODYM SET): A subset of the unit square with unit area, such that for every point in the set there is a line that intersects the set only in this point.

IMPREDICATIVE DEFINITION: In logic, a definition in terms that require quantification over a range that includes that which is to be defined. An example is "Let n be the least natural number such that n cannot be written as the sum of at most four cubes". This definition is impredicative as it generalises over all natural numbers, including n itself.

IMPRESSED: In botany, it refers to anything positioned below the surface, as in some leaf veins.

IMPRESSION: In computer science, it refers to tracking the number of times a webpage or element on a webpage is viewed. For example, if a page shows three ads, that is three impressions. It is one of the standard metrics used in website analytics software.

IMPRINTING: 1. A specialised form of learning in which young animals during critical periods in their early development, learn to recognise and approach some large moving objects nearby. In nature

this is usually the mother, though simple models or individuals of a different species may suffice. 2. The chemical modification of the DNA in some genes that affects how or whether those genes are expressed. One particular kind of DNA imprinting found in mammals is known as parental genomic imprinting, in which the sex of the parent from whom a gene is inherited determines how the gene is modified.

IMPROPER FRACTION: A fraction in which the numerator is greater than the denominator, for example, $\frac{7}{4}$, or a polynomial of higher degree, for example, $\frac{x^2 + 3}{x + 1}$.

IMPROPER INTEGRAL: A definite integral with one or both limits of integration infinite or having an integrand that becomes infinite

between the limits of integration. The improper integral $\int_a^\infty f(x) dx$ is said to converge if $\int_a^b f(x) dx$ exists and to diverge otherwise.

Similarly, if the integrand has a singularity at an endpoint of the interval of integration, then the integral is defined as the limit of the proper integrals as the limit of integration tends to that endpoint from within the interval; if the singularity is in the interior of the interval, then the integral is the sum of the two improper integrals on the subintervals above and below the singularity.

IMPROPER POINT: An ideal element of the augmented Euclidean geometry; a point at which parallel lines meet.

IMPROPERLY POSED PROBLEM (ILL-POSED PROBLEM): In mathematics, a problem that can be solved in one or more ways. Alternatively, a problem whose solutions depend discontinuously upon the initial data.

IMPULSE: 1. The product of force, F and the time t for which it acts. From Newton's second law of motion, Force (F) = change in momentum divided by time (t) = $F = \frac{mV - mU}{t}$ therefore, $Ft = mV - mU$,

where Ft is the impulse and $mV - mU$ is the change in momentum. Hence, impulse equals the change in momentum produced by the force (F) in this time interval. If a graph is drawn of force (F) against time (t), then the impulse equals the area under the curve, i.e., it is the integral of a force F between times t_1 and t_2 , $\int_{t_1}^{t_2} F dt$.

2. An electrochemical signal that travels along the length of a nerve fibre, serving as the means by which information is transmitted through the nervous system. 3. A force that acts for a very short time. 4. A surge of electrical power in one direction.

IMPURE: 1. A substance that contains a small amount of another substance, usually unwanted. 2. An **impure spectrum** is one in which various bands of colours have no sharp well defined boundaries, but merge in each other

IMPURITY: 1. A small amount of a substance added to a pure semiconductor to improve its electrical conductivity. 2. Substances inside a confined amount of liquid, gas or solid, which differ from the chemical composition of the material or compound.

IMPUTATION: An efficient, feasible and individually rational payoff in a game involving coalitions.

IN SITU HYBRIDISATION: The base pairing of a sequence of DNA to metaphase chromosomes on a microscope slide.

IN SITU MICRO-X-RAY DIFFRACTION (KOSSEL-TECHNIQUE): Any technique which utilises the diffraction of X-rays generated in a microstructural domain of a solid under bombardment with a finely focused electron beam, thus providing an X-ray diffraction pattern of this microstructural domain. The pattern can be recorded with a film either on the reflection or transmission side of the specimen (in the latter case the crystalline sample has to be a thin film or a small particle).

IN SITU MICROANALYSIS: A term used for direct analytical investigation of the microstructural domains of a solid by focused

beams of particles and radiation. Analytical characterisation includes obtaining information about type, quantity and distribution of the elements, their state of chemical bonding, morphology and crystalline (geometric) and electronic structure of the individual phases. The combination of this information serves as a basis for the property-relevant characterisation of solids.

IN THE LARGE (GLOBAL; IM GROSSEN): Of a mathematical relation, property, etc., holding for all values of the variables without restriction.

IN UTERO: The state of the uterus or embryo of a mammal during pregnancy.

IN VITRO: Any biological process that is made to occur outside the living body, in a laboratory apparatus such as in a test tube or Petri dish. For example, **in vitro fertilisation** (IVF) is the process in which mature egg cells are removed from the ovary of infertile women who cannot conceive naturally and fertilised by sperm outside the womb in a laboratory dish or test tube. The resultant blastocyst is planted in her uterus with the intention of establishing a successful pregnancy.

IN VIVO: Any biological process that is carried out inside a living organism. For example, **in vivo fertilisation** is an artificial procedure of carrying out fertilisation within the uterus of a fertile donor female, rather than in an artificial medium, for subsequent nonsurgical transfer to an infertile recipient.

IN-LABORATORY PROCESSING: The selection, removal and preparation of the test (or analytical) portions from the laboratory sample. The processing may include a reduction in the size of the units (division) and particle size (reduction), as well as mixing to achieve homogeneity.

INACCESSIBLE CARDINAL: A cardinal σ with the property that $\sigma > \aleph_0$ (Aleph-null) and such that $\sigma > \tau$ implies $\sigma > 2^\tau$ while the union of every subset of σ with cardinality less than σ also has cardinality less than σ .

INBOX: In computer science, it refers to the main folder or the electronic folder for storing incoming e-mails or text messages. For example, when a mail is checked from a POP3 account using an e-mail program, the messages are downloaded to the inbox on the local hard drive. However, if an IMAP mail server is used, an inbox is created on the server and therefore, the messages are stored on the server as well.

INBREEDING: The mating of two genetically related parents of a species, especially over many generations. It may be used to enhance desired traits in plants or animals but is avoided in humans as it increases the risk of undesirable inherited characteristics. Inbreeding in plants also occurs naturally in the form of self-pollination.

INCANDESCENCE: The emission of light from a substance in consequence of its high temperature as a result of raising it to a high temperature. In incandescent lamp, an electric current is passed through a filament to heat it to a temperature of about 3000 °C (3273 K) causing the filament to glow white hot to provide light.

INCANDESCENT LAMP: An electric lamp consisting of a thin glass bulb filled with an inert mixture of nitrogen and argon gas. It contains a filament made of fine tungsten wire. When electricity is passed through the filament, it heats it to a temperature of about 3000 °C (3273 K) causing the filament to glow white hot, producing light. They have been largely replaced by energy efficient lights such as compact fluorescent lamps (CFLs), light emitting diodes (LEDs) and fluorescent lights.

INCANOUS: Covered with fine, soft, white hairs.

INCENTRE (INCENTER): The point formed at the intersection of the three angle bisectors of a triangle. It is also the centre of the incircle of a figure.

INCH: A unit of length in the U.S. Customary and British Imperial systems, which is equal to 2.54 cm. It is one-twelfth of a foot (30.48 cm) with the symbol (").

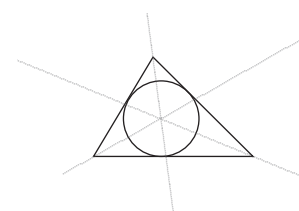
INCIDENCE: 1. The source of a ray of light that strikes a mirror. An angle of incidence is formed when this happens. 2. The partial

coincidence of two geometric figures; for example, a line through a point in the interior of a closed curve is incident to the curve at at least two points. 3. Number of occurrences of illness commencing or of persons falling ill, during a given period in a specific population, usually expressed as a rate.

INCIDENCE MATRIX: A matrix whose rows and columns correspond to vertices and edges of a graph, with ij^{th} entry 1 if edge i meets vertex j and 0 otherwise.

INCINERATOR: Equipment in which solid, liquid or gaseous combustible wastes are ignited and burned. The types include flue-fed and multiple-chamber incinerators with several stages of combustion.

INCIRCLE (INSCRIBED CIRCLE): A circle drawn inside a triangle with its circumference touching all three sides of the triangle within which the circle is drawn, or a polygon (a circle that is tangent to each of the polygon's sides). The centre is called the **incentre** and is where each angle bisector meets, and the radius of the circle is called the **inradius**.

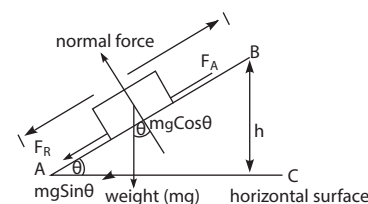


Incircle of a triangle

INCISOR: Each of the sharp chisel-shaped, teeth at the front of the mammalian mouth used for biting or nibbling.

INCLINATION: 1. The angle between a straight line or its extension and the horizontal (x -) axis. By convention it is measured in the anti-clockwise direction from the axis (between 0 and 180 degrees). 2. The smaller angle between a line and its projection onto a plane. 3. The smaller dihedral angle between two planes. 4. The angle between the orbital plane and the plane of the Earth's ecliptic. The inclination of the plane of the Moon's orbit is 5° 9' with respect to the plane of the ecliptic (the plane of the Earth's orbit around the Sun). The inclination of the plane of the ecliptic relative to the plane of the Earth's equator is 23° 27' 8.26"; this angle is called the **obliquity of the ecliptic**.

INCLINED PLANE: A sloping surface with a uniform gradient. One of the four simple machines along with the lever, the wheel and axle and the pulley, it consists of a ramp or a similar wedge-shaped device, which requires less effort than lifting a load vertically straight upwards, yet the total amount of work done is the same in both cases. An example of an inclined plane is a staircase, which allows people to move within a building from one floor to another with less effort than climbing straight up a ladder would require. Referring to the diagram, velocity ratio (V_r), which is the ratio of distance moved by the effort



An example of an inclined plane indicating the forces acting on the body of the object

to the distance moved by the load of an inclined plane is expressed as $V_r = \frac{1}{\sin \theta}$, where θ is the angle between the sloping surface and the horizontal surface. It can also be expressed as follows: $V_r = \frac{l}{h}$,

where l is the distance moved by the effort (AB) and h is the distance moved by the load (CB).

INCLUDED: In botany, contain within, without projecting beyond the mouth of the corolla as seen in a stamen or a style.

INCLUSION: 1. The relation between two sets that occurs when all the members of one are members of the other. **Strict inclusion**, usually written $S \subset T$, is if there exists an element a of T that is not a member of S ; and weak inclusion, denoted by $S \subseteq T$, is if the two sets may also be identical. Sometimes, $S \subset T$ is used for weak

inclusion and S for the strict inclusion. 2. A fragment of older material, including minerals and rock that has been enclosed within another rock.

INCLUSION BODY: A mass of virus particles formed in plant and animal cells due to viral infection. They become embedded in cellular protein.

INCLUSION COMPOUND (INCLUSION COMPLEX): A complex in which one component (the host) forms a cavity or, in the case of a crystal, a crystal lattice containing spaces in the shape of long tunnels or channels in which molecular entities of a second chemical species (the guest) are located. There is no covalent bonding between guest and host, the attraction being generally due to van der Waals forces. If the spaces in the host lattice are enclosed on all sides so that the guest species is 'trapped' as in a cage, such compounds are known as **clathrates** or **cage compounds**.

INCLUSION-EXCLUSION PRINCIPLE: The principle that the number of elements in a finite set that possess at least one out of k possible properties is equal to the number possessing exactly one of the properties, less the number possessing exactly two properties, plus the number possessing exactly three properties and so on, up to the number possessing all k . For example, in two sets, it is the number of elements in its union, given by $|A \cup B| = |A| + |B| - |A \cap B|$, where A and B are two finite sets.

INCLUSIVE: Of a mathematical property defined on intervals or sets, such that if it holds for a given interval or set, it also holds for all its subintervals or subsets. In other words, including the endpoints of an interval, for example, the interval from 1 to 2, inclusive is the closed interval written $[1, 2]$.

INCLUSIVE DISJUNCTION: A relationship that holds between two propositions of which at least one and possibly both can be true. Assuming p and q are statements, then the statement p or q represented by $p \vee q$ is referred to as disjunction of p and q .

INCLUSIVE FITNESS: The quality that organisms try to maximise as a result of natural selection acting on genes that are influential in controlling their behaviour and physiology. It includes the individual's own reproductive success and also the effect of the individual's action on reproductive success of its relatives, because relatives have high probability of sharing identical gene with the individual than do other members of the population.

INCOHERENT SCATTERING: Scattering of particles such as neutrons or photons in which the scattering elements act independently of one another, so that there are no definite phase relationships among the different parts of the scattered beam. It may be used as a ground-based technique for studying the Earth's ionosphere. The scatter signal allows measurement of electron density, ion temperature and electron temperatures, ion composition and plasma velocity.

INCOMMENSURABLE: 1. Of two numbers, having an irrational ratio or not in rational proportion to each other; for example, 2 and $\sqrt{2}$. 2. Not having units of the same dimension. 3. Unrelated to another measurement by integral multiples.

INCOMMENSURATE LATTICE: A lattice that has a long range periodic order of two periodicities in which an irrational number gives the ratio between the periodicities. An example is certain magnetic systems in which the ratio of the magnetic period to the atomic lattice is an irrational number.

INCOMPARABLE: Not able to be compared using a given relation. In a partial order some pairs of terms are incomparable; for example, in the Cartesian product order, $1, 1 > 0, 0$, but $0, 1$ and $1, 0$ are incomparable.

INCOMPATIBLE: 1. Of a tissue transplant or blood transfusion,

inducing a marked immune response that is rejected by the body. 2. Of plants, not capable of being successfully pollinated by or grafted onto each other. 3. Of a set of propositions, statements, equations, etc., not capable of all being true at the same time or under the same conditions; inconsistent.

INCOMPLETE: 1. A flower is said to be incomplete when it lacks one of the floral parts such as sepals, petals, calyx or corolla, stamen or pistils. 2. Of a formal theory, not so constructed that the addition to the axioms of a proposition that is not a theorem renders the whole theory inconsistent; thus the theory does not contain as many theorems as it may without inconsistency. 3. Of an expression, meaningless or unanalysable by itself, requiring some specific type of context in order to constitute a meaningful expression.

INCOMPLETE DOMINANCE: A genetic condition in which the two alleles inherited from the parents are neither dominant nor recessive, but instead blend to produce a physical intermediate trait. For example, when a gene for curly hair is inherited from one parent and a gene for straight hair is inherited from the other parent, it will give the offspring a wavy hair texture that is a blend of the two. **Codominance** is when neither parental trait is dominant nor recessive or two types of alleles that are equally dominant. The two traits both appear in the offspring, often showing up in different parts of the plant or animal. For example, the genes for A and B blood groups are codominant and give rise to the AB blood group if they are both inherited.

INCOMPLETE FLOWER: 1. A flower lacking in one or more of the following structures: sepals, petals, stamens or carpels. Plants such as grasses, produce flowers that are less-conspicuous and lacking in petals. A staminate has no female (carpel) present. 2. A flower with one or two major whorls absent.

INCOMPLETE INDUCTION (FIRST-KIND INDUCTION; SPECIAL INDUCTION): An induction in which the inductive step is from the integer n to $n + 1$.

INCOMPLETE METAMORPHOSIS (HEMIMETABOLISM; HEMIMETABOLY; HEMIMETAMORPHOSIS): A type of development in certain insects, in which there is incomplete or partial metamorphosis, with three distinct stages: the egg, nymph and the adult stage or imago; the nymphs resemble adults but are smaller and have different body proportions. They go through series of molts, each time looking more like adults, until they reach full size, without going through pupal stage. Examples of insect groups showing incomplete metamorphosis include: Hemiptera (scale insects, aphids, whitefly, cicadas, leafhoppers and true bugs) orthoptera (grasshoppers and crickets), Mantodea (praying mantises), Blattaria (cockroaches), Dermaptera (earwigs) and Odonata (dragonflies and damselflies).

INCOMPRESSIBLE FLOW: Fluid motion without any change in density.

INCOMPRESSIBLE FLUID: A fluid which is not reduced in volume by an increase in pressure.

INCOMPUTABLE NUMBER: A real number with an infinite decimal (or binary) expansion that cannot be enumerated by any universal computer.

INCONSISTENT: 1. Inconsistent equations are two or more equations that cannot be solved simultaneously by using one set of values for the variables. For example, the equations $x + 2y = 5$ and $x + 2y = 6$ as well as $2x - 3y = 1$ and $4x - 6y = 1$ are inconsistent as there are no solutions to the systems. If the graphs of equations do not intersect and are parallel, there are no solutions that are true for both equations and are therefore, inconsistent. 2. Of a set of propositions or statements, incapable of all being true at the same time, incompatible. 3. Of a set of well-formed formulae of a formal theory, such that an explicit contradiction may be derived from them by the derivation rules of the theory. 4. Of a formal theory, having an explicit contradiction as a theorem or more generally having an atomic statement as a theorem, whence excluding nothing from its theorems.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Truth table of inclusive disjunction

INCRASSATE: 1. Swollen or thickened. 2. In pharmacology, it is the ability of a liquid to become thicker by the addition of another substance or by evaporation.

INCREASING: Of a function of a single variable, having the property that $f(x) \geq f(y)$ for all pairs of arguments such that $x > y$. If $f(x) > f(y)$ for all $x > y$ the function is strictly increasing. The symbol $\nearrow$ is sometimes used to denote either the strict or weak property.

INCREMENT: 1. For bulk materials and large units, it is an individual portion of material collected by a single operation of a sampling device. Increments may be reduced individually or tested either individually or combined with other increments with the resulting composite reduced in size and tested as a single unit. Increments are created by the sampling operation and are usually taken from parts of a lot separated in time or space. Increments of a bulk population correspond to units of a packaged population. 2. In mathematics, it is a small but finite change in the value of a variable or function, written Δx or δx . A negative increment can be called **decrement**.

INCUBATION: 1. The process of the development of the embryo within the egg of certain oviparous or egg-laying animals to hatch their eggs. The most vital factor of incubation is the constant temperature required for its development over a specific period, which requires sitting on the egg(s) for some time to warm them. In domestic fowl, the act of sitting on eggs to incubate them is called **brooding**. Examples of incubation period for some birds are chicken (20-22 days), duck (26-28 days), eagle (35-36 days) and ostrich (35-45 days). 2. The development of an infectious disease inside the body to the point at which the first signs of disease become apparent. The incubation period or the time during which the infecting organism enters the body until the appearance of the first symptoms varies for different microbes and can be as short as a few hours for cholera or as long as a decade in the case of leprosy. 3. The maintenance of cells or microorganisms under a controlled temperature and humidity in or on a medium so that they can multiply. 4. Maintaining a premature or sick baby in an environment of temperature, humidity and oxygen levels.

INCUBATION PATCH: An area of bare, featherless skin located on the lower abdomen of, especially female birds, consisting of a network of blood vessels in the skin that develops by the shedding of feathers and the consequent thickening of the skin, which serves to transfer warmth from the adult's body to incubate the eggs.

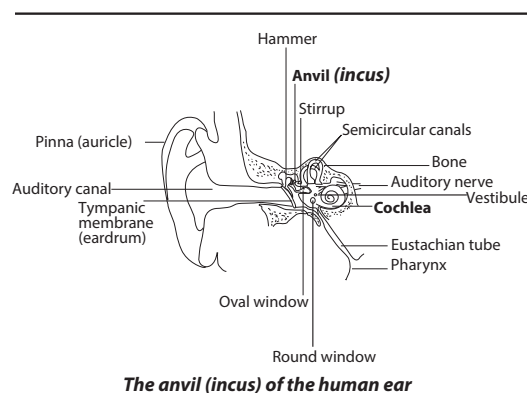
INCUBATOR: 1. An apparatus in which environmental conditions, such as temperature and humidity, can be controlled, often used for growing bacterial cultures, hatching eggs artificially or providing suitable conditions for a chemical or biological reaction. 2. An apparatus for maintaining an infant, especially a premature infant, in an environment of controlled temperature, humidity and oxygen concentration. 3. A place such as an academic/research centre or institution that permits or encourages the formation and development, as of new ideas. 4. A **business incubator** is an organisation that provides new businesses with technical and support services and usually low-cost office space or infrastructure. 5. **Apache Incubator**, created in October 2002 is the entry point for projects and software seeking to become part of the Apache Software Foundation's (ASF) efforts. The ASF is a non-profit organisation that oversees the development of Apache software. Membership is based on contribution to the foundation's open source projects. 6. **Virtual incubator** is a business that does not necessarily have a physical office space but choose to provide service to its 'fledglings' or inductees over the Internet using specially created software packages.

INCUBOUS: Describing the leaf arrangement in leafy liverworts, where the front edges of the leaves lie above the back edges of the leaves in front.

INCUMBENT COTYLEDONS: A form of orientation in an embryo in which the cotyledons lie face to face and folded sideways so that the radicle lies against the face of the cotyledons.

INCURVED: Curved inwards or adaxially.

INCUS (ANVIL): The middle of the three-ear ossicles, which is an anvil-shaped small bone in the mammalian middle ear that transmits sound vibrations from the malleus to the stapes.



INDEFINITE: 1. In botany, very numerous or not easily counted, as in stamens. 2. Of an inflorescence, indeterminate. 3. Of a quadratic form or matrix, neither positive (semi-) definite nor negative (semi-) definite.

INDEFINITE GROWTH: The ability of certain plants or plant organs to show unlimited growth. For example, trees continue to grow each year and the leaves of monocotyledons grow throughout the life of the plant. The term is also used for racemose inflorescences, in which growth of the inflorescence is not terminated by the production of an apical flower bud but continues indefinitely. Meristems may also be said to exhibit indefinite growth.

INDEFINITE INFLORESCENCE (RACEMOSE INFLORESCENCE): An inflorescence in which meristematic activity continues at the apex of the main stem and primary laterals and flowers are developed from the axillary meristems. The inflorescence is often pyramidal in shape with older flowers at the base or, in the case of a flat-topped arrangement, on the outside. Different types of racemose inflorescence include the capitulum, corymb, panicle, raceme, spadix, spike and umbel.

INDEFINITE INTEGRAL (ANTIDERIVATIVE; PRIMITIVE INTEGRAL): An indefinite integral is the integral $\int f(x)dx = F(x) + c$, where c is an arbitrary constant.

INDEHISCENT: A fruit that does not open up to release seeds or anthers that do not open to discharge pollen shed seeds. They disperse their fruits into the environment or attach them to an animal for dispersal. Examples are drupes such as coconut, mango, peach and olive; berries such as guava, tomato and blueberry; pepo such as watermelon and cucumber; pomes such as apples and pear. Some others are oranges, wheat, maize, oats, etc and nuts such as groundnut, palmnut and peanut.

INDEPENDENT: 1. Of a system of equations, not able to solve a single equation necessarily using a solution to the others. 2. Of two or more random variables, distributed in such a way that the value taken on by one variable leaves all others unaffected. 3. Of two or more events, such that the occurrence of one does not affect the probability of any of the others. Consequently, the probability of any set of independent events occurring equals the product of their individual probabilities. 4. Of a set of statements, propositions or formulae, not proved from another logical axiom or proposition.

INDEPENDENT ASSORTMENT: The principle that genes are inherited independently of one another, although genes close together on the same chromosome have a higher likelihood of being inherited together.

INDEPENDENT EVENT: An event whose occurrence does not depend on the occurrence of another one. Two events, A and B, are independent if the fact that A occurs does not affect B occurring and the probability of their occurrence together is the product of the probabilities of their events. Some examples of independent events

are: (i) landing on heads after tossing a coin and rolling a 5 on a single 6-sided die; and (ii) choosing a 3 from a deck of cards, replacing it and then choosing an ace as the second card.

INDEPENDENT VARIABLE: 1. The variable representing the value being manipulated or changed. For example, if y is a function of x , i.e., $y = f(x)$, then x is the independent variable of the function and y the dependent variable (its value depends on the value of x). 2. Also known as **predictor**, in statistics, the variable that defines a set of distinct experimental conditions or that an experimenter deliberately manipulates in order to observe its relationship with some other quantity.

INDEPENDENT-PARTICLE MODEL: A model for electrons in a many-electron system in which the correlation between electrons is either ignored or taken into account by regarding an electron as moving in an averaged-out potential that represents the interactions between the electron and all the other particles in the system.

INDETERMINACY: Heisenberg's original name for his principle 'the Heisenberg uncertainty principle', which states by precise inequalities that certain pairs of physical properties, such as position and momentum, cannot be simultaneously known to arbitrarily high precision. That is, the more precisely one property is measured, the less precisely the other can be measured.

INDETERMINATE: 1. In botany, it refers to an inflorescence in which the outer flowers bloom first, followed by the inner ones, as shown in the diagram. Hence the main axis is not terminated by a flower. **Indeterminate growth** is a type of growth characteristic of plants, in which the organism continues to grow as long as it lives. For example, in racemose inflorescence, a plant axis is not limited by development of a terminal flower bud or other reproductive structure and so continues to elongate indefinitely. 2. In mathematics, having no numerical meaning, as 0.00 or 0/0. In an equation, it is defined as having more than one variable and an unlimited number of solutions. 3. A variable or place-marker in an expression. 4. A purely formal, uninterpreted, symbol. Sometimes a specific contrast is intended with variables for which certain identities may hold by virtue of the structural properties of a given domain; for example, in the integers modulo a prime p , the element $x^p = x$ for all x , permitting cancellation; however, regarded as an indeterminate, x^p is a distinct entity from x . 5. In physics, of an effect, not obeying the law of causality. 6. In engineering, of a structure, framework, etc., comprising forces that cannot be fully analysed, especially by vector analysis.

INDETERMINATE CLEAVAGE: A type of embryonic development in deuterostomes, in which each cell produced by early cleavage divisions produces blastomeres with the potencies of the entire zygote.

INDETERMINATE EQUATIONS: Simultaneous equations which, for some reason, have an infinite number of solutions. Examples are $x + y = 4$ and $2x + 2y = 8$, where the two equations are equivalent and $x + y = z \frac{1}{2}x + 6y + 3 = 8z$, where there is insufficient data.

INDEX: 1. Also known as **exponent**, it is a number or symbol that indicates the power to which another number or expression is raised or a number that indicates the number of times a term is multiplied by itself. For example, $x^2 = x \cdot x$ and $5^4 = 5 \times 5 \times 5 \times 5$. For example, in the expression x^n the letter n is the exponent of the base x . 2. One of a specified set of symbols, usually written subscript to the right of a variable, used to distinguish each in turn of a set or sequence of otherwise identical symbols such as variables, x_1, x_2, x_3 ; the index of x_3 is 3. 3. Of a set of symbols, to distinguish the elements of a set or sequence onto which they are mapped by associating each with a distinct element of an index set. To make this explicit, the



Indeterminate flowers

sequence x_1, x_2, x_3 can be written x_i for $i \in \{1, 2, 3\}$. 4. A number or expression written within a radical and indicating which root is to be extracted, for example, $\sqrt[3]{8} = 2$. 5. Of a subgroup H of a finite group G , the number of left cosets of the subgroup in the group, written $[G : H]$. When E is the unit subgroup, $[G : E]$ is the order of G . 6. In programming, it is a method for accessing data in a table. 7. In computer science, also called database index or indexed file, it is a list of data in which each entry contains the name of the item and its location, used to quickly access and retrieve specific data.

INDEX CONTOUR: On a topographic map, it is a thicker line, usually shown as every fifth contour that has numbers written on it showing the elevation that it represents.

INDEX FOSSIL (OZONE FOSSIL): The ancient remains and traces of a plant or animal that existed continuously during a particular span of geological times and can therefore, be used to date the rocks in which it is found. It is essential in comparing the geological ages of sedimentary rocks which originated in diverse environments such as open oceans, tropical lagoons, coral reefs, beaches, lakes and rivers. Examples of index fossils are ammonites and graptolite.

INDEX LAW (INDICES LAW): A rule for manipulating index numbers when powers are multiplied or divided. Certain index laws must be followed. For example, when multiplying powers of the same term, add indices, for example, $a^3 \times a^2 = a^{(3+2)} = a^5$; when dividing powers of the same term, subtract indices, for example, $a^5 \div a^2 = a^{(5-2)} = a^3$; when raising a power of a term to another power, the indices must be multiplied, for example, $(a^5)^2 = a^{10}$.

INDEX NOTATION: A shorter way of recording different products of numbers, for example a^2 means $a \times a$, the index being 2. It is used to represent powers. Certain index notations give specific results.

INDEX NUMBER: A measure of the magnitude of a variable at one point (in time, for example), to its value at another. In other words, it is a measure of the change relative to some specified base period in some variable such as price, volume, the value of a commodity, the gross national product or the general level of prices. The main application of such numbers are in economics.

INDEX SET: A set whose elements are used to index those of some other set, such as A in $\bigcup_{\lambda \in \Lambda} A_\lambda$, the union of all the sets A_λ , one for each λ in Λ .

INDEX THEOREMS: Mathematical theories that relate the solutions of an equation to topology associated with the equation. Index theorems are applied to certain problems in the theory of condensed matter, string theory, gauge theories, quantum gravity and super symmetry.

INDEXED COLOUR: In computer science, it is a technique that identifies and stores the index of most frequently used colours of digital images in a separate piece of data called a palette. This technique reduces the number of colours in the image to a maximum of 256. Indexed colour images are 8-bit images (8 bits per pixel) but enables its quality to improve to that of 24 bits per pixel. This technique saves a lot of memory, storage space, and transmission time.

INDEXED SEQUENTIAL FILE METHOD (ISAM): In computing, it is a type of file access developed at IBM in which an index is used to obtain the address of the block containing the required record. Each index defines a different ordering of the records.

INDIAN ALMOND TREE (SEA ALMOND; TROPICAL ALMOND; Terminalia catappa): A deciduous, monoecious tree belonging to the genus Terminalia of the Combretaceae family that grows prominently in the tropical regions of Asia, Africa and Australia. The tree is upright with symmetrical crowns and grows to about 35 metres tall with horizontal branches that are distinctively arranged in tiers. The leaves are large, long and broad, ovoid and glossy dark green and leathery. These leaves turn pinkish-reddish or yellow brown before they are shed during the dry season. The leaves and bark are rich in chemicals such as flavonoids, tannins, saponins and phytosterols, which are used to treat various diseases such as liver disease, dysentery and diarrhoea. The nuts taste like almonds and are edible.

INDIAN INK: Black ink containing a suspension of carbon, a variety of soot known as lampblack combined with water, binder such as gelatin and shellac to make it durable once it is dried up. It is used in drawing and technical graphics.

INDIAN PLUM (COFFEE PLUM; INDIAN CHERRY; *Flacourtia jangomas*): A small deciduous shrub or tree in the Salicaceae or Willow family that grows to about 10 metres in height. It is widely cultivated in Southeast and East Asia and is naturalised in the warmer coastal districts of eastern Australia. It grows in waterways, closed forests, roadsides, disturbed sites and waste areas in wetter tropical and sub-tropical regions. It usually has a greyish or pale brown flaky bark. The trunks and branches of younger trees are usually densely armed with simple or branched woody thorns but are commonly thornless when older. Its leaves are glossy green in colour, with toothed margins and are narrowly-ovate to ovate-oblong and usually have pointed tips. Flowers are dioecious, with separate male and female flowers borne on different trees. The male flowers are solitary or in clusters; the female flowers are solitary. The flowers are white to green and exude fragrance. The fruits are sub-globose and turn from green to dark-red or purple when mature. The fruits are eaten raw or cooked and can be processed into jam. The bark is sometimes used as medicine.

INDICATION: A medical sign or symptom that shows the presence of a syndrome or disease or remedy for it and for which reason to use a certain test, medication, procedure or surgery.

INDICATOR: A substance or compound such as litmus or phenolphthalein that changes its structure and colour in response to its environment. It may also be defined as a weak complex organic acid or base that gives different colours in its molecular and ionic forms depending on the pH of the solution. It is used to indicate the presence, absence or concentration of another substance and the degree of reaction between two or more substances. For example, litmus paper is red in acidic condition and blue in alkaline condition.

INDICATOR ELECTRODE: An electrode that serves as a transducer responding to the excitation signal (if any) and to the composition of the solution being investigated, but that does not affect an appreciable change of bulk composition within the ordinary duration of a measurement.

INDICATOR FUNCTION (CHARACTERISTIC FUNCTION): The extended real-valued function defined on a given set that takes value 1 for every element of the set and value 0 for every element not contained in the set. It is denoted i_c or δ_c .

INDICATOR ORGANISM: Organisms which survive only in certain environmental conditions and whose presence provides information about the environment in which it is found. For example, the presence of certain bacterium (*E. coli*), which lives in animal gut, in water, indicates sewage pollution.

INDICATOR PLANTS: Plants that are characteristic of specific soil or site conditions and react to certain viruses or environmental factors with production of specific symptoms that can be used for detection and identification of these factors.

INDICATOR SPECIES: A plant or animal species that is sensitive to a particular environmental factor and sometimes act as an indicator of environmental condition such as a disease outbreak, pollution, species competition or climate change. For example, some lichens are sensitive to the concentration of sulfur dioxide in the atmosphere, thus their presence can indicate the concentration of sulfur dioxide.

INDICIAL EQUATION: An equation that determines the index that is used in Frobenius method of solving a second-order ordinary differential equation. It is obtained by noting that, by definition, the lowest order term x^k (that corresponding to $n = 0$) must have a coefficient of zero.

INDIFFERENCE: 1. In mathematics, the fact that some given ordering is indifferent between the members of some set. Also, the

relation that holds between such elements with respect to such an ordering. 2. **Principle of indifference** is the principle that in the absence of any reason to the contrary, each possible outcome of an experiment is to be treated as equiprobable.

INDIFFERENCE CURVE: A set of points for which a given utility function maintains constant value and which are therefore indifferent to one another with respect to that function. For example, an isotherm, which passes through on a map where the temperature is equal, is an indifference curve for any utility function that depends solely upon temperature; a contour map likewise shows a set of indifference curves with respect to height.

INDIFFERENCE SETS: The equivalence classes of points that are pairwise preferred to one another under a given ordering. In other words, they are the sets of points between which the weak ordering is indifferent and are all singletons exactly when the order is antisymmetric.

INDIFFERENT: 1. In mathematics, of an ordering, giving no precedence to any element of a given set in the sense that it orders each of these elements above the other or others. In other words, the ordering holds between any pair of members of this set, but is symmetric upon it. For example, in the set of combinations of k elements chosen from n , cardinality is an indifferent ordering, since all elements have cardinality k . 2. Of elements of an ordering, mutually preferred one to the other. This is not the same as incomparable, when neither can be said to be preferred to the other; the set of such elements is an indifference set. 3. In statistics, not showing or having preferences, not rendering any of the possible outcomes more probable than the others.

INDIFFERENT ABSORBING ION: An ion which absorbs through Coulomb forces only and which is hence only repelled by surfaces of like sign, attracted by surfaces of opposite sign and is not absorbed on an uncharged surface.

INDIFFERENT ELECTROLYTE (SUPPORTING ELECTROLYTE): An electrolyte solution, whose constituents are not electroactive in the range of applied potentials being studied and whose ionic strength (and, therefore, contribution to the conductivity) is usually much larger than the concentration of an electroactive substance to be dissolved in it.

INDIGENOUS: 1. Originating in and characteristic of a particular region or habitat. Every natural organism has its own natural range of distribution which is its native environment, but human activity may introduce other species into that environment, which are called **introduced species**. 2. Innate; inherent; natural.

INDIGO: 1. A tropical plant, genus *Indigofera* of the pea family with fronds of pointed leaves and red or purple flowers in spikes, a source of indigo dye, with the chemical formula $C_{16}H_{10}O_2N_2$. It is soluble in hot aniline and hot chloroform. It is used as a reagent and a dye. 2. A deep purplish blue colour that lies between blue and violet on the visible spectrum.

INDIRECT AEROSOL EFFECT: Aerosols may lead to an indirect radiative forcing of the climate system through acting as condensation nuclei or modifying the optical properties and lifetime of clouds. Two indirect effects are distinguished: **First indirect effect** also known as the **Twomey effect** is a radiative forcing induced by an increase in anthropogenic aerosols which cause an initial increase in droplet concentration and a decrease in droplet size for fixed liquid water content, leading to an increase of cloud albedo. **Second indirect effect**, also known as the **cloud lifetime effect** or **Albrecht effect** is a radiative forcing induced by an increase in anthropogenic aerosols which cause a decrease in droplet size, reducing the precipitation efficiency, thereby modifying the liquid water content, cloud thickness, and cloud lifetime.

INDIRECT AMPLIFICATION: A type of amplification reaction where the constituent required is associated with some other constituent which is then amplified and measured.

INDIRECT MEASUREMENT: The determination of heights, distances, and other physical objects that cannot be measured directly. There are several methods to do this which include the following: (i) for similar figures, using proportions with a mirror on the ground, or using shadow lengths; (ii) Pythagorean theorem ($a^2 + b^2 = c^2$) method which includes measurement using right-angled triangle; (iii) Thales theorem (if an angle is drawn from the diameter of a circle to the point on its circumference, then, the angle formed is a right angle) by using proportions. Thales derived the following proportion to figure out the height of the Great Pyramid at Giza in Egypt: $\frac{\text{height of man}}{\text{shadow of man}} = \frac{\text{height of pyramid}}{\text{shadow of pyramid}}$. Indirect measurement

can be used to measure so many things such as the width of a river, and the height of tall objects such as trees, pyramids, buildings or mountains.

INDIRECT PROOF: In mathematics, a term for *reductio ad absurdum*, which is proof of a conclusion by showing its negation to be self-contradictory or to contradict known axioms. In other words, the negation of a true statement is also true. It is often used when the given geometric statement is NOT true.

INDIRECT PROPORTION (INDIRECT VARIATION; INVERSE PROPORTION): A mathematical relationship in which an increase in one variable by a given factor brings about a decrease by the same factor in another.

INDIRECT REACTION (COMPLEX-MODE REACTIONS): A chemical process in which the reactive complex has a lifetime that is longer than its period of rotation. In a molecular-beam experiment the products of such reactions are scattered at random with reference to the centre of mass of the system.

INDISCERNIBLE: In mathematics, being qualitatively identical; not distinguishable by any property or relation defined by a formula and defined, usually by only first-order formulae.

INDISCRETE TOPOLOGY: The topology on a given space whose only open sets are the whole space and the empty set. In other words, for any non empty set X , the collection equal to $\{\phi, X\}$ is an indiscrete topology on X and the space (X, τ) is called the **indiscrete topological space** or an **indiscrete space**.

INDIUM: A soft, malleable and easily fusible silvery-white metallic element of Group 13 (IIIA) of the Periodic Table. Its chemical symbol is In, atomic number 49, melting point 156.61 °C (429.76 K), boiling point 2080 °C (2353 K) and specific gravity 7.31. Its primary source is zinc ore. It is used in making mirrors and semiconductors.

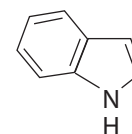
INDIVIDUAL: 1. In biology, a single organism capable of independent existence. Also, a member of a compound organism or colony. 2. In logic, a distinct object as opposed to a property or class. 3. Any element in the domain of a given theory.

INDIVIDUAL DISTANCE: In animal behaviour, the distance from an individual animal at which the presence of another individual of the same species incites avoidance or aggression.

INDIVIDUAL PERCEPTION THRESHOLD (IPT): A term used in odour testing which signifies the lowest concentration of a particular species at which a subject indicates both an initial positive and repeated response.

INDIVISIBLE: 1. Usually of integers or polynomials, not exactly divisible, prime. 2. Not divisible by a given number or quantity without a remainder, relatively prime. For example, 9 is indivisible by 5 without a remainder.

INDOLE (2,3-BENZOPYRROLE, KETOLE; 1-BENZAZOLE): A yellow solid with the molecular formula C_8H_7N , density 1.1747 g/cm³ (solid), melting point 52 °C (325 K) and boiling point 253-254 °C (526 K). Its molecules consist of a benzene ring fused to a nitrogen-containing five-membered ring. It occurs in some plants and in coal tar and is produced in faeces by bacterial action. It is used in making perfume and as a reagent.



Structure of Indole

INDOLE ALKALOIDS: A group of alkaloids which contain an indole nucleus. They are derived either from tryptophan or phenylalanine. Examples are strychnine and yohimbine. Certain alkaloids containing quinoline are derived from indole compounds, an example being quinine, obtained from the bark of Cinchona species.

INDOLE-ACETIC ACID (IAA): The principal auxin of most plants that acts as a growth hormone or in plants, where it controls cell enlargement and, through interaction with other plant hormones. It also influences cytokinesis.

INDUCED CURRENT: The electric current produced due to the electromotive force (emf) produced in a closed circuit when there is relative movement of its conductor in a magnetic field. The effect is known as the dynamo effect and is used in all dynamos and generators to produce electricity. The magnitude of the induced current depends upon the rate at which the conductor cuts the magnetic flux and its direction is given by Fleming's right-hand rule.

INDUCED DIPOLE: The separation of positive and negative charges in a neutral atom caused by the proximity of an ion or a polar molecule.

INDUCED ELECTROMOTIVE FORCE: The emf induced in a conductor in a changing magnetic flux around a conductor. It can also be defined as the emf induced in a conductor, when it is rotated around a stationary magnetic field of force. According to Faraday's Law, the magnitude of the induced emf is proportional to the rate of change of the magnetic flux that cuts the conductor.

INDUCED EMISSION (STIMULATED EMISSION): The emission of a photon by an atom in the presence of electromagnetic radiation.

INDUCED ENZYME (ADAPTIVE ENZYME): An enzyme that is synthesised in response to high concentrations of its substrate. When the substrate is absent the genes governing the enzyme's synthesis are repressed.

INDUCED MAGNETISM: The magnetism acquired by a magnetic material when placed in a magnetic field.

INDUCED MEASURE: The measure induced by the outer measure, μ^* , on the μ^* -measurable sets.

INDUCED/STIMULATED NUCLEAR REACTION: A reaction in which a stable nuclide is bombarded with high speed subatomic particles (protons, neutrons and electrons) or small nuclei resulting in the formation of radioisotopes or elements. **Induced or stimulated nuclear reaction** is of two kinds: fission and fusion. **Fission** is the decay of the nucleus of a heavy nuclide as a result of the introduction of different particles such as alpha particles, protons, neutrons and deuterons into the nuclide by bombarding it with particle accelerators. Examples of particle accelerators are the linear accelerator or linac, storage ring collider accelerators, synchrotron and betatron. **Fusion** is the integration of two or more light atomic nuclei to make a more massive one with the simultaneous release of energy. This process can be achieved by using extremely high temperature plasmas, which may eventually lead to a new readily available source of energy.

INDUCED REACTION: A reaction which occurs or is accelerated only if another reaction is simultaneously occurring in the same system. The substance, which causes an induced reaction, is termed inductor. The inductor, unlike the catalyst, is used up irreversibly during the process.

INDUCED TOPOLOGY (RELATIVE TOPOLOGY): The topology defined on a subset of the underlying set of a given topological space by taking its open sets to be intersections of the open sets of the given topology with the subset. For example, the open sets of the induced topology on a line in the Euclidean plane are the segments of the line in the interior of the open disks.

INDUCED-FIT MODEL: A proposed mechanism of interaction between an enzyme and a substrate. It states that exposure of an enzyme to a substrate causes the active part of the enzyme to change shape in order to allow the enzyme and substrate to bind. This hypothesis is generally referred to as the lock-and-key mechanism. **Induced fit** is the change in shape of the active site of an enzyme so that it binds more snugly to the substrate, induced by entry of the substrate.

INDUCER: In biochemistry, it describes a small molecule that triggers gene transcription by binding to a regulator protein. It acts by combining with the corresponding repressor protein to bring about an allosteric change so that the repressor is made incapable of combining with the operator of the system.

INDUCIBLE ENZYME: An enzyme generated in response to cells being exposed to a substrate. Example of such enzyme is β -galactosidase, found in *Escherichia coli*. β -galactosidase catalyses the hydrolysis of lactose to yield D-Glucose and D-Galactose.

INDUCIBLE NITRIC OXIDE SYNTHASE (INOS): An enzyme found in macrophages and certain other tissues that catalyses the conversion of the amino acid L-arginine to the highly reactive molecule nitric oxide (NO). The three types are the brain (or neuronal) and epithelial nitric oxide synthases, which are always present in cells and inducible nitric oxide synthase, which is produced as needed. Macrophages that have ingested harmful bacteria or pathogens are prompted by helper T cells to destroy these pathogens. This occurs more widely in wound healing and inflammations.

INDUCTANCE: 1. A measure of the ability of an electric circuit or circuit component to form a magnetic field or store magnetic energy when carrying a current. The S.I. unit of inductance is henry (H). 2. The property of an electric circuit or compound that causes an electromotive force (emf) to be generated in it as a result of a change in the current flowing through the circuit (self-inductance) or by the current flowing through a neighbouring circuit with which it is magnetically linked (mutual inductance). The most common application of mutual inductance is the transformer.

INDUCTION: 1. The ability of natural stimuli to cause unspecialised embryonic tissue to develop into specialised tissue. 2. A deliberate intervention to initiate labour before it starts naturally, for example by injecting the hormone oxytocin; or the administration of anaesthetic agents and the establishment of a depth of anaesthesia adequate for surgery. 3. The process of initiating or increasing the production of an enzyme, as in genetic transcription. 4. An alteration in the physical properties of a body that is brought about by the influence of a field or generating an electric current in a wire by varying magnetic flux through the circuit. Induction is used in generators and transformers. 5. Also known as **mathematical induction** or **finite induction**, a two-part method of proving a theorem involving an integral parameter that all integers have a certain property, P , by proving the following: (i) the base clause $P(1)$ and (ii) the recursion clause, if $P(n)$ for an integer n , then $(n + 1)$. The universal generalisation is a consequence of Peano's axioms. 6. Also known as **inductive reasoning**, a form of reasoning in which a conclusion is made about the probability of some state of affairs, based on the available evidence and past experience.

INDUCTION COIL: A type of transformer used to produce a high voltage alternating current or pulses of high voltage current from a low voltage direct current source. A typical induction coil has a core of soft iron, a primary coil and a secondary coil. The primary coil consists of a few turns of fairly heavy wire around the core; the secondary

consists of many turns of fine wire around the primary. The primary coil forms part of a circuit called the primary circuit that includes a direct current source and a circuit breaker or interrupter. The coils are used chiefly in the electrical systems of automobiles and to operate X-ray tubes. When the primary circuit is closed, direct current flows through the primary coil, producing a magnetic field. As the magnetic field builds up, it induces an electric current in the secondary coil. At the same time, the iron core becomes magnetised. The magnetised core draws the interrupter away from a metal contact, breaking the primary circuit. The direct current in the primary coil ceases and the coil's magnetic field collapses, again inducing an electric current in the secondary coil, only in the opposite direction.

INDUCTION HEATING: The heating of an electrical conducting material by eddy currents induced by a varying electromagnetic field, which is similar to that of a transformer in principle. It may produce an undesirable effect leading to power loss in transformers and other electrical devices but it is useful for melting, forging, rolling and heat treating of metals.

INDUCTION MOTOR (ASYNCHRONOUS MOTOR): An alternating current motor where power is supplied to the rotor or secondary winding by means of electromagnetic induction or induced in the secondary windings by the magnetic field of the primary winding or the stator. An electric motor turns because of magnetic force exerted between a stationary primary winding called the stator and a rotating secondary winding called the rotor. The current in the primary side creates an electromagnetic field which interacts with the electromagnetic field of the secondary side to produce a resultant torque, thereby transforming the electrical energy into mechanical energy. An induction motor operates slightly below synchronous speed and also called an **asynchronous motor**, which implies that it is not a synchronous motor. Induction motors are efficient, have good speed regulation, have no brushes which wear out and relatively cheap to construct.

INDUCTION PERIOD: The initial slow phase of a chemical reaction which later accelerates. Induction periods are often observed with radical reactions, but they may also occur in other systems, for example, before steady state concentration of the reactants is reached.

INDUCTIVE: In mathematics, of, relating to or using induction. For example, inductive proof or inductive argument is argument by induction.

INDUCTIVE DEFINITION (RECURSIVE DEFINITION): The definition of a sequence by specifying its first term by the base clause and an algorithm or the recursion clause to derive any term from its predecessor.

INDUCTIVE EFFECTS: The electronic effect of the transmission of charge through a chain of atoms in a molecule by electrostatic induction, resulting in polarisation of the bond. Electron donating species such as alkyl groups exert positive inductive effect ($+I^{\text{effect}}$), while electron withdrawing groups such as carbonyls and nitro groups exert negative inductive effect.

INDUCTIVE ORDER: In mathematics, an order in which every non-empty subset has at least one minimal element.

INDUCTIVE REACTANCE (X_L): The ratio of the potential difference (V_L) across the inductor to the amplitude of current (I_0) through it. It is expressed as $X_L = \frac{V_L}{I_0}$. It can also be expressed as $X_L = \omega L = 2\pi fL$,

where L = inductance and ω is angular frequency. Its unit is ohm and the symbol is Ω .

INDUCTIVE STEP (RECURSION CLAUSE; RECURSION FORMULA): The formula for determining the next term of a sequence from one or more of the preceding terms. For example, the recursive formula for the sequence 5, 20, 80, 320, ... is $a_n = 4a_{n-1}$.

INDUCTOMERIC EFFECT: A molecular polarisability effect occurring by the inductive mechanism of electron displacement.

INDUCTOR: A type of passive circuit element, usually a coil, in which an electromotive force is generated by inductance and which needs no power supply. The inductance of a coil is directly proportional to the number of turns in the coil.

INDUMENT: The overall appearance of the epidermal appendages such as hairs or scales.

INDUPLICATE: Petals or sepals having the entire length of their edges or margins turned or rolled inward without overlapping. Examples are *Bismarckia nobilis* and *Sabal palmetto*.

INDURATED LAYER: A soil horizon compacted, generally by cementation of soil particles to the extent that it limits rooting depth and impedes drainage.

INDUSIUM: 1. The kidney-shaped enclosing membrane of the sorus of certain ferns that protects the developing sporangia, which withers when the sorus ripens to expose the sporangia. 2. In zoology, it is a case enclosing an insect larva or pupa.

INDUSTRIAL MELANISM: The increase of darkness in the skin, feathers or fur pigment developed by a population of animals living in an industrial region where the environment is soot-darkened. Peppered moth is the example often quoted. The melanisation of a population increases the probability that its members will survive and reproduce because it offers protection in the form of camouflage.

INDUSTRIAL WASTE (HAZARDOUS WASTE): Liquid, solid or gas wastes that pose substantial or potential threats to public health, destruction of the environment or death to people if improperly stored, treated, transported or discarded. The four factors that determine whether or not a substance is hazardous are ignitability (inflammable), reactivity, corrosivity and toxicity.

INDUSTRY STANDARD ARCHITECTURE BUS (ISA BUS): In computing, it is a 16-bit data bus introduced in 1984 with the IBM PC AT and still in common use in PCs, alongside the superior PCI bus. It is not compatible with plug and play and in 1993, Intel and Microsoft introduced a PnP ISA bus that allowed the computer to automatically detect and setup computer ISA peripherals, such as a modem or sound card.

INELASTIC: 1. Not elastic; lacking flexibility or resilience; unyielding. 2. Of a function, having elasticity less than unity. In economics, demand for a good is said to be inelastic if an increase in the price results in a decrease in revenue.

INELASTIC COLLISION: A collision in which some of the kinetic energy of the colliding bodies is not conserved but is converted into other forms of energy. A **perfectly inelastic collision** is a collision in which the two bodies stick together and move with a common final velocity. **Totally inelastic collision** is a situation, in which all the energy is lost in a collision.

INELASTIC NEUTRON SCATTERING (NEUTRON SPECTROSCOPY): A technique for investigating the motion of molecules by scattering neutrons using beams of slow neutrons, from nuclear reactors or nuclear accelerators. The neutrons pick up or lose energy as they move through a sample of liquid.

INDEPENDENT-PARTICLE MODEL: A model for electrons in a many-electron system in which the correlation between electrons is either ignored or taken into account by regarding an electron as moving in an averaged-out potential that represents the interactions between the electron and all the other particles in the system.

INEQUALITY: A mathematical relationship that makes use of the way in which numbers are ordered or a statement indicating that two quantities are not equal. It is represented by the symbols $\neq$ (not equal to), $\leq$ (less than or equal to), $\geq$ (greater than or equal to), $>$ (greater than), $<$ (less than) $\approx$ (almost equal to). A conditional inequality is false for some values of a variable while an unconditional inequality is one that is true for all values of the variable. For example, the notation

$a < b$ means that a is less than b ; the notation $a > b$ means that a is greater than b and the notation $a \neq b$ means that a is not equal to b , but does not say that one is greater than the other or even that they can be compared in size.

INERMOUS (UNARMED): An animal or plant that lacks either spines, prickles, claws, scales or thorns.

INERT: 1. A term describing a substance that does not easily take part in chemical reactions or is inactive because the electron shells (valence shells) are completely filled, so that they have no room to acquire or lose electrons. Examples are the noble gases and metals such as gold which are low in the reactivity series of metals. 2. A pesticide that does not have a toxic effect on the species it is meant to combat.

INERT COMPLEX: A complex ion that undergoes very slow ligand exchange reaction.

INERT GASES (NOBLE GASES): A group of six elements: helium, neon, argon, krypton, xenon and radon, belonging to Group 18 in the Periodic Table (VIIIA) of the elements, whose outermost shells are completely filled, so that they have no room to acquire or lose electrons and therefore unreactive. Inert gases are not completely unreactive, especially for the four heaviest inert gases: argon, krypton, xenon and radon. These gases absorb and give off electromagnetic radiation in a much less complex way than other substances, a property exploited in their use in fluorescent lighting devices and discharge lamps. They glow with a characteristic colour when confined in a transparent container at low pressure with an electric current passing through. Their very low boiling and melting points make them useful as refrigerants for low-temperature research.

INERT S-PAIR EFFECT: A characteristic of the post-transition minerals, with the tendency of their outermost s electrons to remain nonionised or unshared in compounds. Examples of post-transition minerals are aluminium, indium, tin, bismuth, lead and thallium.

INERTIA: 1. In mathematics, the ordered triple of which the entries are the number of positive, negative and zero characteristic roots of the matrix. 2. In physics, the property of matter that causes it to resist any force that would cause a change in its motion or rest; or the tendency of an object to remain in a state of rest or uniform motion until an external force is applied (Newton's first law of motion). The inertia of a body can be measured by its mass, which governs its resistance to the action of a force or by its moment of inertia about a specified axis, which measures its resistance to the action of a torque about the same axis.

INERTIA ELLIPSOID (POINOT ELLIPSOID): An ellipsoid used in describing the motion of a rigid body. It is fixed in the body and the distance from its centre to its surface in any direction is inversely proportional to the square root of the moment of inertia about the corresponding axis.

INERTIA TENSOR: For a set of axes, the tensor represented by the symmetric matrix the entries of whose principal diagonal are the moments of inertia of the rigid body and whose other elements are the negative of the products of inertia for the respective coordinate planes.

INERTIAL CONFINEMENT (INERTIAL CONFINEMENT FUSION-ICF): A process used to initiate nuclear fusion reactions, in which a fuel of high-density pellet that most often contains a mixture of deuterium or tritium enclosed within a tiny glass sphere is bombarded on several sides by laser or heavy ion beam. This causes a rapid implosion of the high-density pellet setting off a thermonuclear reaction that ignites the fuel. Typically, fuel pellets are about the size of a pinhead and contain around 10 milligrams of fuel. In practice, only a small proportion of this fuel will undergo fusion, but if all this fuel were consumed it would release the energy equivalent to burning a barrel of oil. ICF has been proposed as a method for generating power from controlled thermonuclear reactions. Beams of heavy ion accelerators, such as barium ions, rather than lasers are also used to trigger inertial-confinement fusion because they produce intense

ion pulses at high repetition rates; are very efficient at converting electric power into ion beam energy, hence reducing the amount of input power; and also because the ion beams can penetrate the glass sphere and fuel better than lasers.

INERTIAL FORCE: 1. The force produced by the reaction of a body to an accelerating force, equal in magnitude and opposite in direction to the accelerating force. An inertial force lasts only as long as the accelerating force does. 2. A force that must be added to the equations of motion when Newton's laws are used in a rotating or otherwise accelerating frame of reference. It is sometimes described as a fictional force because when the same motion is solved in the frame of the external world, the inertial force does not appear.

INERTIAL GUIDANCE: A type of guidance system used by airplanes, ships, submarines, space vehicles and guided missiles, effected by means of mechanisms that automatically adjust the vehicles heading to follow a given flight path. These mechanisms measure inertial forces during periods of acceleration, integrate the data obtained with already known position and velocity and signal the controls to effect the desired direction, altitude, etc.

INERTIAL MASS (MASS MOMENT OF INERTIA; ROTATIONAL INERTIA; POLAR MOMENT OF INERTIA OF MASS; ANGULAR MASS): A measure of an object's resistance to changes to its angular speed or **rotation** speed. It is the inertia of a rotating body with respect to its rotation. To change the rate of rotation of an object by a certain amount, a relatively large force is required for an object with a large rotational inertia and a relatively small force is required for an object with a small rotational inertia. In automobile engines, flywheels attached to the crankshaft have a large rotational inertia. The engine delivers power in surges; the large rotational inertia of the flywheel absorbs these surges and keeps the engine delivering power smoothly. The moment of inertia plays much the same role in rotational dynamics as mass does in linear dynamics, describing the relationship between angular momentum and angular velocity, torque and angular acceleration and several other quantities. Its symbol is I and its SI unit is kg m^2 . It is calculated as the sum of the products of the component masses of an object multiplied by the square of their distance from the axis.

INERTIAL PLATFORM: A gyro-stabilised reference platform that forms part of a guidance system.

INERTIAL REFERENCE OF FRAME (INERTIAL REFERENCE FRAME): A frame of reference for measurements of motion, space and time, in which bodies move in straight lines with constant speeds unless, acted upon by external forces.

INERTIAL SEPARATOR: Any dry type collector which utilises the relatively greater inertia of particles to effect their removal from a gas stream. Examples are cyclonic and impingement separators, gravity settling chambers and high velocity gas reversal chambers.

INESSENTIAL: In mathematics, not central, not difficult, not important or removable, etc.

INEXTENSIBLE STRING: A string with constant length, irrespective of the tension in the string.

INFEASIBLE: Of or pertaining to an optimisation problem, not satisfying all the constraints; possessing inconsistent constraints; having an empty feasible set. Alternatively, of a point, not lying in a prescribed feasible set.

INFECTION: The invasion of the body by any living disease-causing micro organisms including bacteria, fungi, protozoans, viruses and worms, which proceed to establish themselves, multiply and produce various symptoms in their host. The body responds with an attack on the invader by leukocytes, production of antibodies or antitoxins and often a rise in temperature.

INFECTION THREAD: The cellulosic tube in root hair, through which rhizobia can reach and infect roots.

INFECTIOUS DISEASE (COMMUNICABLE/CONTAGIOUS DISEASE): Any of the diseases that can be transmitted from one person to another by physical contact, a carrier or by inhalation from the air. Examples are ebola, flu, measles, rabies, chlamydia, gonorrhoea, HIV/AIDS and tuberculosis.

INFECTIOUS MONONUCLEOSIS (GLANDULAR FEVER; MONONUCLEOSIS): An acute viral disease that mostly affects young adults and caused by the Epstein-Barr virus that can spread through close, personal contact and is transmitted by saliva. It is characterised by fever, sore throat, fatigue, swollen lymph nodes and glands.

INFERENCE: A process of reasoning in which a conclusion is reached based on certain facts, logic, statistics, etc., obtained by experiment or observation.

INFERENTIAL STATISTICS (DEDUCTIVE STATISTICS): Mathematical methods that employ probability theory for deducing (inferring) the properties of a population from the analysis of the properties of a data sample drawn from it. It is concerned also with the precision and reliability of the inferences it helps to draw. It is complementary to descriptive statistics because the goal of most research is not to establish a certain number of parameters characterising the population associated to the treated sample. In inferential statistics, hypotheses are formulated in the parameters of a population. These hypotheses are then tested on the basis of observations made on a representative sample of the population.

INFERIOR: 1. A structure that is positioned below or lower than another structure in the body. The ovary is described as inferior in flowering plants because of its position below the other organs in flowering plants such as the calyx. 2. Orbiting or taking place between Earth and Sun. For example, Venus and Mercury are known as inferior planets. 3. Written or printed as a subscript, for example, 4 as written in SO_4 .

INFERIOR VENA CAVA (POSTERIOR VENA CAVA): The large vein through which deoxygenated blood passes and empties into the right atrium of the heart. It is posterior to the abdominal cavity and runs alongside of the vertebral column on its right side. It enters the right atrium at the lower right, back side of the heart.

INFERNO: In physics, it is a unit used to describe the temperature inside a star. One inferno is 1 billion K.

INFEST: To inhabit in numbers or quantities large enough to be harmful, threatening or unwanted; or to live as a parasite such as tapeworms or lice in or on something. Infestation is anything harmful, dangerous or unwanted, present in a place in large numbers or to a great degree.

INFILTRATION: 1. To make a liquid or gas pass through a substance such as soil by diffusion. Infiltration may be controlled by factors such as cracks, cultivation, freezing, the intensity and type of precipitation and the porosity of the soil. 2. A substance such as fat that gradually accumulates in cells and tissues.

INFILTRATION RATE: In soils, it is the maximum rate at which rainfall or irrigation water can enter the soil. It is usually measured by the depth (in mm) of the water layer that can enter the soil in one hour. Different soil types have different base rates of infiltration, for example, clay is 10 to 20 mm per hour; for loam to sandy soils, it is 50 mm per hour; for sandy and calcareous (limestone), it is 150 to over 200 mm per hour; and for bare soils, it is 3 to 5 mm per hour. It is lowest when there is no vegetation on the surface. The rate decreases as the soil becomes saturated.

INFILTRATION VELOCITY: The actual rate at which water is entering the soil at a given time. It may be less than the infiltration rate, because of limited supply of rainfall or irrigation water. It is measured in millimetres per hour (mm/h), just like that of infiltration rate.

INFILTRMETER: A device for measuring the rate of entry of fluid into a porous body, for example, water into soil. The most commonly used types are single ring and two rings, which has a smaller inner ring inside a larger outer ring.

INFIMUM, INF (GREATEST LOWER BOUND, GLB): The largest member of the set of lower bounds for some given set that is equal to its minimum if the given set has a least member. The infimum of a set T may be defined as satisfying the two conditions, that $\tau \leq t$ for all t in T and that for all $\epsilon > 0$ there is a $t < \tau + \epsilon$ in T . For example, the geometric sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ has as a lower bound every real number less than or equal to zero; it has no least member and so no minimum but its infimum is 0.

INFINITE: 1. Having no boundaries or limits. 2. Existing beyond or being greater than any arbitrarily large value. 3. Unlimited in spatial extent, for example, a line of infinite length. 4. Of or relating to a set capable of being put into one-to-one correspondence with a proper subset of itself.

INFINITE DESCENT: A method of proof by contradiction that may be used to prove that a result known to be false for some, normally small, positive integer p is false for all positive integers or to prove that a result that is assumed to be true for all integers $n \geq n_0 > 0$ and false for all $n < n_0$ is in fact false for all positive integers.

INFINITE DIMENSIONS: In mathematics, it is a vector space with an infinite basis or a space with infinitely many coordinates.

INFINITE PRODUCT: A product comprising the product of sequence of variables, expressions or numbers. It is denoted by $\prod_{k=1}^{\infty} x_k$ which means that $x_1 \times x_2 \times x_3 \times \dots \times x_{\infty} = \prod_{k=1}^{\infty} x_k$.

INFINITE SERIES: A series of numbers consisting of infinite sequence of terms. An infinite series is a sum with infinitely many terms, for example, $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots + \frac{1}{n^2} + \dots$. An infinite series that results in a finite sum is known as **convergent series**.

INFINITE SET: A set in which all its elements cannot be listed. For example, if C is the set of numbers which are also multiples of 3 then $C = \{3, 6, 9, \dots\}$.

INFINITE-ORDER: Of an element a of a group with identity e , such that there is no finite integer n such that $a^n = e$.

INFINITESIMAL: An immeasurably or incalculably minute amount or quantity but not zero or values arbitrarily close to but greater than zero.

INFINITESIMAL CALCULUS (CALCULUS): The branch of mathematics that permits the manipulation of continuously varying quantities, by two fundamental operations integral and differential calculus.

INFINITY: 1. A quantity having a value that is greater than any assignable value or any quantified concept; its symbol is $+\infty$. Also, **negative infinity** ($-\infty$) is used to denote a value that is less than any assignable value. When two non-zero numbers such as $5 + 7$, are added, the result is 12. But when a large positive number (∞) is added to any positive number (n), the answer is still infinity: $\infty + n = \infty$. A negative infinity ($-\infty$) minus any positive number is still negative infinity: $-\infty - n = -\infty$, where n represents any positive number. Subtracting a negative number such as $-n$ from negative infinity ($-\infty$) gives negative infinity ($-\infty$). A positive or negative infinity (∞ or $-\infty$) multiplied by any number, regardless of size, is still infinity, taking the signs positive or negative into consideration. For example, if the number n is more than zero, then the result is positive infinity (∞): (∞) (n) $= \infty$, $n > 0$; if n is less than zero ($n < 0$), then the result of multiplying positive infinity (∞) by negative number ($-n$) is negative infinity ($-\infty$). Some other examples are: $(-\infty)(\infty) = -\infty$; $(-\infty)(-\infty) = \infty$. With division, $\frac{\infty}{n} = \infty$, if $n > 0$, $n \neq \infty$; $\frac{-\infty}{n} = -\infty$, if $n < 0$, $n \neq -\infty$, etc. 2. A part of

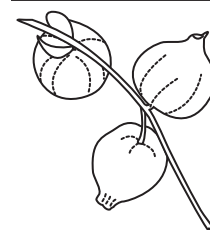
a geometric figure situated at an infinite distance from the observer, for example, the hypothetical point at which parallel lines meet in Euclidean geometry. 3. In optics, it is a point sufficiently far from a lens or mirror that the light emitted from it falls in parallel rays on the surface.

INFIX NOTATION: 1. The notation for binary operators which is written between their arguments; for example, $P \vee Q$ represents disjunction and $x + y$ addition. This requires bracketing to avoid ambiguity in complex expressions such as $2 + g \times 5$. 2. A notation for binary relations in which the symbol for the relation is written between its arguments; thus x has R to y is written xRy . This is less common for most relations than prefix notation, but identity is always written infix as $x = y$.

INFLAMMATION: The end result of the different bodily processes produced in an area of the body as a response to an injurious agent such as viruses, microbes, mechanical or chemical trauma. Some symptoms of inflammation include pain, redness, heat and swelling.

INFLAMMATORY RESPONSE: A defence mechanism in the body to an injurious agent such as viruses, microbes, mechanical or chemical trauma, which triggers physiological responses such as proximal vasodilation, distal constriction of blood vessels, increased leukocytic migration and activity, see page of plasma proteins, increased sensitivity to pain with the increased release of bradykinin and other chemicals by specialised cells.

INFLATED: In botany, it is a plant part that is swollen or unduly enlarged, as shown in the diagram.



Inflated

INFLATIONARY UNIVERSE (COSMIC INFLATION; COSMOLOGICAL INFLATION; INFLATION): A model of

the universe in which the early universe undergoes a period of exponentially rapid expansion, by a factor of at least 10^{78} in volume, required to develop the high degree of homogeneity shown by the present-day universe. The inflationary epoch comprises the first part of the electroweak epoch following the grand unification epoch. It lasted from 10^{-36} seconds after the Big Bang to sometime between 10^{-33} and 10^{-32} seconds. Following the inflationary period, the universe continues to expand.

INFLECTION POINT (INFLEXION): A point on a curve at which the sign of the curvature changes from positive to negative or from negative to positive. In other words, it is a point on a curve at which the curve is oriented concave-upward on one side of the point and concave downward on the other side of the point. If the curve represents the function $y = f(x)$, then the second derivative $\frac{d^2y}{dx^2}$ is equal to zero at the point of inflection.

INFLEXED: In botany, petals or sepals that are bent or curved inward or downward toward the axis.

INFLORESCENCE: A branch or system of branches bearing two or more individual flowers or a group or cluster of flowers or florets on a simple stem or the flowers and ultimately the seed area of the plant.

INFLUENZA (FLU): An infectious viral disease caused by the influenza virus, which when inhaled, attacks cells in the upper respiratory tract. It is similar to the common cold but more serious. Symptoms include high temperature, sore throat, runny nose, headache, dry cough and muscle pain. In domestic animals, it is characterised by respiratory problems and fever.

INFORMATION: 1. A mathematical abstraction of the content of any meaningful statement or data that enables the study of the most efficient way of recording or transmitting them. Its unit is bit, which records no more than the presence or absence of a single feature. 2. Also known as **uncertainty**, it is a real-valued function of events in a probability space that depends only on the probability of events. It is such that events of probability one have zero that increases as

probability drops and that uncertainty simultaneous occurrence of two independent events is the sum of their individual uncertainties and the only measurable functions meeting these requirements are of the form $I(E) = -c \log[P(E)]$ for positive constants c . For a measurable partition ξ has the corresponding information function

$$I(\xi) = -c \sum_{E \in \xi} \log[P(E)]x_E.$$

INFORMATION AND COMMUNICATION TECHNOLOGIES

(ICT): Technologies that provide access to information through telecommunications. It is similar to Information Technology (IT), but focuses primarily on communication technologies. This includes the Internet, wireless networks, cell phones and other communication mediums.

INFORMATION APPLIANCE: A computing device that is used to access, manipulate or transmit information but is not traditionally thought of as a computer. Cellular phones and personal digital assistants (PDAs) are examples of information appliances.

INFORMATION CONTENT EXCHANGE (ICE): In information technology, it is an XML-based standard protocol developed in 1998 for the transfer of data to partnering Web sites for an agreed-upon price.

INFORMATION RETRIEVAL: Recovery of information, especially in a database stored in a computer, which may use microfilm, tapes, computer storage devices, etc.

INFORMATION SUPERHIGHWAY: The global information and communications network that includes the Internet and other networks and switching systems such as telephone networks, cable television networks and satellite communication networks. The term was first used in 1993 by US vice-president AI Gore in a speech outlining plans to build a high-speed national data communications network.

INFORMATION TECHNOLOGY (IT): The use of modern techniques to electronically input, process, store, output, transmit and receive data and information, including text, graphics, sound and video, as well as the ability to control machines of all kinds electronically.

INFORMATION THEORY: The branch of mathematics based on probability theory that analyses information mathematically, to find fundamental limits on signal processing operations such as compressing data and reliably storing and communicating data. In other words, it is concerned with methods of coding, decoding, storing and retrieving information and the likelihood of a given degree of accuracy in the transmission of data through a channel that is subject to probabilities of failure that are described by a probability law. The theory has been used in the study of learning processes and acquisition of skill. One of its most valuable contributions has been in the analysis of the processes associated with selection, perception, memory and decision making in skilled performances. It was developed by Claude E. Shannon (1916-2001) was an American mathematician, electrical engineer and cryptographer.

INFRA SPECIFIC: Occurring within a species such as the variation occurring between individuals, of the same species.

INFRA-: A prefix meaning below or beneath.

INFRA-AXILLARY: Situated below the axil, as seen in a bud.

INFRADIAN RHYTHM: A biorhythm such as menstruation, breeding, tidal or seasonal rhythms with regular recurrence in cycles of more than 24 hours, regardless of conditions of illumination.

INFRAMEDIAL: Situated below the middle.

INFRARED (INFRARED RADIATION, IR): The region of the electromagnetic spectrum extending from 780 nm to about 20000 nm. Any hot object emits radiant heat energy as infrared radiation. Bright sunlight provides an irradiance of just over 1 kilowatt per square metre at sea level. Of this energy, 527 watts is infrared radiation, 445 watts is visible light and 32 watts is ultraviolet radiation.

INFRARED ASTRONOMY: The study of radiation from space in the infrared region of the spectrum. The infrared portion of the spectrum

spans the range from the red limit of human vision (approximately 0.7 micrometre) to the shortest wavelengths accessible to heterodyne radio receivers (several hundred micrometres).

INFRARED CHEMILUMINESCENCE: A technique for studying chemical reaction mechanisms by measuring and analysing weak infrared emissions from product molecules formed in certain chemical reactions.

INFRARED DATA ASSOCIATION (IrDA): Founded in 1993 by around 50 companies, it is an industry-sponsored organisation that sets the standard by which wireless infrared communications may be used to beam data between two devices with built-in IrDA ports, for example, between two computers or a computer and a printer. For two IrDA devices to beam data between them, they must be about 1 metre from each other and must have a clear line of sight. Portable devices such as mobile telephones, laptops, cameras and printers, have IrDA ports.

INFRARED DETECTOR: A device responding to infrared radiation, used in detecting fires or overheating in machinery, planes, vehicles and people and in controlling temperature-sensitive industrial processes.

INFRARED GAS ANALYSER: An instrument used for the quantitative measurement of the proportion of a particular gas in a mixture. Different gases absorb infrared at different characteristic frequencies. Infrared analysers are used for simple measurements, especially the continuous monitoring of air for a given pollutant. They are also used to measure rates of respiration or photosynthesis by recording changes in carbon dioxide level.

INFRARED RADIOMETER: An instrument that measures the intensity of infrared energy coming from source.

INFRARED SPACE OBSERVATORY (ISO): The European Space Agency (ESA) orbiting infrared observatory for studying the Universe at infrared wavelengths of 2.5 to 240 micrometres. The ISO was launched on 17 November 1995 from the ELA-2 launch pad at the Guiana Space Centre near Kourou in French Guiana by an Ariane 44P rocket.

INFRARED SPECTROSCOPY (IR SPECTROSCOPY): A technique for chemical analysis and the determination of structure. It is based on the principles that molecular vibrations occur in the infrared region of the electromagnetic spectrum and functional groups have characteristic absorption frequencies.

INFRASOUND: Sound waves with frequencies lower than 20 Hz and too low to affect the human ear or to detect. Earthquake waves, for example, cause infrared waves that animals can detect but not humans. The vibrations may, however, be felt by the body as a whole or parts of it such as the rib cage.

INFRASPECIFIC VARIATION: Differences in appearance exhibited by different members of the same species or breeding population. The variation may be a result of environmental influences, such as nutrition, crowding, seasonal effects, etc. It may also be due to genetic factors, principally the effects of genetic recombination and mutation. The term may be applied to characteristics only apparent by advanced laboratory techniques, as well as to those readily visible to the naked eye.

INFRASTAMINAL SCALES: Staminode-like formations that surround the ovary in the flowers of dodders.

INFRASTIPULAR: Situated below the stipules. This is usually applied to outgrowths such as spines. This is seen in some roses. An example is the Swamp rose (*Rosa palustris*).

INFUNDIBULUM: Having the shape of a funnel. Any of various funnel-shaped bodily passages, openings, structures or parts, in vertebrates. Examples are the opening of a fallopian tube into the ovary and the stalk connecting the pituitary gland to the brain.

INFUSION: 1. Flavour extracted from an ingredient such as tea leaves, herbs or fruit by steeping them in water or oil. For example, when tea is prepared from leaves or when a liquid is extracted by steeping a plant substance in water. 2. Introduction of fluids,

medication or nutrients into a patient's circulatory system, with the aid of infusion pump, done usually by intravenous but can also by subcutaneous, arterial or epidural.

INGESTION: The process of taking in food by an organism. The method of food capture varies but may involve biting, sucking or filtering. It is a means by which pathogens or harmful substances can be introduced into the body.

INGUINAL: Of, relating to or located in the groin. For example, the inguinal canal which contains the spermatic chord lies in the abdominal cavity of a male foetus.

INHALATION (INSPIRATION): The process by which gas is drawn from the external environment into the lungs through the trachea into the alveoli. In mammals, the rib cage is raised by contraction of the intercostal muscles and the muscles of the diaphragm causing the thorax to enlarge, reducing the pressure in the lung cavity below atmospheric pressure, which causes an influx of air until the pressures are equalised.

INHERITANCE: 1. In genetics, it is the transmission of traits from generation to generation by means of the genetic code which is transferred to offspring through gametes. Each member of the parental generation transmits only half its genes to the offspring and different offspring of the same parents receive different combinations of genes. 2. In object-oriented programming, it refers to the situation/case where a new class absorbs the characteristics of an existing class.

INHIBITION: A reduction in the rate of an enzyme-catalysed reaction, in which a substance called **inhibitor** interacts with the enzyme so that the resulting complex cannot undergo the usual reaction or cannot form the usual product or the slowing down or prevention of a chemical reaction.

INHIBITOR: A substance which will slow down or stop a chemical reaction. Such compounds are usually most effective in low concentrations, usually under 1%. Free-radical chain reaction, such as vinyl polymerisation and autoxidation are most easily inhibited. Corrosion, especially in aqueous system reactions is also a type of reaction for which inhibitors have been found. In enzyme-catalysed reactions an inhibitor frequently acts by binding to the enzyme, in which case it may be called an **enzyme inhibitor**.

INHIBITORY CONCENTRATION (IC): Concentration of a substance that causes a defined inhibition of a given system: IC_{50} is the median concentration that causes 50% inhibition.

INHIBITORY DOSE (ID): Dose of a substance that causes a defined inhibition of a given system: ID_{50} is the median dose that causes 50% inhibition.

INHIBITORY POSTSYNAPTIC POTENTIAL (IPSP): The electric potential that is generated in a postsynaptic neuron when an inhibitory neurotransmitter is released into the synapse, causing a slight increase in the potential difference across the postsynaptic membrane. This makes the neuron less likely to transmit an impulse.

INHOMOGENEITY ERROR: An error due to inhomogeneous distribution of absorbing species in the analytical sample. It may be observed by probing the sample volume with an incident beam of radiation of reduced cross-section.

INHOMOGENEOUS: In mathematics, not homogeneous, especially of a system of linear equations (often differential equations) or inequalities.

INITIAL CONDITION: A boundary condition involving a differential equation at the beginning of the relevant time period, such as the initial velocity and acceleration of a particle of which the position vector is sought.

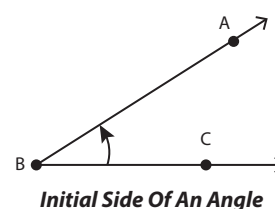
INITIAL LINE: The single axis for polar coordinates.

INITIAL RATE METHOD: A mode of measurement in a kinetic method of analysis, in which the initial reaction rate is determined, usually by the slope method and utilised for the measurement of concentration.

INITIAL SEGMENT: A finite subsequence consisting of successive terms of infinite sequence beginning with the first term; that is all elements of the sequence with indices less than a given one. Similarly,

an initial segment of an ordered set is the set of all elements less than or equal to a given one in terms of that ordering.

INITIAL SIDE (TERMINAL RAY) OF AN ANGLE: The ray on the x-axis where measurement of an angle starts. Where the measurement of an angle stops is called the **terminal side of an angle**. In the diagram, BC is the initial side or terminal ray of the angle and BA is the terminal side or terminal ray of the angle.



INITIAL/FINAL STATE CORRELATIONS: Correlations

between the states of the products of a reaction and the initial states of the reactants. The correlations may relate to energy levels, quantum numbers and symmetries.

INITIALISE: In mathematics, to set variables or parameters at the commencement of an algorithm. For example, when computing the determinant of a matrix by partial pivoting, one initialises the value of the determinant to unity and then iteratively updates that value during the computation.

INITIATING CELL (INITIAL; MERISTEMATIC CELL): One of a group of cells that actively divide to produce the cells of a plant tissue or organ. Examples are the cells of the apical meristem and cork cambium

INITIATION: 1. The first step in translation which occurs when a messenger RNA molecule, a ribosomal subunit and a transfer RNA molecule carrying the first amino acid bind together to form a complex. It begins at the start codon on mRNA. 2. A reaction or process generating free radicals (or some other reactive reaction intermediates) which then induce a chain reaction. For example, in the chlorination of alkanes by a radical mechanism the initiation step is the dissociation of molecular chlorine.

INITIATION CODON (AUG): Three-base sequence on the messenger RNA that codes for the amino acid methionine (Met) in eukaryotes and a modified Met (fMet) in prokaryotes. It is the start command for protein synthesis.

INITIATION FACTOR (IF): Any of a group of proteins that bind to the small subunit of the ribosome required for initiating the translation stage of protein synthesis. Each ribosome is assembled on the messenger RNA. The three major groups are prokaryotic initiation factors, archaeal initiation factors and eukaryotic initiation factors.

INITIATOR: A substance introduced into a reaction system in order to bring about an initiation reaction.

INJECTION: 1. The process of administering drugs by needle and syringe, usually for drugs that would be destroyed by the digestive processes if taken by mouth. Common routes for injection are below the skin called **subcutaneous injection**, for example, for insulin; into a muscle called **intramuscular injection** for drugs that are slowly absorbed; and into a vein called **intravenous injection**, for drugs to be rapidly absorbed. Enemas (introduction of fluid into the rectum) are also regarded as injections. 2. In mathematics, a mapping or function between two sets such that the domain or input of the mapping consists of all the elements of the first set, the range or output consists of some subset of the second set and each element of the first set is mapped to a different element of the second set (one-to-one). A function f is **injective**, if and only if anytime $f(x) = f(y)$, $x = y$.

INJECTION MOULDING: A technique used in forming thermoplastic materials. The hot, molten plastic material is forced into a mould cavity, which is chilled, usually by water, in which it sets. Injection moulding is capable of producing large numbers of parts to very high levels of precision. Holding tolerances of less than 0.0025 mm is easily accomplished with the right combination of material, part design and mould design.

INJECTIVE: 1. In mathematics, of a function or mapping, associating two sets in such a way that different members of the domain or input are paired with different members of the codomain although not all members of the codomain need to be members of the specified range. For example, $f(x) = x^2$ is not injective on the set of real numbers since $f(x) = f(-x)$ but is injective on the positive real number, since an injective mapping is supposed to be one-to-one. A function f is injective, if and only if anytime $f(x) = f(y)$ $x = y$. 2. Of a left R-module Q , having the property that whenever there is a left R-module A with a submodule B such that there is a homomorphism f of B into Q , then f can be extended to a homomorphism g of A into Q .

INJECTIVE MAPPING (INJECTION; ONE-TO-ONE MAPPING):

A mapping $f: S \rightarrow T$ is one-to-one if, anytime s_1 and s_2 are distinct elements of S , their images $f(s_1)$ and $f(s_2)$ are distinct elements of T . So f is one-to-one if $f(s_1) = f(s_2)$ implies that $s_1 = s_2$.

INK: A liquid or paste, containing dyes or pigments, used for writing or printing. The most common is blue-black permanent ink, made by dissolving gallic acid, iron(II)sulfate and tartaric acid in water. Since the blue-black colour is produced only when the ink dries, a dye is usually added to colour the ink during writing. In ballpoint inks, dyes are dissolved in glycols and other liquids and wetting agents added. Black waterproof India ink is a suspension of carbon particles stabilised by gelatin, glue, etc. Carbon black is also the pigment used in black printing ink. Printing inks, which are diverse in their composition and uses, are viscous pastes made by grinding pigments with varnishes or petroleum solvents and contain various additives for printability and drying speeds.

INK-JET PRINTER: A type of computer printer that reproduces digital image using particles or droplets of electrically charged ink from a matrix of tiny ink-jet.

INMARSAT: A network of communications satellites operated by the International Maritime Satellite Organization (IMSO) to support global mobile transmissions at sea, in the air and on land.

INNATE: In botany, it defines anthers that are joined to the very tip of the filament by the base only.

INNATE BEHAVIOUR: An inherited pattern of behaviour that appears in a similar form in all normally reared individuals of the same sex and species.

INNATE IMMUNITY: An immunity consisting of cells and proteins that are already present in the body, which it mobilises to fight micro-organisms that invade the body. The innate immune system consists of physical epithelial barriers (such as the skin, tear fluid, acid in the stomach and mucus in the airways), phagocytic leukocytes, dendritic cells and a type of lymphocyte called a natural killer (NK) cell that destroys cells infected by viruses and cancerous cells and circulating plasma proteins.

INNATE RELEASING MECHANISM: In ethology, a circuit within an animal's brain that is hypothesised to respond to a specific stimulus, setting in motion or "releasing", the sequence of movements that constitute a fixed action pattern.

INNER: Describing a chemical compound formed by reaction of one part of a molecule with another part of the same molecule. Thus, a lactam is an inner amide; a lactone is an inner ester.

INNER AUTOMORPHISM: A function in which an initial operation is applied, then another operation and then the initial operation is reversed. An inner automorphism of a group G is a function: $f: G \rightarrow G$ defined for all r in G by an element a in the group or ring by $r \mapsto a^{-1}ra$, where a is a given fixed element of G and the operation $a^{-1}ra$ is the conjugation. The set of all inner automorphisms of a group form a normal subgroup of the automorphism group that is isomorphic to the factor group of the given group by its centre.

INNER CELL MASS: A cluster of cells in a mammalian blastocyst that protrudes into one end of the cavity, which develops into the body of the embryo.

INNER CORE: The solid material found in the centre of some planets at extremely high temperature and pressure. For Earth, inner core is composed of a nickel-iron alloy and its hottest part, even hotter than the surface of the Sun.

INNER EAR (INTERNAL EAR; *Auris interna*): Part of the ear containing organs of hearing and equilibrium or balance surrounded by the temporal bone of the skull. The bony labyrinth has three sections: semicircular canals, vestibule and cochlea. Within each structure is a corresponding part of the membranous labyrinth: semicircular ducts, two saclike structures in the vestibule and cochlear duct. Sound vibrations are transmitted from the middle ear through the membrane-covered oval window to fluid in the snail-shell-shaped cochlea, whose motion stimulates hair cells in the cochlea. The hair cells trigger nerve impulses that travel to the brain, which interprets them as sound. The vestibule and semicircular canals also have organs with hair cells. Those in the vestibule indicate the head's position with respect to the rest of the body. The three semicircular canals, at right angles to each other, signal motion of the head in three-dimensional space. Continued stimulation after motion stops causes a mismatch with visual input, experienced as dizziness or motion sickness.

INNER JORDAN MEASURE (JORDAN INNER CONTENT; JORDAN INNER MEASURE): The supremum of the volumes of finite collections of disjoint hypercubes lying inside the set; thus, a type of internal quadrature.

INNER MEASURE: The supremum of the measures of Lebesgue measurable subsets of a set in Euclidean space.

INNER ORBITAL COMPLEX: Valence bond designation for a complex in which the metal ion utilises d orbitals for one shell inside the outermost occupied shell in its hybridisation.

INNER PLANET (TERRESTRIAL PLANET): Any of the four highly dense planets whose orbit lies nearest the Sun, or with orbit lying within the asteroid belt. The planets are Mercury, Venus, Earth, and Mars. The inner planets have similar size and composed primarily of silicate rocks or metals.

INNER PRODUCT (DOT PRODUCT; SCALAR PRODUCT): The dot product of U with unit vector V , denoted $U \cdot V$, is defined as the projection of U in the direction of V or the amount that U is pointing in the same direction as unit vector V . This operation that can be defined algebraically as $U \cdot V = U_1V_1 + U_2V_2 + U_3V_3$ or geometrically as $U \cdot V = |U||V|\cos\theta$, where $|U|$ and $|V|$ are the magnitudes of U and V respectively and θ is the angle between them.

INNER PRODUCT SPACE: A complex or real vector space endowed with an inner product. This inner product associates each pair of vectors in the space with a scalar quantity known as the inner product of the vectors.

INNERVATION: 1. In biology, it is the distribution of a nerve fibre to and from an organ. 2. The amount of nerve stimulation received by a muscle organ or other parts of the body.

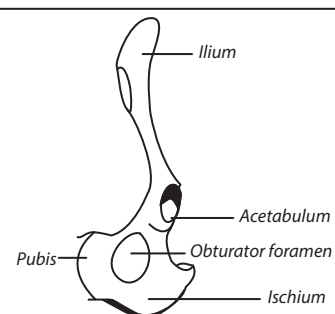
INNERVATION RATIO: Number of muscle cells per motor neuron. It is variable from muscle to muscle. It is low for fine motor control such as hand muscles and high for gross motor control such as big leg muscles.

INNOCUOUS: Producing no injury; lacking thorns or spines.

INNOMINATE ARTERY (BRACHIOCEPHALIC TRUNK; BRACHIOCEPHALIC ARTERY): A short artery that branches from

the aortic arch to divide into the subclavian artery and the right carotid artery. It supplies blood to the right arm and the head and neck.

INNOMINATE BONE (OS INNOMINATUM): One of the two large flat bones that form each half of the pelvic girdle in adult vertebrates consisting of the fused ilium, ischium and pubis.



Innominate bone

INNOMINATE VEINS (BRACHIOCEPHALIC VEIN): The wide vessels that collect blood from the head and neck and the upper limbs. They consist of a pair of veins, each formed by the union of the internal jugular and subclavian veins that join to form the superior vena cava.

INNOVATION: In botany, it is a short basal shoot or offset from the base of a stem.

INOCULATION (VACCINATION; IMMUNISATION): The prevention of infection by artificially introducing a pathogen into the body to stimulate resistance in the human body to specific diseases using microorganisms that have been modified or killed. This triggers the body's immune system to build a defence mechanism that guards against the disease.

INOCULUM: A small amount of material containing bacteria and other microorganisms that is used to start a culture. In botany, under artificial conditions, this may be a suspension of spores, which is sprayed onto the plant. In nature, the inoculum is carried to a healthy plant from its source by, for example, an infected plant or an agent, such as wind or insect.

I INORGANIC: Substances that do not form part of living things and have no life. Examples are rocks, water, etc.

INORGANIC CHEMISTRY: The branch of chemistry concerned with properties and behaviour of compounds of elements other than carbon. Certain simple carbon compounds, such as carbon monoxide (carbonous oxide, CO), carbon dioxide (CO₂), carbon disulfide (CS₂) and carbonates and cyanides, are usually treated in inorganic chemistry.

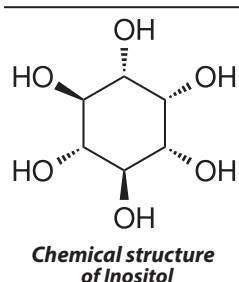
INORGANIC COMPOUND: Chemical substances or compounds within cells or found in organisms which are derived from the external physical environment and are not organic or typically biological or any substance in which two or more chemical elements other than carbon are combined, as well as some compounds containing carbon but lacking carbon-carbon bonds. Examples are water, common salt or sodium chloride, potassium and tetraoxosulfate(VI) acid (H₂SO₄).

INORGANIC FERTILIZER (MINERAL FERTILIZER, COMMERCIAL FERTILIZER): Fertilizer manufactured from synthetic compounds or mined from mineral deposits. Inorganic nitrogen fertilizers include ammonium nitrate, potassium nitrate, calcium nitrate and urea; inorganic potassium fertilizers include potassium sulphate (sulfate), potassium nitrate and potassium chloride; inorganic phosphorus fertilizers include rock phosphate and superphosphate. The two categories of fertilizers are organic and inorganic. Organic fertilizers undergo little processing and include ingredients such as compost and manure while inorganic fertilizers are synthetic and typically made from petroleum products.

INORGANIC POLYMER: Polymer or polymer network with a skeletal structure that does not include carbon atoms. Examples include polyphosphazenes, polysilicates, polysiloxanes, polysilanes, polysilazanes, polygermanes, polysulfides, etc.

INORGANIC-ORGANIC POLYMER (IOP): Polymer or polymer network with a skeletal structure comprising inorganic and organic units. Examples include poly(carbasilane)s, poly(phenylenesilanediy)s, poly(phenylenesulfide)s, etc.

INOSITOL (CIS - 1, 2, 3, 5 - TRANS - 4, 6 - CYCLOHEXANEHEXOL; CYCLOHEXANEHEXOL; NUCITE; PHASEOMANNITE): A cyclic alcohol that is a constituent of certain cell phosphoglycerides with the chemical formula C₆H₁₂O₆, density 1.752 g/cm³ and melting point 225 to 227 °C (498-500 K). The most important isomer is myo-inositol, a water-soluble crystalline compound. It is sometimes classified as



a member of the vitamin B complex but can be synthesised by many animals and is regarded as essential nutrients in humans.

INOSITOL TRISPHOSPHATE (IP3; INOSITOL 1,4,5-TRISPHOSPHATE): 1. The second messenger, which functions in signal transduction and lipid signalling in biological cells. Its molecular formula C₆H₁₅O₁₅P₃. 2. A rise in cytoplasmic Ca²⁺ concentration.

INOX STEEL (CORROSION-RESISTANT STEEL, CRES; STAINLESS STEEL): A widely used alloy of iron, chromium and nickel that resists rusting. It contains a minimum of 10.5 or 11% chromium content by mass. Stainless steel does not stain, corrode or rust as easily as ordinary steel, but it is not stain-proof. There are different grades and surface finishes of stainless steel to suit the environment to which the material will be subjected in its lifetime. It is used where both the properties of steel and resistance to corrosion are required. Stainless steels are used in petroleum refineries and chemical plants for the pipes and tanks, for jet planes and for space capsules, for surgical instruments and equipment, for replacing broken bones because the steels can withstand the action of body fluids. It is also commonly used for cutlery and kitchen fittings because it does not taint food and can be easily cleaned.

INPUT: 1. A number inserted into an imaginary function machine, which applies a rule to pair the input with an output. 2. The values for x in a function consisting of ordered pairs (x,y) . 3. Numbers or other information entered into a calculator or computer.

INPUT DEVICE: Any peripheral part of a computer such as a keyboard that feeds it with data and program instructions. To **input** a data is to enter data into a computer. Input can be text typed in a word processing document, keywords entered in a search engine's search box or data entered into a spreadsheet. Input can also be moving the mouse or clicking the mouse button or it can be scanning a document or downloading photos from a digital camera.

INPUT SET: In information theory, the set of signals from which the sender selects one to represent a message.

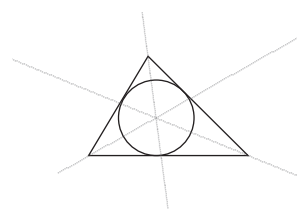
INPUT-OUTPUT MODEL (LEONTIEF MODEL): A matrix of economic production that shows the relationships between different industries in an economy and to determine whether an economy is productive or non-productive by looking at the Leontief models. In the model, one commences with n goods together with a matrix A whose i,j^{th} entry specifies the number of units of good i necessary to produce one unit of good j . The quantity $y = (I - A)x$ then represents the output of the economy corresponding to input x . This determines whether and how a given net output is feasible by solving the matrix system.

INPUT/OUTPUT CHANNEL: 1. In a data processing system, a functional unit that handles transfer of data between internal and peripheral equipment. 2. In a computing system, a functional unit, controlled by a processor, that handles transfer of data between processor storage and local peripheral devices.

INQUILINISM: An association between two different species in which one species, the inquiline, lives on or in the other or inside the hosts home or habitat without causing any detriment to the host. Inquilinism is considered a variety of commensalism.

INROLLED: A leaf or flower that is curled or rolled inward.

INSCRIBE: To construct a geometrical figure inside another figure so that the two have points in common but no intersection. For example, an **inscribed angle** is an angle formed by two chords of a curve that meet at the same point on the curve; and an **inscribed circle** is a circle drawn inside a triangle with its circumference touching all three sides of the triangle.



Inscribed circle in a triangle

INSCRIPTION: In plane geometry, the construction of a circle such that each side of a polygon is a tangent to it. An inscribed angle is one whose sides are chords of a circle and whose vertex lies on the circle. In three dimensions, inscription implies the construction of a sphere such that every side of a regular polyhedron is a tangent plane to it.

INSECT: Members of the class Insecta of the phylum Arthropoda within the arthropods that have a chitinous exoskeleton, jointed legs with body divided into head, thorax and abdomen (three body segments), compound eyes and two antennae. Examples of insects are flies, beetles, wasps and mosquitoes.

INSECT GROWTH REGULATOR: Any of the various substances that interrupts or inhibits the normal development or growth of insects, some of which are used to control insect pest.

INSECTICIDE: Any inorganic or organic chemical pesticide used to kill insects or a chemical substance that kills insects and other arthropods by disruption of vital processes through chemical action. It is used to control pests that infest cultivated plants and crops or to eliminate disease-carrying insects in specific areas. Insecticides are classified according to the type of action such as stomach poisons, contact poisons, residual poisons, systemic poisons, fumigants, repellents, attractants, insect growth regulators or pheromones. Inorganic insecticides include arsenic, lead and copper compounds. Some organic insecticides are natural, such as rotenone, pyrethrins and nicotine (toxin). Others are synthetic, such as chlorinated hydrocarbons, for example, dieldrin and lindane.

INSECTIVORE (INSECTIVORA): An order of small, placental, mainly nocturnal mammals that include the hedgehogs, moles and shrews of the class Mammalia, feeding chiefly on insects. Currently, the order has been reclassified into three orders: Soricomorpha, (shrews and moles), Erinaceomorpha (hedgehogs and Afrosoricida (tenrecs and golden moles). They have long snouts covered with tactile hairs and specialised teeth for seizing and crushing insects and other small prey. **Insectivorous plant** or **carnivorous plant** is a plant that is adapted to obtain food by digesting small animals, particularly insects in addition to feeding by photosynthesis. Such plants are found in regions, where the soil is deficient in certain nutrients, particularly nitrates. Examples are butterworts (*Pinguicula*) and sundews (*Drosera*) growing on heathland and moorland and tropical plants, such as the Venus' flytrap (*Dionaea muscipula*) and the various pitcher plants (families Nepenthaceae and Sarraceniaceae). The plants are adapted in a number of ingenious ways to attract, trap and kill the insects, which are then digested by proteolytic enzymes secreted by the plant. Butterworts trap insects by having sticky infolded leaves. Venus' flytraps have folding leaves with spines along the edges that spring together and trap the insect. The leaves of pitcher plants form a long narrow container shaped like a pitcher, into which the insects fall and drown in the liquid at the bottom.

INSEMINATION: The placement of semen in the female uterus of a mammal or the oviduct of an oviparous during copulation with another animal. Insemination which takes place through sexual intercourse is known as **natural insemination**. Insemination by some means other than that is called **artificial insemination**.

INSERT: 1. In biochemistry, it is a sequence of foreign DNA introduced into a restriction site (insertion site, cloning site) of a vector DNA. 2. A folded section placed in another for binding in with a book; or a printed sheet, such as one bearing advertising, placed loose between the leaves of a book, periodical, etc.

INSERT KEY (INS): In computing, it is a key commonly found on most keyboards near or next to the backspace key. It is often abbreviated Ins. It is primarily used to switch between insert and overtype mode or to insert an object at the current cursor location. The two text-entering modes on a personal computer (PC) or word processor are overtype mode, in which the cursor, when typing, overwrites any text that is present in the current location; and insert mode, where the cursor inserts a character at its current position, forcing all characters past it one position further.

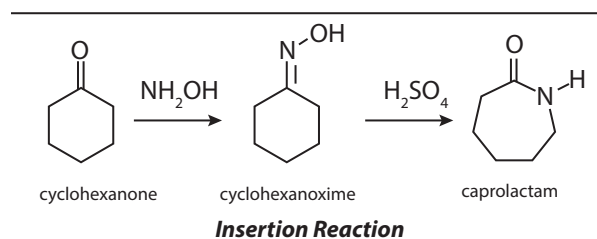
INSERTED: 1. In botany, it refers to parts of flowers joined by natural growth. An example is the tree trunk and vines which have grown together. 2. In anatomy, having an insertion such as a muscle, tendon or ligament; attached, as the end of a muscle that moves a bone.

INSERTION (INSERTION MUTATION): A point mutation in which one or more nucleotidebase pairs is added to DNA sequence or chromosome. When one or more nucleotides are inserted into the genetic sequence, the reading frame is disrupted during transcription and eventually, translation. This may result in an altered sequence of amino acids, extra amino acid in a chain or premature termination. Hence, the newly synthesised protein could be abnormally short, abnormally long and/or contain the wrong amino acids. It will most likely not be functional. Insertion of a larger segment into the chromosome can also lead to mutation. It results when an unequal crossover happens during meiosis. The addition of non-coded nucleotides during recombination by terminal deoxynucleotidyl transferase is called **N region addition**. **P nucleotide** insertion is the insertion of palindromic sequences encoded by the ends of the recombining gene segments. Insertion mutation.

INSERTION LOSS: 1. The attenuation or loss in signal power, resulting from the insertion of a component, such as a connector or splice, in a circuit. It is measured as a comparison of signal power at the point the incident energy strikes the component and the signal power at the point it exits the component. Insertion loss typically is measured in decibels (dB), although it also may be expressed as a coefficient or a fraction. A negative loss is a gain. 2. In a fibre optic transmission system (FOTS), it is a measure of loss across a circuit due to all factors, including absorption, bending loss from both macrobends and microbends, diffusion, dispersion, Fresnel reflection and leaky modes.

INSERTION POINT: The location on the screen where the next character typed will be inserted. This location may be in a text document, a form field, a Web browser address bar or anywhere else on the screen that allows text input. The insertion point is often identified by a flashing cursor.

INSERTION REACTION: A type of chemical reaction in which an atom is inserted between two atoms in a molecule. Usually, molecules such as sulfur dioxide, carbon monoxide and alkenes are inserted into metal-carbon bonds and sometimes into metal-hydrogen bonds. A common example is the Beckmann reaction for the preparation of caprolactam as shown below:



INSERTION SEQUENCE: The simplest kind of a transposon, consisting of inserted repeats of DNA flanking a gene for transposase, the enzyme that catalyses transposition.

INSIDE: Abbreviated ins, with the symbol $\text{ins } \Gamma$, in mathematics, it is the set of all points, not lying on a given contour Γ , for which the winding number is not zero.

INSIGHT LEARNING: A form of learning in which an animal such as chimpanzee responds to a new situation by adapting the experiences gained in other contexts. Insight learning requires an animal to solve problems by viewing a situation as a whole instead of relying wholly on trial and error.

INSIPID: Lacking taste or savour.

ISOENZYMES (ISOZYMES): Enzymes, which are different in amino composition and acid sequence, but act as catalysts for the same reaction, or set of reactions.

INSOLATION (SOLAR INSOLATION): The incident radiant energy emitted by the Sun that is received on Earth. It is measured by the amount of solar energy received per square centimetre per minute.

INSOLUBLE: A term describing a substance, which does not dissolve in a solvent or a given liquid. For example, salt is insoluble in ethanol.

INSPIRATION (INHALATION): The process by which gas is drawn from the external environment into the lungs through the trachea into the alveoli. In mammals, the rib cage is raised by contraction of the intercostal muscles and the muscles of the diaphragm causing the thorax to enlarge, reducing the pressure in the lung cavity below atmospheric pressure, which causes an influx of air until the pressures are equalised.

INSPIRATORY CAPACITY: The maximum volume of gas that can be drawn into the lungs after a normal expiration.

INSPIRATORY CENTRE: The most important sub-centre of the ventilation centre in the brain which controls the regularity of breathing. It is situated in the medulla oblongata and is composed of a group of neurons.

INSTABILITY: The condition of a system when it is easily disturbed by internal or external forces or events, in contrast to a **stable system**, which will return to its previous condition when disturbed. An unstable system is one whose attractors can change; thus, instability is a characteristic of a system that is far from equilibrium or at bifurcation.

INSTAGRAM: A popular freely accessible, online photo-sharing application and social network platform that allows users to edit and upload photos and short videos using a mobile application. It was acquired by Facebook in 2012. So popular is the platform that as of June 2018, the monthly active users of the network had reached 1 billion, up from 800 million in September 2017.

INSTALLATION PROGRAM: A computer program used in installing hardware or applications.

INSTANT MESSAGING (IM): A type of communications service which can be used to create a kind of private chat room with another individual in order to communicate in real time over the Internet. It is analogous to a telephone conversation but using text-based, not voice-based, communication. The IM system alerts you whenever somebody on your private list is online. You can then initiate a chat session with that particular individual.

INSTANTANEOUS: Of any property of a function of time, occurring at or associated with a given instant or as a limit when a time interval approaches a point.

INSTANTANEOUS ACCELERATION: The limiting value of the acceleration calculated over a very short time interval at the appropriate instant of time. It can also be defined as the limit of average acceleration as the time interval approaches zero. It is the slope of the tangent of the velocity-time graph at the same instant of time. Instantaneous acceleration is calculated as $\lim_{\Delta t \rightarrow 0} \Delta v / \Delta t = dv/dt = a$.

Instantaneous velocity is the derivative of displacement or position with respect to time.

INSTANTANEOUS (SPOT) SAMPLING: Obtaining a sample of the atmosphere in a period which is short compared with the duration of the sampling exercise. These are useful for the analysis of hydrocarbons and other complicated mixtures of trace gases which are relatively stable in a stainless steel canister or tank and can be transported back to the laboratory for chromatographic or other analysis which cannot be done satisfactorily in the field.

INSTANTANEOUS BANDWIDTH: The widest contiguous frequency range across which a radio telescope and its associated receiving equipment can make observations at any given time.

INSTANTANEOUS CURRENT: At a dropping electrode, the total current that flows at the instant when a time has elapsed since the fall of the preceding drop. At any other electrode, the total current that flows at the instant when a time has elapsed since the beginning of an electrolysis. The instantaneous current is usually time dependent and may have the character of an adsorption, catalytic, diffusion, double-layer or kinetic current and may include a migration current. A plot of the dependence of instantaneous current on time is commonly called an 'i-t curve'.

INSTANTANEOUS VALUE: The value of any sinusoidal or varying quantity at a specified instant.

INSTANTANEOUS VELOCITY: For a body moving with varying velocities, it is the velocity at a particular instance of time.

IN instantiation: 1. In logic, **instantiation rule** is deriving an individual statement from a general one by eliminating a specified quantifier and replacing the bound variable with a name or other referring expression. 2. A concept in object-oriented programming, it is the production or creation of an object (instance) to give the object a unique identity by the allocation of a structure with the types specified by a template and initialisation of instance variables with either default values or those provided by the class's constructor function.

INSTAR: The developmental stage of arthropods, such as insects, during the period between moults. Arthropods must shed the exoskeleton in order to grow or assume a new form. Differences between instars can often be seen in altered body proportions, colours, patterns or changes in the number of body segments.

INSTINCT: An innate, involuntary response found in all equivalent members of a species or an inherited pattern of behaviour that serves to promote the survival of an animal or species.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE): A USA based international non-profit organisation founded in 1884 that develops, defines and reviews electro technology and allied sciences for the benefit of humanity, the advancement of the profession and the well-being of members.

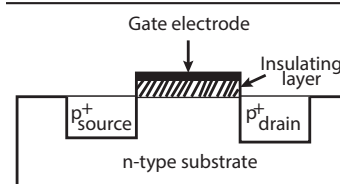
INSTRUMENT: 1. An object used to aid in working, especially where exact details and measurement are required such as in science. 2. A measuring device used to gauge the level, position, speed of something, especially a motor vehicle or aircraft. 3. A mechanical tool or implement, especially one used for delicate or precision work, for example, surgical instruments. 4. A contrivance or apparatus for producing musical sounds, for example, a piano. 5. A mechanical or electronic measuring device, for example, one used in navigation.

INSTRUMENTAL DEPENDABILITY: This relates to the frequency of failures which interrupt the operation of an instrument. It can be quantified, for example, by the infrequency of breakdowns or by the availability of the instrument for use when required.

INSULATED GATE FIELD-EFFECT TRANSISTOR (IGFET; METAL OXIDE FIELD EFFECT TRANSISTOR, MOSFET): A type of field

effect transistor (**FET**), which differs from the JFET by the addition of a silicon dioxide layer over the JFET and then a layer of silicon nitride. The result is a device which has even higher input impedance. It has a source, gate and drain terminals like the FET. However, the gate lead does not make a direct connection

to the silicon compared with the case for the FET. The gate is a metallic or polysilicon layer atop a silicon dioxide insulator. The gate bears a resemblance to a metal oxide semiconductor (MOS) capacitor. When charged, the plates of the capacitor take on the charge polarity of the respective battery terminals. The lower plate is P-type silicon from which electrons are repelled by the negative (-) battery terminal



Insulated Gate Field-effect Transistor (IGFET or MOSFET)

toward the oxide and attracted by the positive (+) top plate. This excess of electrons near the oxide creates an inverted (excess of electrons) channel under the oxide. This channel is also accompanied by a depletion region isolating the channel from the bulk silicon substrate. MOSFETS have several advantages over junction transistors as the basic component in integrated circuits. There are fewer steps in processing and there is a smaller area per device. They, therefore, have a greater functional density and lower cost than junction transistors. Other advantages include their high value resistors and their low power dissipation.

INSULATOR (INSULATION): A layer of material that is a poor conductor of heat or electricity. It reduces energy transfer from one place to another. Used as a layer of material, usually plastic to cover a wire carrying an electric current, it prevents short circuits between wires and protects against electric shock. Common electrical insulators include rubber, plastic, porcelain and mica and thermal insulators, which break up the heat-flow path by absorbing radiant heat. Insulators also include fibreglass, cork and rock wool.

INSULIN: A hormone, which is a small, simple protein secreted by the beta cells of the islets of Langerhans of the pancreas. It helps to regulate the amount of glucose in blood for storage as glycogen in the liver. Insulin is now produced using genetic engineering.

INSULIN-LIKE GROWTH FACTOR (IGF): Any of certain polypeptides that are structurally similar to insulin and promote division and growth of cells. They are part of a complex system that cells use to communicate with their physiologic environment. Growth hormone stimulates the liver to release them.

INTAKE: 1. An amount of a substance taken into an organism, either eaten or absorbed. 2. A hill pasture which has been improved and fenced in. 3. In internal-combustion engines the **intake valve** is a valve in the cylinder head that opens at the proper moment in the cycle to allow the fuel-air mixture to be drawn into the cylinder.

INTEGER: 1. An entity or a whole unit. 2. A negative or positive whole number with no fractional part. It includes the counting numbers $\{1, 2, 3, \dots\}$, zero $\{0\}$ and the negative of the counting numbers $\{-1, -2, -3, \dots\}$; the set can be written as $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ and denoted as $\mathbb{Z}$. It is sometimes called **rational integer** to differentiate it from **algebraic integer**, which is the root of a polynomial with integer coefficients, where the coefficient of the highest power is 1. The integers are the closure of the natural numbers under subtraction and are identified with the rational numbers with denominator 1.

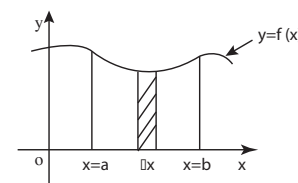
INTEGER LATTICE: A subset of Euclidean n -space that is closed under addition and subtraction; most usually, such a set constructed as all integral combinations of n linearly independent points or generators.

INTEGER PROGRAMMING: The extension of linear programming in which some variables in mixed programming or all variables in pure integer programming are restricted to the integers. Most computing problems are expressible in this form.

INTEGER QUANTUM HALL EFFECT (QUANTUM HALL EFFECT; VON KLITZING EFFECT): A phenomenon exhibited by certain semiconductor devices at low temperatures and high magnetic fields, whereby the Hall resistance becomes precisely equal to $(h/e^2)/n$, where h is Planck's constant, e is the electronic charge and n is either an integer or a rational fraction. It was discovered in 1879 by American physicist Edwin Hall. The German physicist, Klaus-Olaf von Klitzing discovered the quantum Hall effect, which helps explain electrical movements in atoms, for which he won the 1985 Nobel Prize in physics.

INTEGRABLE: 1. Of a function, possessing a finite integral in the sense of Lebesgue or Riemann integration or otherwise. 2. Of a differential equation or form, being the derivative of another form.

INTEGRAL: 1. Being, containing or relating to one or more mathematical integers; or relating to or concerned with mathematical integration or the results of mathematical integration. 2. For a given function, $f(x)$, the limit, evaluated by the integral calculus, of the sum of the rectangular elements $f(x)\delta x$, where δx is a subinterval of a partition of an interval of values of the independent variable. This is sometimes expressed in terms of an infinite sum of infinitesimal quantities. An indefinite integral or antiderivative, written $\int f(x)dx$, is any other function of x whose derivative is $f(x)$ and is unique up to a constant;



Element of area = $f(x) \delta x$
Area enclosed by the curve
 $y=f(x)$, x -axis, the lines $x=a$
and $x=b$ is

$$\int_a^b f(x)dx = \lim_{\delta x \rightarrow 0} \sum_{i=1}^n f(x_i) \delta x_i$$

for example, the indefinite integral

of ax^n is $\frac{ax^{n+1}}{n+1} + c$, where c is a

constant, called **constant of integration**, which is often omitted in the definite integral. Integrals are formally defined in terms of upper and lower Darboux sums; and the definite and indefinite integrals are related by the fundamental theorem of calculus.

INTEGRAL CALCULUS: The branch of calculus that is concerned with the evaluation of integrals and their application to the determination of areas, volumes and lengths and to the solution of differential equations. It is useful in doing calculations that involve a continually varying quantity, for example, to determine the position at any given instant of a space rocket that is accelerating away from Earth.

INTEGRAL DOMAIN: A non-zero commutative ring with a (multiplicative) identity in which the (additive) zero has no zero divisors, for example the integers, but not the integers mod m unless m is prime. A ring is an integral domain if and only if $ax = ay$ implies $x = y$ and if an integral domain is finite it is a field. In algebra, a non-zero commutative ring with no zero divisors, whether or not it possesses a multiplicative identity.

INTEGRAL EQUATION: A functional equation that involves an integral of the function which is to be solved.

INTEGRAL PART (INTEGER PART): The integral part denoted **int** of a real number, is the largest integer not greater than the given number; for example, the integral part of $[3.42]$, written **int** 3.42 is 3 and of -3.42 is -3 .

INTEGRAL POLYNOMIAL: A polynomial with integer coefficients.

INTEGRAL RATIONAL: Of an expression, function, equation, etc., expressible as a ratio of polynomials with integral coefficients.

INTEGRAL TEST: A test for the convergence of an infinite series $\sum_n f(n)$, where f is continuous, non-negative and monotone decreasing on $(1, \infty)$, because the series converges if and only if the improper integral $\int_1^\infty f(x) dx$ converges.

INTEGRAL TRANSFORM: Any transform, often used in the solution of partial differential equations of the form $\int_A k(x,y) f(x) dy$ is an integral transform of f , where A is a range fixed for that transform and k is the kernel. Integral transform is a particular kind of mathematical operator and include Fourier transform, Laplace transform and Mellin transform.

INTEGRAND: The function that is integrated in an integral.

INTEGRATED ASSESSMENT: A method of analysis that combines results and models from the physical, biological, economic, and social sciences, and the interactions between these components, in

a consistent framework, to evaluate the status and the consequences of environmental change and the policy responses to it.

INTEGRATED CIRCUIT; I.C. (CHIP; MICROCHIP): A miniature or small silicon semiconductor crystal that contains electrical components such as transistors, diodes, resistors and capacitors. Their interconnections are fabricated as a single unit on a wafer of semiconducting material, especially silicon. It is used in electronic equipment such as radios and television sets.

INTEGRATED CONTROL: The control of pests using a combination of chemical and biological means.

INTEGRATED CROP MANAGEMENT: An approach to growing crops that combines traditional good farm husbandry with reduction in the use of agrochemicals, taking into consideration the impact of farming practices on the environment.

INTEGRATED DRIVE ELECTRONICS (IDE): A standard interface for mass-storage devices used between a computer motherboard's data paths or bus and the computer's disk storage devices where the controller is integrated into the disk drive (either a hard disk, a high-capacity removable disk drive or a CD-ROM drive). It is the most popular interface used in modem hard disks and is most commonly used to refer to the advanced technology attachment (ATA) specification.

INTEGRATED FARM MANAGEMENT: An approach to farming that combines the best of traditional methods with modern technology, to achieve high productivity with a low environmental impact.

INTEGRATED PEST MANAGEMENT (IPM): The ecologically based approach of controlling insects and other crop pest that combines many pest-control methods such as physical, chemical and biological in an attempt to reduce reliance on chemical pesticides and hence minimise pollution and harmful residue in the product. Some of the methods are breeding of pest-resistant crop varieties, the development of crop culture methods that inhibit pest proliferation, the release of predators or parasites of the pest species and the placement of traps baited with the pest's own sex attractants (pheromones).

INTEGRATED POLLUTION CONTROL: An approach which looks at all inputs and outputs from a process that are likely to cause pollution and tries to regulate them as well as other factors such as emissions.

INTEGRATED RATE EQUATION: An equation giving the concentration of a reactant or product remaining after a specified time. The rate of reaction is proportional to the rates of change in concentrations of the reactants and products; that is, the rate is proportional to a derivative of a concentration. For each integrated rate law, there is a characteristic plot that can be created which produces a straight line.

INTEGRATED RATE LAW: The concentration of a reactant or product in a chemical reaction as a function of time (the curve itself). For a zero order reaction: $A \rightarrow \text{products}$, rate = k , the integrated rate law is $[A] = -kt + \ln [A_0]$. For a first order reaction: $A \rightarrow \text{products}$, rate = $k [A]$, the integrated rate law is $\ln[A] = -kt + \ln[A_0]$. For a second order reaction:

$2A \rightarrow \text{products}$ or $A + B \rightarrow \text{products}$ (when $[A] = [B]$), rate = $k[A]^2$, the integrated rate law is $1/[A] = kt + 1/[A_0]$. The **rate law** is the instantaneous rate (the slope of the curve) as a function of concentration. **Zero-order reactions** are reactions that appear to occur at a rate independent of substrate concentration.

First-order reactions are reactions in which the rate varies with concentration of a single species and the change in concentration is exponential, so that a plot of $\ln(\text{concentration})$ against time is linear. The stoichiometric coefficient is 1. An example of first-order reaction is radioactive decay, but many electron transfer processes and most enzyme mechanisms contain intermediate reactions that

are first order. Second-order reactions are of two types: Second-order reactions of Classes I and II. **Second-order reactions of Class I** reactions are reactions in which the rate varies with concentration of a single species, but the stoichiometric coefficient is 2. The rate varies with the reciprocal of the concentration, so that a plot of $1/(\text{concentration})$ against time is linear. **Second-order reactions of Class II** are reactions in which the rate varies with the concentration of two substrates, each of which has a stoichiometric coefficient of 1. A plot of $\ln(\text{conc.A}/\text{conc.B})$ against time is linear. Examples of second-order reaction are found in many important biological reactions, such as the formation of double-stranded DNA from two complementary strands. **Higher-order** or **Mixed-order reactions** are reactions in which more than two species are involved or one species reacts with stoichiometric coefficient >2 . Many different classes, of which the simplest involves a single species with stoichiometric coefficient greater than 2. Although many biochemical reactions might appear to be higher-order, in general, the enzyme mechanism functions through binding of multiple substrates to a single protein and the empirical order is simpler and often appears to be zero-order.

INTEGRATED SERVICES DIGITAL NETWORK DEFINITION

(ISDN): An international telecommunications standard for digital transfer of simultaneous voice and data signals over digital telephone lines or traditional telephone copper wires that can support data transfer rates of 64 Kbps (64,000 bits per second). Analogue signals such as from traditional telephones are converted into digital signals by an ISDN terminal adapter before being transmitted.

INTEGRATING FACTOR (EULER MULTIPLIER): A function, $m(x,y)$ by which an ordinary differential equation can be multiplied in order to make it integrable. Its form is $y'f(x,y) - g(x,y) = 0$.

INTEGRATING SPHERE: A hollow sphere having a highly reflecting inside surface used as a device to collect, with very high efficiency, light scattered or emitted from a sample contained in it or located outside and near one of the ports. Small ports allow the entrance of light and access to a detector.

INTEGRATION (ANTIDIFFERENTIATION): The reverse process of differentiation or a process of finding and solving an antiderivative for a given function; especially when deriving an indefinite integral or when evaluating a definite integral using the fundamental theorem of calculus rather than as an infinite sum of infinitesimal elements. The symbol $\int f(x)$ indicates the operation of antidifferentiation on $f(x)$, which is the function being antidifferentiated or integrated and known as integrand. The antiderivative of a function $f(x)$ is another function $F(x)$, whose derivative is $F'(x) = f(x)$. The generalised form of the antiderivative of $f(x)$ with respect to x is $\int f(x) - F(x) + C$, where C is called the constant of integration or arbitrary constant. This formula works because the derivative of a constant is zero. For example, x^2 is the antiderivative of $2x$, because $2x$ is the derivative of x^2 .

INTEGRATION BY PARTS: A method of integrating a function of a variable by expressing it in terms of two parts, both of which are differentiable functions of the same variable. Integrating by parts can be evaluated by using the relation $\int u dv = uv - \int v du$, where $u dv$ is given by $u dv = f(x)$ and $f(x)$ is a function. The key is to define u and dv in a fashion such that the integral $\int v du$ will be easier to solve than the original integral ($\int u dv$). For example, $\int \ln x dx$ can be integrated by defining

$u = \ln x$, $dv = dx$, then $v = x$, $\int \ln x dx = x \ln x - \int \frac{x \cdot d}{x} du = x \ln x - x + c$.

INTEGRATION BY SUBSTITUTION: An integration process in which a function of one variable is expressed as a simpler or more integrated function of another variable. For example, to integrate $\cos(ax + b)$ with respect to x ; let $u = ax + b$ then $\frac{du}{dx} = a$, this implies that $du = \frac{du}{a}$, therefore, Substituting $u = ax + b$ into the function the answer becomes $\int \cos u(ax + b) dx = \frac{1}{a} \sin(ax + b) + c$.

INTEGRATION BY TRIGONOMETRIC SUBSTITUTION: Some integrals involving expressions of the form $(1+x^2)$ or $(1-x^2)$ can be solved by making trigonometric substitutions and taking advantage of trigonometric identities, such as $\sin^2\theta + \cos^2\theta=1$ or

$\tan^2\theta + 1=\sec^2\theta$. For example, to evaluate $\int \frac{1}{\sqrt{1-x^2}} dx$, make the substitution $x = \sin\theta$. Then $dx = \cos\theta d\theta$ and $\theta = \arcsin x$ or $\theta = \sin^{-1}x$.

The integral becomes $\int \frac{1}{\sqrt{1-\sin^2\theta}} \cos\theta d\theta = \int \frac{1}{\sqrt{\cos^2\theta}} \cos\theta d\theta = \int \frac{1}{\cos\theta} \cos\theta d\theta = \int d\theta = \theta + c$. Therefore,

$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + c$, where $\theta = \arcsin x$ or $\theta = \sin^{-1}x$.

INTEGRO-DIFFERENTIAL EQUATION: A functional equation involving both integrals and derivatives that can be solved by Laplace transform of integrals and derivatives.

INTEGUMENT: 1. External protective covering of animals such as skin and cuticle, which protects it from the outside world and from drying out. 2. The protective layer around ovules, which after fertilisation develops into the testa or the seed coat.

INTEGUMENTARY SYSTEM: The skin and its derivatives including hair, nails, feathers, horns, antlers and glands, which in multicellular animals protect against invading foreign microorganisms and prevent the loss or exchange of internal fluids. In humans, the integumentary system additionally provides vitamin D synthesis. It serves to waterproof, cushion and protect the deeper tissues, excrete wastes, regulate temperature and is the attachment site for sensory receptors to detect pain, sensation, pressure and temperature.

INTELLECTUAL PROPERTY (IP): The ownership of creativity, such as a manuscript or a design, to which one has rights and for which one may apply for a patent, copyright, trademark, etc. It also refers to digital media, such as audio and video clips that can be downloaded online. By defining and establishing intellectual property rights, innovators and creators can have legal protection of their ideas and creations.

INTELLIGENCE: The coordination of memory, learning and reasoning in animals due to the integrative and adaptive functions of the brain. It is also defined as the ability of an animal to form an associative link between events or objects with which it has had no previous experience. It is expressed as Intelligence Quotient (IQ) in humans.

INTELLIGENT TERMINAL: In computing, a terminal such as a monitor and keyboard that has its own processor that can take some of the processing load away from the main computer. It has a memory, a processor and a printer to perform functions such as display operations, to send data to and receive it from a central processor, perform arithmetic and logic operations, etc.

INTELLIMOUSE SCROLL WHEEL (SCROLL MOUSE): A mouse developed by Microsoft in 1996 that has a wheel between the left and right mouse buttons that enables the user to scroll up and down using the wheel instead of having to use the vertical scroll bar. The wheel is raised slightly, which allows the user to easily drag the wheel up or down using the index finger.

INTENDED INTERPRETATION: For a formal calculus, a mathematical or other theory that is an interpretation of the given calculus when the latter was devised in order to display the formal properties of the theory. For example, the intended interpretation of first-order predicate calculus is a fragment of natural language.

INTENSION: In logic, the set of characteristics or properties by which the referent or referents of a given expression is determined. It is the sense of an expression that determines its reference in every possible world, as opposed to its actual reference. For example, the intension of prime number may be having no non-trivial integral factors, whereas its extension would be the set $\{2, 3, 5, 7, 11, \dots\}$.

INTENSITY: 1. The rate or degree at which radiant energy is transferred per unit area or volume or the rate at which sound energy is transferred as measured relative to some reference value. The intensity of a sound wave relates to its loudness and that of light wave gives the brightness. In sound, it is the amount of energy per second per unit area which is normal to the direction of the sound wave. Its

S.I. unit is bel (B) or decibel (dB). It is expressed as $B = \log_{10} \left(\frac{I}{I_0} \right)$,

where I is one sound and I_0 is the intensity of another sound. 2. The strength of a colour, especially the degree to which it lacks its complementary colour.

INTENSIVE: 1. Involving concentrated effort, usually in order to achieve something in a comparatively short time. 2. Having the same value for any subdivision of a thermodynamic system.

INTENSIVE AGRICULTURE (INTENSIVE FARMING): A method of farming intended to increase the productivity of a fixed area by means of scientific and technological methods or a high level of inputs such as capital and labour. The practice together with erosion, results in the increased rate of loss of soil and fertility. Intensive agriculture is usually found in regions of dense population and high land values.

INTENSIVE PROPERTY (BULK PROPERTY): A property that does not depend on the system size or the amount of material in the system. For example, density is an intensive quantity, which does not depend on the quantity, while mass and volume are **extensive quantities**.

INTENSIVE VARIABLE: A quantity in a macroscopic system that has a well defined value at every point inside the system and which remains approximately constant when the size of the system is increased. Intensive variables are pressure, viscosity, density, temperature and specific heat capacity at constant volume.

INTER: A prefix meaning between or among.

INTERACTING-BOSOM MODEL (IBW): A model of nuclear structure in which nucleons pair up and form bosons from which collective states are formed. It is very useful in describing the spectra of nuclei with even numbers of protons and neutrons.

INTERACTION: 1. An effect involving a number of bodies, particles or systems having an effect upon one another as a result of which some physical or chemical changes take place. In physics, it is the direct effect that one kind of particle has on another, in particular, in inducing the emission or absorption of one particle by another. 2. In biology, it is the effects organisms in a community have on one another. In the natural world no organism exists in absolute isolation and thus every organism must interact with the environment and other organisms. An organism's interactions with its environment are fundamental to the survival of that organism and the functioning of the ecosystem as a whole. In ecology, biological interactions can involve individuals of the same species called **intraspecific interactions**. When interactions involve individuals of different species, it is called **interspecific interactions**. 3. In statistics, it is the variation among the differences between means for different levels of one factor over different levels of t

INTERACTION EFFECT: The result or consequence of the interaction of climate change policy instruments with existing domestic tax systems, including both cost-increasing tax interaction and cost-reducing revenue-recycling effect.

INTERACTION DISTANCE: The farthest distance of approach of two particles at which it is discernible that they will not pass at the impact parameter.

INTERACTIVE COMPUTING: A system for processing data in which the operator is in direct communication with the computer by entering data or commands and receiving immediate responses. In contrast, non-interactive programs such as batch processing and compilers operate without human contact; the necessary data and instructions are prepared in advance and processed by the computer with little or no intervention from the operator.

INTERACTIVE VIDEO (IV): A type of digital video that allows users to have direct exchanges through a CD-ROM or DVD and computer systems for user response and participation.

INTERACTIVE VOICE RESPONSE (IVR): A technology that can read a combination of touch tone and voice input. It gives users the ability to access a database of information via phone and very useful in providing around the clock automated customer service and reducing clerical processing.

INTERACTOME: All the molecular interactions that occur in an organism, in all its cells and tissues and at all stages of its life. Molecular interactions can occur between molecules belonging to different biochemical families such as proteins, nucleic acids, lipids and carbohydrates and also within a given family.

INTERCALARY INFLORESCENCE: An inflorescence with the main vegetative axis of the plant continuing to elongate after the flowers are produced.

INTERCALARY MERISTEM: Describing a meristem that grows in the internode of a stem. Intercalary meristems may contain not only initials, but also relatively mature cells. This prevents it from being a structurally weak part of the organ. The primary function of an intercalary meristem is to facilitate longitudinal growth of a plant organ, independent of activity of the apical meristem. Examples are the meristems in the leaf sheaths and internodes of grasses and horsetails.

INTERCALATED DUCT: One of the minute ducts leading from the acini in glands such as the salivary gland. Together with the striated ducts, they modify salivary fluid by secreting HCO_3^- and K^+ and reabsorbing Na^+ and Cl^- using the Na-K pump and the Cl-HCO_3 pump.

INTERCALATION: Reversible inclusion of a ligand molecule between two other molecules or two layers of a giant molecule. An example is DNA intercalation, whose polycyclic aromatic compounds fit themselves between base pairs. For example, ethidium bromide intercalates between two adenine-uracil base pair in DNA. Another example is potassium graphite, KC_8 , in which the potassium atoms are trapped in-between the layer of graphite, where they are held by electrostatic forces. The compound is a highly conducting, reducing agent and pyrophoric.

INTERCALATION CELL: A type of secondary cell in which layered electrodes usually made of metal oxides or graphite, store positive ions between the crystal layers of an electrode. In one type, lithium ions form an intercalation compound with a graphite electrode when the cell is charged. During discharge, the ions move through an electrolyte to the other electrode, made of manganese oxide, where they are more tightly bound. When the cell is being charged, the ions move back to their positions in the graphite.

INTERCALATION COMPOUND: A type of crystalline or partially crystalline solids in which atoms, ions or molecules are trapped between layers or voids in the lattice. There is no formal chemical bonding between the host crystal and the trapped molecules. The solids are classified as isotropic or three-dimensional lattices, which includes many oxides and zeolites and contain large voids that can accept multiple guest atoms or molecules; layer-type or two-dimensional lattices such as graphite and clays, which swell up perpendicular to the layers when the guest atoms enter; and the chains in one-dimensional structures such as polyacetylene rotate cooperatively about their axes during the intercalation reaction to form channels that are occupied by the guest atoms. In the intercalation family based on solid buckminsterfullerene (C_{60}), the zero-dimensional host lattice consists of 60-atom carbon clusters with strong internal bonding but weak intercluster bonding. These clusters pack together like hard 1-nm-diameter spheres, creating interstitial voids which are large enough to accept most elements in the periodic table.

INTERCALATION REACTION: Reaction, generally reversible, that involves the penetration of a host material by guest species without causing a major structural modification of the host. Intercalation can refer to the insertion of a guest species into a one-, two- or three-dimensional host structure. The guest species is not distributed randomly but occupies positions predetermined by the structure

of the host material. Examples are the insertion of lithium into layered TiS_2 [$\text{LiTiS}_2(0 \leq x \leq 1)$] and of potassium into the layers of graphite (C_8K).

INTERCELLULAR (INTRACELLULAR): Located or occurring between cells. For example, **intercellular fluid** is liquid located between cells.

INTERCEPT: 1. The point at which a curve or straight line intersects the x , y or z axis of a coordinate system. The parabola $y = x^2 - 4$, for example, has a y -intercept at -4 and x intercepts at +2 and -2. The term is also used to refer to the portion of a line lying between two other lines or planes. 2. The difference between the calculated and observed altitude of an astronomical object such as asteroids, moons, planets and the stars; or between pre-computed and sextant altitudes, also called altitude intercept.

INTERCEPT FORM: The intercept form of equation of a line based on the intercepts with both axes is expressed as $\frac{x}{a} + \frac{y}{b} = 1$, where a is the intercept on the x -axis and b on the y -axis; for example, in $\frac{x}{4} + \frac{y}{6} = 1$, x -intercept is 4 and y -intercept is 6.

INTERCHANGE REACTION: Reaction that results in an exchange of atoms or groups between a polymer and low-molar-mass molecules, between polymer molecules or between sites within the same macromolecule. An interchange reaction that occurs with polyesters is called **transesterification**.

INTERCONAL REGION: The region of a Bunsen flame where the combustion zone has the form of a cone.

INTERCONVERTIBLE ENZYME: An enzyme existing in at least two well defined, reversibly convertible forms, produced by covalent modifications of amino acid side chains under biological conditions. Covalent modifications that occur as intermediates in the catalytic process are not included in this definition.

INTERCOSTAL MUSCLE: The three muscles located between the ribs and surrounding the lungs. They are important in lung ventilation. The superficial external intercostals pull the ribs upwards and outwards, increasing the volume of the thoracic cavity and drawing air into the lungs during inspiration. The deeper, internal intercostals draw the ribs downward and inwards, decreasing the volume of the thoracic cavity, forcing air out of the lungs during expiration.

INTERCOURSE: A technical term used to describe mating in individuals that involves the genitalia of at least one person. **Axillary intercourse** is an intercourse with the penis inserted into an armpit; **femoral intercourse** is intercourse with the penis inserted between the thighs; **mammary intercourse** is intercourse with the penis inserted between the woman's breasts; and **coitus a tergo** is sexual intercourse in which the male enters the woman's vagina from behind.

INTERCROPPING: The practice of growing two or more crops simultaneously on the same land. This practice is one type of multicropping, the other type being **sequential cropping**, enabling two or more crops to be tended to at the same time. In order not to have crops competing with each other, for example, for space, nutrients, water, or sunlight it is important to choose the crops carefully. An example is planting maize with peanuts as this reduces competition of light. There are four types of intercropping: (i) **Mixed intercropping** is cultivating two or more crops simultaneously on the same field not in rows but randomly distributed. (ii) **Row intercropping** is cultivating different crops simultaneously in adjacent rows. (iii) **Strip intercropping** is cultivating different crops simultaneously using alternate strips of land with uniform width and on the same field strips of land rather than narrow rows. This practice is suitable for mechanised commercial large-scale farming. (iv) **Relay intercropping** involves the planting of a second crop into a first crop that is at a stage through its growth but is not yet ready for harvesting. Some of the advantages of intercropping include the following: (i) It gives greater yield and income/unit area than sole cropping. (ii)

In case of a particular crop failure it offers an alternative. (iii) Intercrops maintain the soil fertility as the nutrient uptake is made from both layers of soil. (iv) It reduces soil runoff and controls weeds. (v) Secondary crops provide shelter and protect the primary crops. Some of the disadvantages of intercropping include the following: (i) It does not permit some implements be used efficiently. (ii) Incompatibility of some crops which may lead to competition for the same resources resulting in poorer yields. (iii) It is labour intensive.

INTERFACE: 1. A common boundary shared between adjacent regions, bodies, substances or phases. 2. A computer hardware and or software that connects one piece of equipment to another such as a printer or human operator; and also the layout of an application's graphic or textual controls in conjunction with the way the application responds to user activity. 3. Boundary of contact of two adjacent phases, either or both of which may be solid, liquid or gaseous.

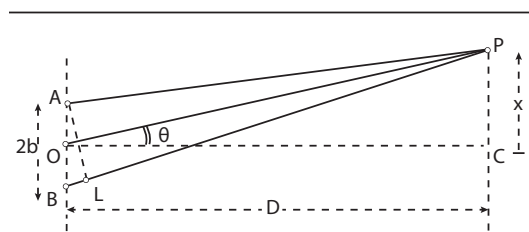
INTERFACIAL ADHESION: Adhesion in which interfaces between phases or components are maintained by intermolecular forces, chain entanglements or both, across the interfaces.

INTERFACIAL LAYER: The inhomogeneous space region intermediate between two bulk phases in contact and where properties are significantly different from, but related to, the properties of the bulk phases. Examples of such properties are compositions, molecular density orientation or conformation, charge density, pressure tensor and electron density. The interfacial properties vary in the direction normal to the surface. Complex profiles of interfacial properties occur in the case of multicomponent systems with coexisting bulk phases where attractive/repulsive molecular interactions involve adsorption or depletion of one or several components. This interfacial region may also be regarded as a distinct, though not autonomous, phase and be called the interphase.

INTERFASCICULAR CAMBIUM: The part of the vascular cambium that forms between the vascular bundles (fascicles) and joins up with the fascicular cambium to form a continuous meristematic ring.

INTERFERENCE: 1. A systematic error in the measure of a signal caused by the presence of concomitants in a sample. 2. The superimposition of two waves of equal frequency or amplitude to form a new wave pattern. When waves reinforce, it is called **constructive interference** and when they cancel out, it is called **destructive interference**. Constructive interference occurs if two components have the same frequency and phase and the wave amplitudes are reinforced. Destructive interference occurs when the two waves are out of phase by one-half period; if the waves are of equal amplitude, they cancel each other. Two waves moving in the same direction but having slightly different frequencies interfere constructively at regular intervals, resulting in a pulsating frequency called a **beat**. Two waves travelling in opposite directions but having equal frequencies interfere constructively in some places and destructively in others, resulting in a **standing wave**.

INTERFERENCE FRINGE: A bright or dark band caused by beams of light that are in phase or out of phase with one another. Light waves and similar wave propagation, when superimposed, will add their crests if they meet in the same phase (the waves are both increasing or both decreasing); or the troughs will cancel the crests if they are out of phase; these phenomena are called constructive and destructive interference, respectively. If a beam of monochromatic light (all waves having the same wavelength) is passed through two narrow slits, the two resulting light beams can be directed to a flat screen on which, instead of forming two patches of overlapping light, they will form interference fringes, a pattern of evenly spaced alternating bright and dark bands. This experiment was first performed in 1801 by Thomas Young, an English scientist, who inferred from the phenomenon the wavelike nature of light. All optical interferometers function by virtue of the interference fringes that they produce.



The diagram shows formation of interference fringes

INTERFERENCE MICROSCOPE: An advanced development of the phase contrast microscope in which the light from the condenser is split into two beams by a prism. The object beam passes through the specimen and the objective. The reference beam passes through a second matched objective without going through the specimen. The two beams are recombined before going through the eye piece. Interference between the beams produces a series of light and dark fringes in the field of view. These are the result of differences in refractive index of the specimen and allow detail of the specimen to be seen. The instrument is more sensitive than the phase contrast microscope and spurious effects (such as halos) are eliminated. With white light different parts of the specimen appear coloured as a result of interference.

INTERFERENCE PATTERN: 1. Alternating light and dark bands, known as fringes that are produced by interference. In general, it is the pattern of peaks and troughs that result when two or more waves that travel on slightly different paths from the same source are brought together. 2. In radio astronomy, it is the pattern that results when the signals picked up by two or more elements of an interferometer are combined. Interference is the basic principle used in interferometry to increase resolving power.

INTERFEROMETER: Any of several optical, acoustic or radio frequency instruments that use interference phenomena between a reference wave and an experimental wave or between two parts of an experimental wave to determine wavelengths and wave velocities to measure very small distances and thicknesses and calculate indices of refraction. Radio interferometers are one of the two basic types of radio telescopes which are used in astronomy.

INTERFERON (IFN): A protein produced by cells of an organism into its bloodstream or intercellular fluid when invaded by viruses to induce healthy cells to manufacture an enzyme that counters the infection.

INTERGENIC SUPPRESSOR: A point mutation that negates or lessens the effects of another mutation in the same gene. For example, the effect of a primary frameshift mutation may be suppressed by a second mutation in which the deletion or insertion of a nucleotide occurs.

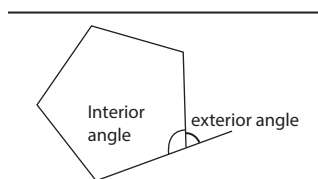
INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE (IPCC): A scientific and intergovernmental body established by the World Meteorological Organisation and the United Nations Environment Programme in 1988 to provide the world with a clear scientific view on the current state of knowledge in climate change and its potential environmental and socio-economic impacts. It regularly assesses the up-to-date evidence from international research into global warming and human induced climate change. Currently member countries number 195.

INTERGRAND: A mathematical function or equation to be integrated in either a definite or indefinite integral. For example, $x^2 \cos 3x$ is the integrand in $\int x^2 \cos 3x dx$.

INTERHALOGEN: A chemical compound formed between two halogens that react with each other to form a series of binary compounds; for example, iodine monofluoride. Interhalogens are highly reactive and volatile, made by direct combination of the elements. They include compounds with two atoms such as ClF , IBr ; four atoms such as ClF_3 , IF_3 ; six atoms such as BrF_5 , IF_5 ; and IF_7 with eight atoms.

INTERIOR (INT): 1. The set of points lying strictly between the end points of an interval. 2. In topology, the set of all interior points of a given set in a topological space. It is the largest open subset of the set, equal to the union of all the open sets contained in the given set. The interior of A is denoted $\text{Int}A$ or A° . 3. Of a simple closed curve, the bounded region determined by the curve, as ensured by the Jordan curve theorem.

INTERIOR ANGLE: 1. The angle formed between two adjacent sides of a polygon and lying in its interior. The sum of the interior angles of any polygon is given by $180^\circ (n - 2)$, where n is the number of sides of the polygon. 2. Any of the four angles formed in the area between two lines by a third line that intersects them transversally.



$\text{Interior angle} + \text{exterior angle} = 180^\circ$

Fig I: A polygon indicating its interior angle

INTERIOR OF A FIGURE: 1. The set of all points in a plane bounded by a closed 2-dimensional figure such as a polygon or circle. 2. The set of all points in space bounded by a closed 3-dimensional figure such as a polyhedron or sphere. The interior is usually not considered to be part of the figure.

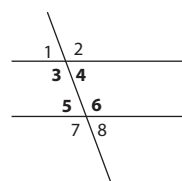


Fig II: Angles 3, 4, 5 and 6 are interior angles

INTERIOR PENALTY FUNCTION

(BARRIER FUNCTION): For a set, S , with non-empty interior, a non-negative function, B that is continuous over the interior of S and approaches infinity as the boundary is approached from the set.

INTERIOR POINT: 1. In topology, a point of a given set contained within an open subset of it. For example, 0.5 is an interior point of the real interval $[0, 1]$, whereas 0 is not. 2. Of a conic in Euclidean geometry, a point not on the conic through which no tangent to the conic runs.

INTERKINESIS (INTERPHASE II): An abbreviated form of interphase or a period of rest that cells enter. It occurs between the first and second nuclear divisions of meiosis in some organisms. The chromosomes begin to uncoil and there is at least partial re-formation of the nuclear envelope.

INTERLACED VIDEO: A technique used to improve the picture quality of a video transmission without consuming any extra bandwidth for television video formats, such as NTSC and PAL. Each full frame of video actually consists of alternating lines taken from two separate fields captured at slightly different times. As the frames are projected on the screen, the video signal alternates between showing even and odd lines. When this is done fast enough, around 60 frames per second, the video image looks much improved. While interlace can improve the resolution of still images, it can cause flicker and various kinds of distortion.

INTERLEUKIN: Any of several cytokines that act specifically as mediators between leucocytes, whose production is stimulated by antigens and microbes, which induce production of various types of lymphocytes in a complex series of reactions that ensure a plentiful supply of T cells that fight specific infectious agents. There are two types: **interleukin I**, which activates any killer T cells that have already bound antigen attached to class I MHC molecules on infected cells, killing any cells displaying antigen attached to class I MHC molecules. **Interleukin II** activates any B lymphocytes that have also bound the antigen. The activated B cells then divide and secrete antibodies. Finally, the secreted antibodies bind the antigen and help destroy it.

INTERMEDIATE: 1. Also called **synthetic intermediate** and **intermediate product**, it is a chemical compound that is formed

during a chemical reaction, which is not itself the final product, but is used in further reactions to produce the final products or a compound to be subjected to further chemical treatment to produce finished products such as dyes and pharmaceuticals. There may be many different intermediates between the starting material and the end product in the course of a complex synthesis. 2. A molecule, free radical or ion that exists for a short time during a chemical reaction or a short-lived species in complex chemical reaction. 3. An igneous rock with a silica content of between 55% and 66%. 4. Lying or occurring between two different points, forms, extremes or states.

INTERMEDIATE COUPLING (JJ-COUPLING): A type of coupling in many-fermion systems or heavier elements with larger nuclear charge, such as electrons in atoms and nucleons in nuclei, in which the spin-orbit interactions become as strong as the interactions between individual spins or orbital angular momenta. Multiplets of many-electron atoms having a large atomic number are characterised by intermediate coupling. Multiplets in the shell model of nuclei characterised by intermediate coupling are invoked to explain the magic numbers of nuclei.

INTERMEDIATE FILAMENT (IFS): Any of numerous microscopic protein fibres that share common structural and sequence features and forms part of the cytoskeleton of eukaryote cells. Intermediate filaments have an average diameter of 10 nanometres, which is between that of actin (microfilaments) and microtubules. Most types of intermediate filaments are cytoplasmic, but one type, the lamins, are nuclear.

INTERMEDIATE FREQUENCY: The frequency that lies between the modulated radio frequency and the audio frequency; and to which a carrier frequency is shifted in super heterodyne receivers as an intermediate step prior to amplification before final detection is done. It is produced by combining the received signal with that of the local oscillator in a super heterodyne receiver, resulting in a signal at the difference or beat frequency. There may be several such stages of intermediate frequency in a super heterodyne, called double or triple conversion.

INTERMEDIATE HOST: An organism that is the host for an adult but not sexually mature parasite, which then moves on to a different host before reproducing.

INTERMEDIATE NEUTRON: A neutron having kinetic energy in the range $10^2 - 10^5$ electron volts ($1.6 \times 10^{-17} - 1.6 \times 10^{-14}$ joules).

INTERMEDIATE VALUE THEOREM (BOLZANO'S THEOREM): A theorem, used in mathematical analysis which states that if a continuous function f having interval $[x, y]$ as its domain takes every value between $f(x)$ and $f(y)$ at each end of the interval, then it also takes values between $f(x)$ and $f(y)$ at some point within the interval. This intermediate value property, which derivatives also possess by virtue of the mean-value theorem, is also called the Darboux property. It was named after the Czech analyst Bernhard Bolzano (1781-1848).

INTERMEDIATE VECTOR BOSON: Any of the three fundamental particles (W^+ , W^- and Z^0 particles) that transmits the weak nuclear force between other elementary particles in the same manner that the photon transmits the electromagnetic force.

INTERMEDIATE-VALENCE COMPOUNDS (MIXED-VALENCE COMPOUNDS): A chemical compound in which energy levels of f electrons coexist with energy bands of s and/or d electrons. It contains an element, usually a metal in more than one oxidation state. Such compounds have usual thermal and magnetic properties.

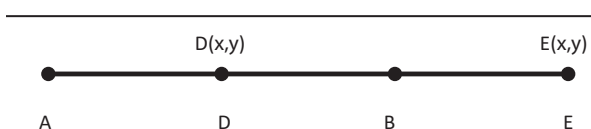
INTERMETALLIC COMPOUND (INTERMETALLICS; INTERMETALLIC COMPOUNDS): A compound of two or more metallic elements present in definite proportions in an alloy. They are homogeneous, composite substances and differ discontinuously in structure from that of the constituent metals. Examples are NaAu_2 , CuSn and CuAl_2 .

INTERMITTENCY: The alternation in time of a dynamical system between nearly periodic and chaotic behaviour.

INTERMOLECULAR: 1. Descriptive of any process that involves a transfer (of atoms, groups, electrons, etc.) or interactions between two or more molecular entities. 2. Relating to a comparison between different molecular entities.

INTERMOLECULAR FORCES: Weak attractive or repulsive forces that occur between all atoms and molecules. They are classified as hydrogen bonding and van der Waals' forces. They are important because they are responsible for many of the physical properties of solids, liquids and gases. These forces are also largely responsible for the three-dimensional arrangements of biological molecules and polymers.

INTERNAL AND EXTERNAL DIVISION (IN FIXED PROPORTION): Coordinates of a point dividing a line segment in a given ratio $m:n$. If $D(x,y)$ is the point which divides AB internally in the ratio $m:n$, where A is (x_1, y_1) and B is (x_2, y_2) , then D is $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$. If $E(x,y)$ is the point at which divides AB externally in the ratio $m:n$, then the coordinates of E are $\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$.



The diagram shows internal division by D and External Division by E

INTERNAL BISECTOR: It is the bisector of the interior angle of a polygon.

INTERNAL COMBUSTION ENGINE: A heat engine such as petrol and diesel engine in which the combustion of a fuel occurs with an oxidiser in a combustion chamber rather than in a separate furnace. This contrasts with an external combustion engine such as steam engine in which fuel is burned in a separate unit. In an internal combustion engine the expansion of the high temperature and pressure gases, which are produced by the combustion, directly applies force to component of the engine, such as the pistons or turbine blades or a nozzle and by moving it over a distance, generates useful mechanical energy.

INTERNAL CONVERSION: A radioactive decay process in which an excited atomic nucleus interacts with an electron in one of the lower atomic orbitals, causing the electron to be emitted from the atom, but without beta decay taking place. It can also be defined as a process in which an excited atomic nucleus decays to the ground state and the energy released is transformed by electromagnetic coupling to one of the bond electrons of that atom rather than being released as a photon. Since no beta decay takes place in internal conversion, the element atomic number does not change and no transmutation of one element to another is seen. Also, no neutrino is emitted in internal conversion.

INTERNAL DIRECT PRODUCT: The internal direct sum of groups under multiplication.

INTERNAL DIRECT SUM: A sum of a finite or infinite number of ideals of a ring, normal subgroups of a group under addition or submodules of a module over a ring, such that the ideals or submodules $J_1 \dots J_2$ satisfy the condition that for all $0 \leq i \leq n$,

$$J_i \cap \sum_{\substack{j=1 \\ j \neq i}}^n J_j = \{0\}, \text{ the sum is written as } J_1 \oplus J_2 \oplus \dots \oplus J_n.$$

INTERNAL ENERGY (THERMODYNAMIC ENERGY): The sum of both the kinetic and potential energies of the particles (i.e. atoms or molecules) in a matter sample or the energy that substances contain in the form of the kinetic energy of their atoms and molecules and

of the potential energy in their bonds. The symbol is U . It does not include the kinetic and potential energies of the system as a whole or their nuclear energies.

INTERNAL ENERGY DENSITY: The scalar field, $\Phi(x, t)$, unique almost everywhere, such that the internal energy of a sub-body of a body with density ρ is given by the integral $\int \Phi(x, t) \rho dv$ over the volume of the configuration of the at time t .

INTERNAL ENVIRONMENT: The condition that prevails within the body of an organism, particularly with respect to the composition of the tissue fluid.

INTERNAL HARD DRIVE: A hard drive that resides inside the computer. Most computers come with a single internal hard drive, which includes the operating system and pre-installed applications. While laptop computers only have room for one internal hard drive, some desktop computers have multiple hard drive bays, which allow for multiple internal hard drives.

INTERNAL LAYING: A condition in hens caused by a fault in the oviduct, which results in the yolks not being passed along the oviduct for covering with membranes and shell.

INTERNAL MODEM: A type of modem contained on a printed circuit board within the computer; this is in contrast to a self-contained modem that sits outside a personal computer (PC) and has lights indicating various modem functions and connected to the PC by a cable via the serial or USB port. There are two types: dial-up internal modem, which requires a connection with a telephone line, network access phone number and username and login ID; and wireless internal modem.

INTERNAL OBLIQUE MUSCLE: One of four pairs of abdominal muscles lying under the external obliques. It is a broad, thin, muscular sheet with the majority of its fibres running at right angles to those of the external oblique. It functions with the external obliques to flex and rotate the trunk and compress the abdomen.

INTERNAL RADIATION: Nuclear radiation resulting from radioactive substances in the body. The substances can enter via inhalation, ingestion, absorption or injection. Once in the body they will enter the systemic system where they will then migrate to the organs of accumulation. The actual radiation dose delivered will depend on the radioactive nature of the material (its half-life and principal emitters) and its physical and chemical properties (its particle size and the biological half-life).

INTERNAL REFERENCE ELECTRODE: An 'inner' reference electrode which is contained inside an ion-selective electrode assembly. The system frequently consists of a silver chloride electrode in contact with an appropriate solution containing fixed concentrations of chloride and the ion for which the membrane is selective. This 'inner' filling solution is in contact with the ion-selective membrane.

INTERNAL RESISTANCE: The opposition to the flow of current within a source of electric current such as a cell generator or the resistance inside a power supply such as a battery or cells that limits the current that it can supply in a circuit. As a result of the resistance from the source, the output voltage is smaller than the true electromotive force. When a battery fails, it is typically because it has built up enough internal resistance that it can no longer supply a useful amount of power to an external load. The internal resistance of a battery is dependent on the specific battery's size, chemical properties, age, temperature and the discharge current. It has an electronic component due to the resistivity of the battery's component materials and an ionic component due to electrochemical factors such as electrolyte conductivity, ion mobility and electrode surface area. The internal resistance, R_{int} of a battery can be calculated from its open circuit voltage, E , load voltage, V_L load resistance, R_L , and the current, I .

$$\text{Thus, } E = I(R_L + R_{int}) \text{ or } R_L = R_{int} \left(\frac{E}{V_L} - 1 \right) R_L.$$

INTERNAL STANDARD: A term used in chromatography for a compound added to a sample in known concentration to facilitate the qualitative identification and/or quantitative determination of the sample components.

INTERNAL SURFACE: With respect to porous solids, the surface associated with pores communicating with the outside space. Since the accessibility of pores may depend on the size of the fluid molecules, the extent of the accessible internal surface may depend on the size of the molecules comprising the fluid and may be different for the various components of a fluid mixture (molecular sieve effect).

INTERNATIONAL ATOMIC TIME: A precisely determined system of measuring time in which a second is defined in terms of atomic events that are known to a high degree of accuracy. The second is defined by the Systeme International (SI) as the duration of 9,192,631,770 cycles of radiation corresponding to the transition between two hyperfine levels of the ground state of caesium 133.

INTERNATIONAL BUSINESS MACHINES (IBM): A US multinational and consulting company, which is the world's largest information technology company. Its revenue in 1914 was USD \$93 billion. It was formed in 1911 as the Computing-Tabulating-Recording Company (CTR) by a merger of The Tabulating Machine Company (formed in 1896 by US inventor Herman Hollerith to produce punch card tabulating machines) and some other companies. IBM adopted its present name in 1924. Its present products include hardware and software, microchips, network hosting and IT consulting services.

INTERNATIONAL CANDLE: A former unit of luminous intensity, now replaced by the candela.

INTERNATIONAL ATOMIC ENERGY AGENCY (IAEA): Set up in 1957, it is an organisation within the United Nations for cooperation in the nuclear field to promote the safe, secure and peaceful use of nuclear technologies. Its headquarters is in Vienna, Austria.

INTERNATIONAL CODE OF BOTANICAL NOMENCLATURE (ICBN): A periodically revised set of rules according to which plants are given their formal botanical names (scientific names). It specifies the standards and forms of names given to each taxon of plants recognised in botany. The code is maintained by the International Association for Plant Taxonomy. The most recent is the 'St Louis code' which was adopted in 1999; which supersedes the earlier versions.

INTERNATIONAL CODE OF ZOOLOGICAL NOMENCLATURE (ICZN): International authority that draws up the regulations governing the scientific naming of animals and the resolution of nomenclatural problems.

INTERNATIONAL DATA ENCRYPTION ALGORITHM (IDEA): In computing, it is an encryption algorithm that operates on 64-bit blocks using a 128-bit key. It is used in the encryption program Pretty Good Privacy (PGP), a very secure public key encryption application for MS-DOS, UNIX and VAX/VMS. It was developed in 1990 by Xuejia Lai and James Massey in Zurich, Switzerland.

INTERNATIONAL DATE LINE: An imaginary line on the Earth's surface that joins the north and south poles and approximately follows the 180° meridian through the Pacific Ocean that arbitrarily separates each calendar day from the next. It deviates to the east through the Bering Strait to avoid dividing Siberia and then deviates to the west to include the Aleutian Islands with Alaska. South of the Equator, another eastward deviation allows certain island groups to have the same day as New Zealand. The line has been agreed upon internationally to mark the beginning and end of a day. A traveller moving towards the east, gains 1 hour for every 15° of longitude but when travelling westward, he loses time at the same rate.

INTERNATIONAL ENERGY AGENCY (IEA): Paris-based energy forum established in 1974. It is linked with the Organisation for Economic Cooperation and Development to enable member countries to take joint measures to meet oil supply emergencies, to share energy information, to coordinate their energy policies, and to cooperate in the development of rational energy programs.

INTERNATIONAL HAPMAP PROJECT: An international, collaborative organisation in Canada, China, Japan, Nigeria, the United Kingdom and the United States, initiated in 2002, to identify and catalogue the common genetic variations among human populations affecting health, disease and responses to drugs and environmental factors. The information produced by the project is made freely available to researchers around the world.

INTERNATIONAL NUCLEOTIDE SEQUENCE DATABASE COLLABORATION (INSDC): A collaborative venture that effectively divides up the task of collecting, updating and storing the nucleotide sequence data reported by researchers throughout the world. It involves the following computerised databases: DNA Data Bank of Japan (Japan), GenBank (USA) and the EMBL (European Molecular Biology Laboratory, Germany). New and updated data on nucleotide sequences contributed by research teams to each of the three databases are synchronised on a daily basis through continuous interaction between the staff at each of the collaborating organisations.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO): An international organisation established to promote development of standards to facilitate international exchange of goods and services and to develop cooperation in intellectual, scientific, technological and economic activities.

INTERNATIONAL PRACTICAL TEMPERATURE SCALE (ITS-90): An equipment calibration standard for making measurements on the Kelvin and Celsius temperature scales, adopted by international agreement in 1968 and revised in 1990, based on thermodynamic temperature and using experimental values to define 16 fixed points. The fixed temperature reference points include the water triple point, the boiling points of oxygen, water, sulfur and the solidification points of silver and gold. ITS-90 offers defined calibration points ranging from approximately 0.65 K (-272.5 °C), which is the triple point of an equilibrium mixture of orthohydrogen and parahydrogen to approximately 1358 K (1085 °C), which is the freezing point of copper. It is subdivided into multiple temperature ranges which overlap in some instances.

INTERNATIONAL STANDARDS ORGANIZATION (ISO): A voluntary non-treaty organisation founded in 1946 in Geneva as a United Nations (UN) agency. It issues standards on a number of subjects including telephone pole coatings, cocoa beans, fishing nets, computers and data communications etc. It does not however, include electrical and electronic engineering, which is governed by the International Electrotechnical Commission (IEC); as well as telecommunications, which is governed by International Telecommunications Union's Telecommunications Standards Sector or ITU-TSS.

INTERNATIONAL TELECOMMUNICATION UNION (ITU): The specialised telecommunication agency of the United Nations that was established to provide standardised communication procedures and practices, including frequency allocation and radio regulations worldwide.

INTERNATIONAL UNION FOR THE CONSERVATION OF NATURE, IUCN (WORLD CONSERVATION UNION): Founded in 1948, it is an international body that is concerned with finding pragmatic solutions to the most pressing environment and development challenges in the world and encourages the preservation of wildlife, natural environments and living resources. It has a membership of about 1,300 government and NGO members and more than 15,000 volunteer experts in 185 countries. Its activities include data gathering and analysis, research, field projects, advocacy, lobbying and education.

INTERNEURVE: A space between two nerves.

INTERNET (NET): The global network that links most of the world's computer networks that use the standard Internet Protocol Suite (TCP/IP) to serve billions of users worldwide. It does not offer services to users, but serves primarily to interconnect other networks on which

services are located. These include basic services for electronic mail, the transfer of computer files, remote log-in and high-level services including the World Wide Web (www). The Internet is informal, with a minimal level of administration by governing bodies.

INTERNET ARCHITECTURE BOARD (IAB): A board of researchers and professionals that manages the engineering and technical development related to the Internet. It has two task forces: the Internet Engineering Task Force (IETF) that defines standard Internet operating protocols such as TCP/IP; and the Internet Research Task Force (IRTF), which works on network technology.

INTERNET CONNECTION FIREWALL (ICF): A Windows XP feature that protects a home or small business network that is connected to the Internet via a cable, DSL or dial-up modem from unauthorised access. When ICF is enabled, Windows keeps a log of incoming requests from other systems on the Internet. If the request is something the user has requested, like a Web page, the transmission will not be affected. However, if the request is unsolicited and is not recognised by the system, the transmission will be dropped. This helps prevent intrusion by hackers or malicious software such as spyware.

INTERNET CONTROL MESSAGE PROTOCOL (ICMP): A network control protocol useful in Internet Protocol (IP) network management and administration. When information is transferred over the Internet, computer systems send and receive data using the TCP/IP protocol. If there is a problem with the connection, error and status messages regarding the connection are sent using ICMP, which is part of the Internet protocol.

INTERNET CORPORATION FOR ASSIGNED NAMES AND NUMBERS (ICANN): A non-profit corporation that is responsible for allocating IP addresses and managing the domain name system. The ICANN allocates blocks of IP addresses to companies, educational institutions and Internet service providers, which in turn allocate IP addresses to computers that use their Internet connections.

INTERNET MESSAGE ACCESS PROTOCOL (IMAP): A protocol that allows users to retrieve e-mail, but still uses SMTP for sending e-mail messages. IMAP4 is the latest version of IMAP. IMAP includes operations such as creating, deleting, renaming mailboxes, checking for new messages, permanently removing messages, searching, etc. IMAP was first developed by Mark Crispin while at Stanford University in 1986.

INTERNET PROTOCOL (IP): A standard set of rules for sending and receiving data through the Internet, which may be a wide area network (WAN) such as the public Internet or a private, local network (LAN) using the Transmission Control Protocol/Internet Protocol (TCP/IP). IP takes care of handling the actual delivery of the data by moving data packets between nodes using an address such as 166.70.10.23 of a computer or other network device on a network using IP or TCP/IP. TCP ensures the reliability of data transmission across Internet connected networks, that all data arrive accurately at the other end.

INTERNET PROTOCOL MULTICAST (IP MULTICAST): A method of sending data between a single sender and multiple receivers on a network that enables many recipients to share the same source. It is often employed for streaming media applications on the Internet and private networks.

INTERNET RELAY CHAT (IRC): A protocol that allows users that connect to the Internet anywhere to join in live discussions, using IRC client, a program in the computer that sends and receives messages to and from an IRC server.

INTERNET SERVICE PROVIDER, ISP (SERVICE PROVIDER): A company or organisation that provides internet access. ISPs often provide e-mail capability, Web hosting and other services in addition to connectivity. The ISP connects to its customers using a data transmission technology appropriate for delivering Internet Protocol datagrams, such as dial-up, DSL, cable modem, wireless or dedicated high-speed interconnects.

INTERNET SOCIETY (ISOC): Founded in 1992, it is an international non-profit organisation composed of companies, government agencies and foundations concerned with Internet standards, education and policy development.

INTERNEURON (INTERNEURONE; RELAY NEURON): A nerve cell found within the central nervous system of vertebrates that transmits messages from the sensory neurons to motor neurons, allowing information to pass directly to the spinal cord, without reaching the brain. Interneurons are involved in reflex actions, which are extremely rapid, causing the sudden removal of a limb from a painful stimulus.

INTERNODE: The part of a plant stem between two nodes. In botany, a node is a point or joint on a stem where a leaf is attached or has been attached.

INTEROCEPTOR: A specialised sensory nerve receptor that detects stimuli from the internal environment of an organism. An example is chemoreceptors that detect changes in the level of oxygen concentration in the blood.

INTEROPERABILITY: 1. In computer science, it is the ability of different information technology systems and software applications to share and use resources between them through local area networks (LANs) or wide area networks (WANs). **Semantic interoperability** is the ability of two or more systems to automatically interpret the information exchanged between them meaningfully and accurately even if the interface and the programming language are different in order to produce useful results as defined by the end users. It may also be defined as the characteristic of a product or system to work effortlessly with other products or systems. **Syntactic interoperability** is the ability of two or more systems to communicate and exchange data. It allows different software components to cooperate, even if the interface and the programming language are different. 2. In health information systems, it is the ability to work together within and across organisational boundaries in order to advance the effective delivery of healthcare for individuals and communities. There are three levels of health information technology interoperability: (i) **Foundational interoperability** allows data exchange from one information technology system to be received by another and does not require the ability for the receiving information technology system to interpret the data. (ii) **Structural interoperability** is an intermediate level that defines the structure or format of data exchange (i.e., the message format standards) where there is uniform movement of healthcare data from one system to another such that the clinical or operational purpose and meaning of the data is preserved and unaltered. Structural interoperability defines the syntax of the data exchange. It ensures that data exchanges between information technology systems can be interpreted at the data field level. (iii) **Semantic interoperability** provides interoperability at the highest level, which is the ability of two or more systems or elements to exchange information and to use the information that has been exchanged.

INTERPETIOLAR: A stipule that is located between the bases of two opposite leaves.

INTERPHASE: The period of the cell cycle in which a cell is not actively dividing and other activities such as DNA synthesis take place. In preparation for cell division, it increases its size and makes a copy of its DNA. It is also the phase of the cell, in which the cell obtains nutrients, grows, reads its DNA and conducts other cell functions. The majority of eukaryotic cells spend most of their time in interphase.

INTERPHASE TRANSITION: A transition that occurs at boundaries between phases. Example is the precipitation of second phases or the initiation of new phase growth at an interface.

INTERPLANETARY SPACE: The space between the Sun and the planets within the solar system. It extends beyond near-space away from Earth to the other planets in the solar system.

INTERPOLATION: A process in mathematics used to estimate the values of a function, for example, $f(x)$ for any value of the variable x that lies between those for which the function is known.

INTERPOLATION HYPOTHESIS: A hypothetical account that a saprophyte generation (organism that lives on decay) emerge during the evolution of vascular plants. It postulates that early plants were exclusively haploid gametophytes and that at a point, a zygote developed mitotically instead of meiotically, producing a rudimentary diploid sporophyte. Hence, a sporophyte generation was interpolated, into the life cycle. The sporophyte evolved to become increasingly elaborate in a more prominent form such as in modern seed plants, while the gametophyte became progressively reduced.

INTERPRETATION: 1. In logic, an allocation of significance to the terms of a purely formal system; an assignment of ranges to the bound variables, denotations to the individual constants and extensions to the predicate constants of a formal calculus, so that the closed well-formed formulae of the calculus have a truth-value in the interpretation. 2. In mathematics, a function from the formal calculus to possible worlds that yields such an assignment. 3. Also, in mathematics, the formation of understanding of a concept or the solution to a problem, such as relating the solution to the problem's context.

INTERPRETER: In computer science, it refers to a program that reads the source code of another computer program and executes that program. This includes source code, pre-compiled code and scripts. Examples of interpreters include Perl, Python and Ruby, which execute Perl, Python and Ruby code respectively.

INTERQUARTILE RANGE: A statistical measure of the spread of a sample data, which is the difference between one quarter and three quarters of a sample. It is a truncated mean and very similar to the scoring method used in sports that are evaluated by a panel of judges, which discard the lowest and the highest scores and calculate the mean value of the remaining scores.

INTERRUPT: In computing, a program interrupt refers to the transfer of control from a currently running program to another service program as a result of an externally generated control signal.

INTERRUPTED PINNATE: A compound leaf which has leaflets arranged on opposite sides of an enlarged axis, with small leaflets intermixed with large ones.

INTERSECTING LINES (LINE SEGMENTS): Lines or line segments that have at least one point in common.

INTERSECTION: 1. A set of points or a point common to two or more intersecting geometric figures. 2. A set that consists of all the elements common to two or more other sets, which can be written for two sets S and T as $S \cap T$ or $T \cap S$. 3. Common members (parts) of curves or surfaces, which may be points, curves or parts of a plane.

INTERSEX: A human that has biologically several variations in sex characteristics that do not apply to the typical definitions of a female or male. Such differences in characteristics include chromosomes, gonads, sex hormones, or genitals.

INTERRUPT REQUEST (IRQ): In computing, it is a signal in the form of a unique value sent by a device such as a printer to a computer to interrupt its operation. This allows multitasking so that other programs and devices can function while the computer is performing other tasks. A PC-compatible computer devices such as the mouse, keyboard, serial port, sound card, floppy disk controller and the primary/secondary advanced technology attachment (ATA) channels used for hard disk devices have interrupt request levels of IRQ0 to IRQ15.

INTERSECTION GRAPH: Of a family of sets, a graph in which a pair of vertices is joined by an edge if and only if the intersection of the sets represented by those vertices is non-empty; or in the case of three sets S , T and R , $S \cap T \cap R$.

INTERSEX: An organism displaying characteristics that are intermediate between those of a typical male and a typical female of its species. In humans, it is the intermediate or typical combinations of physical features that usually distinguish female from male.

INTERSPECIES HYDROGEN TRANSFER: The process in which hydrogen (H_2) is transferred between hydrogen-producing fermenting bacteria and hydrogen-consuming bacteria, occurring by the interaction of various microorganisms such as methanogens, acetogens, and sulfate-reducing bacteria.

INTERSPECIFIC: Occurring between or involving members of different species. The term is applied, for example, to some types of competitions and communications.

INTERSTELLAR SPACE: The space between the stars or any region of space beyond limits determined with reference to the boundaries of a celestial body or system, for example, the region of space immediately beyond Earth's atmosphere.

INTESTINE: Long, continuous tube running from the stomach to the anus. Most absorption of nutrients and water happen in the intestines. It includes large and small intestines and rectum.

INTERSTITIAL: A term describing a small atom that is in a position other than a normal lattice place. Examples of small atoms are hydrogen, boron, carbon and nitrogen.

INTERSTITIAL AD: An advertising screen interposed in a separate window while a webpage is loading.

INTERSTITIAL CELL: A cell that is not peculiar to or characteristic of a particular organ or tissue but forms part of a connective tissue between other tissues and structures, especially any of the cells of the testis that lie between the seminiferous tubules and secrete androgen in response to stimulation by interstitial-cell-stimulating hormones. Examples are neuroglial cells and Leydig cells.

INTERSTITIAL COMPOUND (INTERSTITIAL ALLOY): A compound in which ions or atoms of sufficiently small radius of a nonmetal occupy interstitial positions in a metal lattice. Such compounds often have metallic properties. Examples of small atoms are hydrogen, boron, carbon and nitrogen. These compounds are important industrially, for example as transition metal carbides and nitrides.

INTERSTITIAL FAUNA (MEIOFAUNA): Small invertebrate animals that live between individual substrate particles, such as sand grains, in the bed of a sea, lake or river.

INTERSTITIAL FLUID (TISSUE FLUID): Fluid that bathes and surrounds the cells in body tissues, providing a path through which nutrients, gases and wastes can travel between the capillaries and the cells.

INTERSTITIAL SKIN: The skin found between the scales of a snake.

INTERSTRATIFIED CLAY MINERAL: An aggregation composed of random or regular intergrowths of two or more clay minerals.

INTERSYSTEM CROSSING: A radiationless process involving a transition between two electronic states with different spin multiplicity. In this process a single excited electronic state makes a transition to a triplet excited state at the point where the potential energy curves for the excited singlet and triplet states cross. This transition is forbidden in the absence of a spin-orbit coupling but occurs in the presence of spin-orbit coupling.

INTERTIDAL ZONE: 1. The shallow zone of the ocean where land meets water. 2. A region at the sea shore which is covered by water at high tide and exposed to air at low tide.

INTERTROPICAL CONVERGENCE ZONE, ITCZ (EQUATORIAL CONVERGENCE ZONE; INTERTROPICAL FRONT): An area of low atmospheric pressure near the Equator that goes around Earth and where trade winds from the Northern and Southern hemispheres meet, forming thunderstorms. It is involved in the global circulation system.

INTERVAL: A set of points or real numbers between two specified points or real numbers. A closed interval, $[a, b]$, includes the endpoints and so is a set of the form $\{x: a \leq x \leq b\}$, where a and b are the endpoints, while an open interval, (a, b) , does not and so of the form $\{x: a < x < b\}$. On the real line, the half-open or half-closed interval from a to b is written $[a, b)$ or $(a, b]$, where square brackets indicate the inclusion of the endpoint (corresponding to a weak inequality the

set notation) and round parentheses its exclusion (corresponding to a strong inequality). Reversed brackets, $]a, b[$ are also used for the exclusion. Unbounded intervals are also written in this notation; for example, $[a, \infty)$ is the unbounded interval $x \geq a$ is regarded as closed, while (a, ∞) is the open interval $x > a$; the real line, $(-\infty, \infty)$, is both open and closed.

INTERVAL ANALYSIS: A method for estimating uncertainties in the results of a least-squares procedure. The input data are primary values plus limits of intervals expressing their uncertainty. The output consists of derived values plus limits of intervals expressing a pessimistic estimate of maximal uncertainty.

INTERVAL ESTIMATE (CONFIDENCE INTERVAL): In statistics, it is the probability that the true value of a population parameter will fall within an upper and lower bound of a probability distribution. For example, if $[F, G]$ is a 95% confidence interval of some parameter and F and G are both functions of a random sample from the given distribution, then 95% of such samples yield intervals that include the true value of the parameter. In other words, if the same sampling method is used to select different samples and an interval estimate for each sample is computed, one would expect the true population parameter to fall within the interval estimates 95% of the time. It is used to express the degree of uncertainty associated with a sample statistic.

INTERVAL GRAPH: An intersection graph based on a family of intervals of a partial order, especially the real line. It has one vertex for each interval in the family and an edge between every pair of vertices corresponding to intervals that intersect.

INTERVAL NOTATION: A notation for representing an interval as a pair of numbers. The numbers are the endpoints of the interval. Parentheses and/or brackets are used to show whether the endpoints are excluded or included. For example, $[3, 8)$ is the interval of real numbers between 3 and 8, including 3 and excluding 8.

INTERVAL OF CONVERGENCE: A power series of the form $\sum_{n=0}^{+\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots, c_n;$ where $a, c_0, c_1, c_2, \dots, c_n$ are given real numbers, the interval with endpoints $a-r$ and $a+r$ on which the power series converges.

INTRACELLULAR: Existing or taking place within a cell.

INTRACELLULAR DIGESTION: A form of digestion in which food nutrient is taken into cells by phagocytosis. It is found in sponges and most protozoa and coelenterates.

INTRACELLULAR PARASITES: Organisms which can only feed and live within the cell of a different animal. Examples are viruses that enter a host cell and take over the host's cellular machinery to produce new viral particles.

INTRACELLULAR ROUTE: A path taken by water through the cells of the root between the epidermis and the xylem, moving through plasmodesmata.

INTRADERMAL: Between or within layers of the skin. For example, in intradermal injection, the drug is injected into the upper layers of the skin. This is not the same as subcutaneous injection, which is injected under the skin.

INTRAFAVICULAR CAMBIUM (FASCICULAR CAMBIUM): The part of the vascular cambium that originates within the vascular bundles (fascicles) between the xylem and phloem.

INTRAFUSAL MUSCLE FIBRES: A type of muscle fibre that occurs in the muscle spindles in voluntary muscles. Two to twelve intrafusal fibres enclosed in a capsule are found in each capsule spindle, lying parallel to the main fibres. They are proprioceptors that detect the amount and rate of change of length in a muscle. There are two types of intrafusal muscle fibres: nuclear bag and nuclear chain fibres, bearing two types of sensory ending, known as annulospiral; and flower-spray endings.

INTRAMOLECULAR: 1. Descriptive of any process that involves a transfer (of atoms, groups, electrons, etc.) or interactions between

different parts of the same molecular entity. 2. Relating to a comparison between atoms or groups within the same molecular entity.

INTRAMOLECULAR CATALYSIS: The acceleration of a chemical transformation at one site of a molecular entity through the involvement of another functional ('catalytic') group in the same molecular entity, without that group appearing to have undergone change in the reaction product. This applies to cases for which analogous intermolecular catalysis by chemical species bearing that catalytic group is observable. Intramolecular catalysis can be detected and expressed in quantitative form by a comparison of the reaction rate with that of a comparable model compound in which the catalytic group is absent or by the measurement of the effective molarity of the catalytic group.

INTRAMOLECULAR CHARGE TRANSFER (ICT): Process that changes the overall charge distribution in a molecule.

INTRAMUSCULAR: Within or into a muscle or a drug, nutrient solution or other substance administered into the muscle. For example, intramuscular fat also known as seam fat is fat located between muscle systems.

INTRANET: A privately maintained computer network of an organisation or company that conforms to the same standards as the Internet but is only accessible to those members or employees who have authorisation to use it.

INTRAPERITONEAL: Within abdominal cavity or the peritoneal cavity, located between the peritoneum and the abdominal wall. For example, kidneys, adrenals and pancreas are intraperitoneal organs.

INTRAPHASE TRANSITION: A transition that occurs within a phase. An example is initiation of phase growth at an intragranular feature, such as twin planes and dislocations in steels.

INTRARUMINAL: Situated within, occurring within or administered by entering the rumen.

INTRANSITIVE: In mathematics, of a relation, having the property that if it holds between one element and a second and between that second and a third, then it must fail to hold between the first and third. For example, the relation mother of is intransitive, as no-one's mother can be his or her mother's mother or the successor function, since no integer can be its own successor. An intransitive relation is never transitive for any set of arguments whatever.

INTRASPECIFIC: Occurring between members of the same species. The term is applied to some types of competition and communication. For example, **intraspecific selection** is a natural selection among individuals of the same species. **Intraspecific competition** is the competition between organisms of the same species.

INTRASTAMINAL: Located within the stamen whorl of a flower. This usually refers to the location of the nectary disk.

INTRAVENOUS: Within the vein or a drug, nutrient solution or other substance administered into a vein.

INTRAZONAL SOILS: Soils that have well-developed structure different from that expected for its climatic and vegetational zone. This may be caused by the dominant influence of some resident factor of relief or parent material over the classic zonal effects of climate and vegetation. Examples of such soils are found in bogs and alkali flats.

INTRINSIC: 1. In anatomy, of certain muscles, nerves, etc., belonging to or lying within a given part. 2. In mathematics, of a property, relating only to the bearer of the property and not to the space in which it is embedded; thus a left-hand and a right-hand glove are intrinsically distinct but extrinsically homeomorphic. Being compact is an intrinsic property of a subset of a topological space, while being an open set is not.

INTRINSIC DETECTOR EFFICIENCY: The ratio of the number of particles or photons detected to the number of similar particles or photons which have struck the envelope limiting the sensitive volume of a radiation detector.

INTRINSIC FULL ENERGY PEAK EFFICIENCY: The detector efficiency when considering only events where the total energy of the radiation is absorbed in the sensitive volume of the detector.

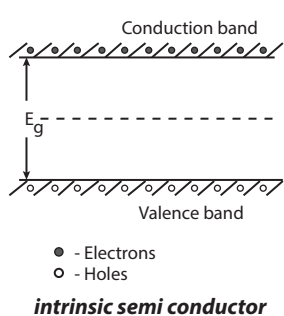
INTRINSIC PERMEABILITY: The ease with which gases or liquids can pass through a porous material under a hydraulic or potential gradient. It is independent of the nature of the liquid or the potential field.

INTRINSIC RATE OF INCREASE: 1. The difference between the number of births and the number of deaths. It is a measure of the instantaneous rate of change of population size (per individual). Its symbol is r , where r is expressed in numbers per unit time per individual and has the units of 1/time. In a closed population the intrinsic rate of increase is defined as the instantaneous per capita birth rate, b , minus the instantaneous per capita death rate, d . Similarly, in an open population $r = (\text{births} + \text{immigration}) - (\text{deaths} + \text{emigration})$. When per capita births exceed per capita deaths $b > d$, the population is increasing and r is positive; when deaths exceed births $b < d$, r is negative and the population is decreasing. 2. The maximum population growth rate.

INTRINSIC REACTION COORDINATE: A minimum-energy reaction path on a potential energy surface in mass-weighted coordinates, connecting reactants to products via the transition state.

INTRINSIC SEMI-CONDUCTOR (UNDOPED SEMICONDUCTOR, I-TYPE SEMICONDUCTOR): A

pure semiconductor without any significant dopant species present. It is an undoped semiconductor, meaning that holes in the valence band are vacancies created by electrons that have been thermally excited to the conduction band. Thus, the number of charge carriers is determined by the properties of the material itself instead of the amount of impurities. On the other hand, doped semiconductors refer to a situation where holes or electrons are supplied by a foreign atom acting as an impurity.



INTRODUCED SPECIES: A species that does not occur naturally in a given area, but has been brought to it from another area either deliberately or accidentally. An example is the Sweet clover (*Melilotus sp.*) that was introduced and naturalised to the U.S. from Eurasia as a forage and cover crop. Some introduced species are detrimental to the ecosystem while others are beneficial.

INTRODUCTION RULE: In logic, any syntactic rule of inference specifying the conditions under which a formula or statement containing a specified operator may be validly derived from others that may not contain it. For example, **conjunction introduction** is the rule in sentential calculus that permits the inference of the conjunction $A \& B$ from the two separate premises A and B ; **universal introduction** is the rule in predicate calculus that permits, under certain conditions, the inference of the universally quantified statement $(\forall x)Fx$ from an arbitrary instance Fa .

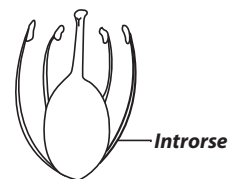
INTROGRESSION (INTROGRESSIVE HYBRIDISATION): The insertion of gene of one species into the gene pool of another. This can occur when two species interbreed to produce fertile hybrids, which can back-cross with individuals of one of the species.

INTROMITTENT ORGAN: An external copulatory organ like a penis or hemipenis in mammals and reptiles, respectively that delivers sperm.

INTRON (INTERVENING SEQUENCE; JUNK DNA): An intervening section of DNA occurring almost exclusively within an eukaryotic gene but which is not translated to amino acid sequences

in the gene product; it contains no genetic information. The introns are removed from the premature mRNA through a process called **splicing** to form an active mRNA.

INTRORSE: Facing inward or toward the axis of growth such as anthers that release their pollen to the inside of the flower to promote self pollination.



INTRUSION: 1. In botany, it is a protrusion as seen in placentae in plants when it protrudes into a cell of an ovary. 2. In geology, it is the movement of magma from within the earth's crust into spaces in the overlying strata to form igneous rock. Intrusive rocks are formed from the underground cooling of magma and are generally coarse-grained.

INTRUSION DETECTION SYSTEM (IDS): A type of security management system for computers and networks that gathers and analyses information from various areas within a computer or a network to identify possible security breaches such as attacks from outside the organisation and attacks from within the organisation. ID uses vulnerability assessment (sometimes referred to as scanning), which is a technology developed to assess the security of a computer system or network.

INTRUSIVE: Descriptive of igneous rocks that have formed underground. Intrusive or plutonic, igneous rock forms when magma is trapped deep inside the Earth. Great globs of molten rock rise toward the surface. Some of the magma may feed volcanoes on the Earth's surface, but most remains trapped below, where it cools very slowly over many thousands or millions of years until it solidifies. Slow cooling means the individual mineral grains have a very long time to grow, so they grow to a relatively large size. Intrusive rocks have a coarse-grained texture. A common intrusive rock is granite.

INTRUSIVE GROWTH (INTERPOSITIONAL GROWTH): The type of growth exhibited by certain cells when they intrude between other cells that are either not growing or growing at a slower rate. For example, certain members of the Liliaceae have vascular tracheids that elongate at both ends to achieve a length of between 15 and 40 times that of the original. These tracheids insinuate themselves amongst the surrounding cells, disrupting established plasmodesmata.

INTUBATION: The insertion of a tube into the larynx through the throat to permit breathing when the larynx becomes closed through swelling.

INTUSSUSCEPTION: 1. The growth of the surface area of a cell wall by the insertion of new cellulose fibre in the space between existing fibres in the wall of an elongating plant cell. 2. A sliding or prolapsing of a section of a tubular organ such as the intestine into another portion, particularly if part of the intestine invaginates or prolapses into another section of the intestine, causing swelling that results in obstruction. 3. Also known as **splitting angiogenesis**, it is the creation of a new blood vessel by the splitting of an existing one into two, one of the three ways that blood vessels are known to be formed in the human body; the others are angiogenesis and vasculogenesis. The section of the tubular organ such as the intestine that prolapses or invaginates into the other is called the **intussusceptum** and the part that receives it is called the **intussusciptions**.

INTUITIONISM: 1. In mathematics, the philosophical doctrine that mathematics cannot intelligibly comprehend the properties of most infinite sets and that only what can be shown to be provable by finitary methods can be justifiably asserted. For example, it follows classically from the identity $((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} = 2$ that there exists a pair of irrational numbers α and β for which α^β is rational. 2. The reconstruction of mathematics in accordance with this doctrine. **Intuitionistic** refers to a logical or mathematical system that is constructed in accordance with the principles of intuitionism, so that, for instance, the law of excluded middle does not hold since it may have no justification for asserting either a given statement or its negation.

INULIN: A group of naturally occurring polysaccharides, made from fructose molecules and produced by many plants. It is stored as food reserve in the root or rhizome as energy. Other forms of energy such as starch are not stored by plants.

INVAGINATION: 1. The local infolding of a layer of tissue, especially in animal embryos, so as to form a depression or pocket opening to the outside. 2. One of the methods by which the various germinal layers of the ovum are differentiated. In **embolic invagination**, one half of the blastosphere is pushed in towards the other half, producing an embryonic form known as a gastrula. In **epibolic invagination**, a phenomenon in the development of some invertebrate ova, the epiblast appears to grow over or around the hypoblast.

INVALID: Of an argument, not valid. The validity of a logical argument is whether or not the conclusion follows logically from the premises. An argument may be shown to be invalid either by constructing a counterexample or else by devising a possible world in which the premises are true but the conclusion is false.

INVAR: Nickel-steel alloy composed of 36% nickel, 0.4% manganese, 0.1% carbon and 63.5% iron developed by Charles-Édouard Guillaume (1861-1938), Swiss-born French physicist, for which he was awarded the 1920 Nobel Prize in physics. The length of invar does not vary with temperature: it is **invariable**, hence the name. It is used mainly in computer monitors.

INVARIABLE: Not changing or subject to change; constant.

INVARIANCE OF DOMAIN THEOREM: A theorem in topology about homeomorphic subsets of Euclidean space $\mathbb{R}^n$, which states that if U is an open subset of $\mathbb{R}^n$ and $f: U \rightarrow \mathbb{R}^n$ is an injective continuous mapping, then $V = f(U)$ is open and f is a homeomorphism between U and V .

INVARIANT: 1. The mathematical property that remains constant under certain transformation. For example, the distance between two points on a number line is not changed by adding the same quantity to both numbers. Multiplication does not have this property, so distance is not invariant under multiplication. Angles and ratios of distances are invariant under scalings, rotations, translations and reflections. 2. A quantity or set of quantities that characterise the relevant properties of an object within a given class.

INVARIANT PLANE (INVARIABLE PLANE): The plane in the Solar System defined by the combined angular momentum of the planets and the Sun. It is tilted by about $1^\circ 35'$ to the plane of the ecliptic.

INVARIANT SUBGROUP (NORMAL SUBGROUP): In mathematics, a subgroup that is left invariant by all inner automorphisms of the group or equivalently, has identical left- and right-cosets in the group. This occurs as the kernel of some homomorphism of the given group into another group (this may also define monoids). For example, the centre of a group is a normal subgroup.

INVARIANT SUBSPACE CONJECTURE: The conjecture that every continuous linear operator on an Hilbert space possesses a nontrivial invariant proper linear subspace. This clearly holds for matrices in more than one dimension as is shown by consideration of the span of an eigenvector. It has recently been shown that the analogous conjecture is false in the Banach space of absolutely summable sequences.

INVARIANT THEORY: The study of quantities that are associated with polynomial equations and that are left invariant under transformations of the variables. For example, the discriminant $b^2 - 4ac$ is an invariant of the quadratic form $ax^2 + bxy + cy^2$.

INVASIVE SPECIES: An introduced species that invades natural habitats. Its introduction may cause economic or environmental harm that can be detrimental to human health.

INVERSE: 1. In multiplication, it is also known as **reciprocal**, the reciprocal of a number is 1 divided by that number. For example, the reciprocal of x is $1/x$. 2. Of an element, related to a given element of a set on which an operation is defined in such a way that its product

with the given element under that operation is the identity element, e . The additive inverse of x is $-x$ and the multiplicative inverse or reciprocal of x is $1/x$. For non-commutative non-associative operations, an element, x , may have distinct left and right inverses, x_l and x_r , respectively, such that $x_l x = e = x x_r$. 3. Of a relationship, relating two quantities in such a way that an increase in one corresponds to a decrease in the other and vice versa; thus speed is in inverse proportion to the time taken to cover a given distance. The inverse of a function, f is denoted by f^{-1} .

INVERSE COMPTON EFFECT (COMPTON EFFECT): The decrease in energy and increase in wavelength of photons high-energy electromagnetic radiation when they are scattered by free electrons with much higher energy. As a result, the high-energy electrons lose energy. It was discovered by Arthur Compton (1892-1962), American physicist and Nobel laureate.

INVERSE ELEMENT: An element which gives an identity element when operated or combined with another element. For example, 1 is an identity element for multiplication because $2 \times 1 = 2$, this means that the inverse element for 2 is $\frac{1}{2}$ because

$2 \times \frac{1}{2} = 1$, $2 + (-2) = (-2) + 2 = 0$ is the additive identity element.

This shows that $\frac{1}{2}$ is the multiplicative inverse element of 2 and -2 as additive inverse element of 2. Every real number has its own inverse element.

INVERSE FUNCTION: In mathematics, if $y = f(x)$ and a function can be found such that $x = g(y)$, then $g(x)$ is said to be the inverse function of $f(x)$. Also, if the function is trigonometrical and of angle h , i.e. $y = \sin h$, then h is the inverse trigonometrical function of y .

INVERSE FUNCTION THEOREM: The special case of the implicit function theorem stating that a continuously differentiable function on Euclidean space possesses an inverse function in a neighbourhood of a point at which it has a non-singular Jacobian.

INVERSE IMAGE (COUNTER IMAGE): The set of elements whose image under a mapping lies in a given set. It is denoted by $f^{-1}(S)$ and is well defined even if the inverse mapping is not; for example, the counter image of the square root function on the non-negative reals is all the reals, even though this is a set-valued function.

INVERSE IMAGE SET: 1. **Weak or lower inverse image set** is the set in which each element has an image under a given correspondence F with a nonzero intersection with a given set B expressed as $\{x: F(x) \cap B \neq \emptyset\}$, and denoted $F^{(B)}$ or $F^{\cap}(B)$. 2. **Strong or upper inverse image set** is the set in which each element has an image under a given correspondence F that is contained in a given set B given as $\{x: F(x) \subseteq B\}$, denoted by either $F^{+(B)}$ or $F^{\subseteq}(B)$. If the given correspondence is one-to-one both inverse image sets may be identified with the counter image.

INVERSE MATRIX: The inverse of a matrix A is denoted by A^{-1} .

The inverse of a 2×2 square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is given by

$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Note, a matrix multiplied by its inverse gives the identity matrix. That is $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.

INVERSE MULTIPLEXING: In computing, a technique for spreading high-speed analogue and digital signals across multiple low-speed analogue and digital signals that are transmitted at the same time across separate channels to speed up voice, video and other data transmission which require the transmission of huge quantities of data. A device that performs this process is called a demultiplexer (DEMUX or DMX).

INVERSE NORMAL DISTRIBUTION: The probability distribution,

$f(x)$ which is given by $f(x) = \left(\frac{\gamma}{2\pi x^3}\right)^{\frac{1}{2}} \exp\left[-\frac{\gamma(x-\mu)^2}{2\mu^2 x}\right]$, where $x > 0$,

mean = μ , variance = $\frac{\mu^3}{\gamma}$. It is a member of the exponential family which is skewed to the right.

INVERSE OF A COMPLEX NUMBER: A complex number, denoted by $\frac{1}{z}$ or z^{-1} , which is the inverse of the complex number

$z = x + iy$, that is, it is the number z^{-1} for which $zz^{-1} = 1$. It is given by $z = \frac{x}{x^2+y^2} - i \frac{y}{x^2+y^2}$.

INVERSE PROPORTION (INDIRECT PROPORTION; INDIRECT VARIATION): A mathematical relationship in which an increase in one variable by a given factor brings about a decrease by the same factor in another.

INVERSE RATIO: A reciprocal ratio, for example, the inverse $\frac{c}{d}$ which is $\frac{1}{\frac{c}{d}} = \frac{d}{c}$.

INVERSE SQUARE LAW: A law in which one of the quantities is inversely proportional to the square of another. Examples are the law of electrostatic attraction $F_E \propto 1/r^2$ and gravitational force of attraction $F_G \propto 1/r^2$. An example is sound intensity and its inverse square law $1/r^2$.

INVERSION: 1. Also known as **chromosome inversion**, it is a structural change in chromosome, which occurs when a segment of the chromosome breaks off and reattaches in the reverse direction so that the genes within that part are in reverse order. It usually occurs during crossing over in meiosis. 2. A chemical reaction in which the direction of an optically active compound is reversed, from the dextrorotatory (rotating the plane of polarisation of light passing through it clockwise or to the right) to the levorotatory or from the levorotatory (rotating the plane of polarisation of light passing through it anticlockwise or to the left) to the dextrorotatory form. 3. Also known as **basin inversion**, it is the relative uplift of a sedimentary basin or similar structure as a result of crustal shortening. 4. A typical positioning of an organ, especially the turning inward or inside out of an organ. 5. A stable atmospheric condition in which the air temperature increases with increasing altitude, holding surface air down and preventing dispersion of pollutants. 6. In mathematics, a transformation in Euclidean geometry under which the image of each point, P, on a half-line through the origin, O, of a given circle or norm sphere with radius r , is the point, Q, on the same half-line such that $OP \cdot OQ = R$.

INVERSION HEIGHT: The height above ground level at which there is change in sign of the normal temperature 'lapse rate' (the rate of change of the temperature with height). Several temperature inversions may be present in the air over a given site at different altitudes as result of various meteorological factors.

INVERSION LAYER (DEPLETION REGION; DEPLETION LAYER; DEPLETION ZONE; JUNCTION REGION; SPACE CHARGE REGION): 1. A layer in a semiconductor material, formed from a conducting region by the removal of all free charge carriers and therefore impossible for current to flow. The only elements left in the depletion region are ionised donor or acceptor impurities. 2. In meteorology, it is a layer within which an inversion occurs. Temperature inversion is an increase of air temperature with altitude. It may occur during the passage of a cold front. It may also occur as a result of the invasion of sea air by a cooler onshore breeze. Such an increase is a reversal of the normal temperature condition of the troposphere, where temperature usually decreases with altitude. Inversions play important roles in determining cloud forms, precipitation, visibility and also acts as a lid, preventing the upward movement of the air below it.

INVERSION POINT: The temperature at which one polymorph of a solid phase changes into another at constant pressure.

INVERSION TEMPERATURE: 1. The critical temperature below which a non-ideal gas is expanded at constant enthalpy resulting in the decrease of temperature and above which a temperature increase occurs. This temperature change is known as the Joule-Thomson effect and is exploited in the liquefaction of gases, such as in refrigeration and air conditioning systems. 2. The temperature to which one junction of a thermocouple must be raised in order to make

the thermoelectric electromotive force in the circuit equals to zero, when the other junction of the thermocouple is held at a constant low temperature.

INVERT SUGAR: Natural disaccharide sugar that consists of a mixture of fructose and glucose, found in many fruits. It is also formed by the hydrolysis of sucrose, which can be brought about by the enzyme invertase. It can also be made by combining a sugar syrup with a small amount of acid such as cream of tartar or lemon juice and by heating, breaks down the sucrose into its two components, glucose and fructose, thereby reducing the size of the sugar crystals. Because of its fine crystal structure, invert sugar produces a smoother product and is used in making candies such as fondant and some syrup. The process of making jams and jellies automatically produces invert sugar by combining the natural acid in the fruit with granulated sugar and heating the mixture.

INVERTASE (SACCHARASE; SUCRASE; BETA-FRUCTOFURANOSIDASE; INVERTIN): An enzyme derived from yeast that hydrolyses sucrose into glucose and fructose. Biologically, many microorganisms produce invertase and can, thus, utilise sucrose as a nutrient. Commercially, invertase is biosynthesised chiefly by yeast strains of *Saccharomyces cerevisiae* or *Saccharomyces carlsbergensis*.

INVERTEBRATE: An animal without a backbone or vertebral column, generally soft-bodied and have an external skeleton for muscle attachment and protection. Examples are protozoans, annelids, cnidarians, echinoderms, flatworms, nematodes, molluscs and arthropods.

INVERTER: An electronic device that converts direct current into alternating current. It performs the opposite function of a rectifier.

INVERTIBLE: 1. Possessing an inverse. 2. Of an ideal, X , in a ring, R , such that in an over-ring of R , if $A = \{s \in S : sX \subseteq R\}$, $B = \{s \in R : Xs \subseteq R\}$, then $AX = XB = R$.

INVESTIGATE: To examine, study, or inquire into something systematically. For example, scientists and researchers use a systematic approach to answer questions about many things such as the environment.

INVISCID FLUID: An ideal fluid that is not a viscous fluid. The flow of fluids with low values of viscosity agree closely with inviscid flow everywhere except close to the fluid boundary where the boundary layer plays a significant role.

INVISIBLE INK (SECURITY INK): An ink, used for secret communication, that is colourless when written but that can be developed by heat or chemical reaction. For example, potassium hexacyanoferrate(II) ink can be developed by an iron(III) salt, giving Prussian blue. Fluorescent inks fluoresce in ultraviolet light. Invisible ink's uses include anti-counterfeiting, property marking, hand stamping for readmission and marking for the purpose of identification in manufacturing.

INVOLUCRE: A protective structure in some flowering plants such as zinnias and strawflowers and bryophytes such as moss. It consists of a ring of bracts arising beneath the flower cluster of those species with a capitulum or an umbel. In mosses and liverworts the involucre is a projection of tissue from the thallus that arches over the developing archegonium.

Involute is a plant that processes an involucre. **Involucel** is a whorl of bracts (involucre) that forms at the base of a flower cluster within a compound inflorescence.

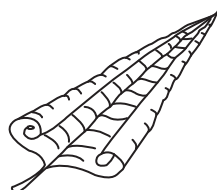


INVOLUNTARY ACTION: An action that occurs without conscious control. For example, breathing and urination reflexes are involuntary, although both can be controlled to some extent. If it occurs specifically in response to a stimulus, it is known as a **reflex**.

INVOLUNTARY MUSCLES (SMOOTH MUSCLES): A muscle that acts independently of the will, especially in reflex functions or a non-striated muscle associated with internal tissues and organs in mammals, which is not normally under voluntary, conscious control.

Unlike the voluntary muscle, it has no cross stripes; consists of narrow spindle-shaped cells with a single, centrally located nucleus; and contracts slowly and automatically. Its presence in the wall of the alimentary canal, for example, allows rhythmic movements known as peristalsis, which causes food to be mixed and forced along the gut.

INVOLUTE: 1. Also known as **evolvent**, it is a curve traced by the end of a taut thread that cannot be extended as it is being wound or unwound from another curve. An example is a cylindrical spool that is not free to rotate. 2. In botany, having petals or leaves that roll inward at the edges, as shown in the diagram. 3. In mathematics, any of a family of curves that is the locus of the free end of a thread that is held taut as it is unwound from a given curve, the evolute; the thread is thus always tangential to the given curve.



Involute leaf

INVOLUTION: 1. A decrease in the size of an organ or the body. It may be associated with functional decline as occurs in the ageing process or following an enlargement as when the uterus returns to its normal size after childbirth. 2. The turning or rolling inward of cells that occur during the development of some invertebrate embryo. 3. An operation, such as negation, which, when applied to itself, returns the original number. For example, a correspondence between the points on a line that is its own inverse, is given algebraically by $x' = \frac{ax+b}{cx-a'}$, where $cx + a' \neq 0$.

IODATED SALT: Table salt mixed with a small amount of potassium iodide or sodium iodide. The ingestion of iodine prevents iodine deficiency that can cause thyroid gland problems, including goitre, an enlargement of the thyroid gland.

IODIC ACID (IODIC(V) ACID): A colourless or very pale yellow solid, with the chemical formula HIO_3 , density 4.62 g/cm^3 (solid) and melting point 110°C (383 K). It gives a mixture of iodine, oxygen and lower oxides of iodine. It is soluble in water but insoluble in pure ethanol and other organic solvents. The compound is obtained by oxidising iodine with concentrated nitric acid, hydrogen peroxide or ozone. It is a strong acid and a powerful oxidising agent.

IODIC(VII) ACID (PERIODIC ACID): A hygroscopic white solid, which is an oxoacid of iodine with the molecular formula H_5IO_6 and decomposes at 140°C (413 K). It is very soluble in water, ethanol and ethoxyethane. Iodic(VII) acid may be prepared by electrolytic oxidation of concentrated solutions of iodic(V) acid at low temperatures. It is a weak acid but a strong oxidising agent.

IODIDE: A compound of iodine and another element. An example is a salt of hydriodic acid (HI). Metallic iodides such as silver, sodium or potassium iodide are used in photography and in iodised table salt.

IODINE: A shiny soft greyish black, corrosive, poisonous halogen element belonging to Group 17 (VIIA) of the Periodic Table. Its symbol is I, atomic number 53, atomic weight 126.9045, melting point 113°C (386 K), boiling point 184°C (457 K), specific gravity 4.93 and valence 1, 3, 5, 7. It sublimates (when heated) to a purple vapour. It is used in the manufacture of some antiseptics, germicides, dyes and medicines.

IODINE NUMBER (IODINE VALUE; IODINE ABSORPTION): A measure of the amount of iodine absorbed in a given time by a chemically unsaturated substance, such as a vegetable oil or a rubber. It gives a measure of the number of unsaturated bonds present. The higher the iodine number, the more unsaturated fatty acid bonds are present in a fat.

IODINE(V) OXIDE (IODINE PENTOXIDE; IODIC ACID ANHYDRIDE): A white solid with the chemical formula I_2O_5 which dissolves in water to give iodic(V) acid. It is insoluble in absolute alcohol, ether and chloroform and also acts as an oxidising agent. It decomposes at 275°C (548 K). It is used as an oxidising agent to oxidise carbon monoxide to dioxide at ordinary temperatures and in organic synthesis.

IDOETHANE (ETHYL IODIDE; HYDROIODIC ETHER): A colourless liquid with the molecular formula $\text{C}_2\text{H}_5\text{I}$ and boiling point 72°C (345 K). It is made by reacting ethanol with a mixture of iodine and red phosphorous. It is used in medicine and in organic synthesis.

ODOFORM (TRIODOMETHANE): A yellow, volatile, sweet-smelling solid with the chemical formula CHI_3 , density 4.008 g/mL^{-1} and melting point 121°C (394.25 K). It is synthesised by the reaction of iodine and sodium hydroxide with a methyl ketone, acetaldehyde, ethanol and certain secondary alcohols (CH_3CHROH , where R is an alkyl or aryl group). It is occasionally used as a disinfectant.

ODOMETHANE (METHYL IODIDE): A colourless liquid with the chemical formula CH_3I , density 2.28 g mL^{-1} , melting point -66°C (206.7 K) and boiling point $42-43^\circ\text{C}$ (315.5-315.9 K). It is made by reacting methanol with a mixture of iodine and red phosphorus. It has been used as an intermediate in the manufacture of some pharmaceuticals and pesticides. Iodomethane is a pre-plant biocide used in the control of insects, plant parasitic nematodes, soil borne pathogens and weed seeds.

ION: An electrically charged particle, which forms from an atom or group of atoms, when it loses or gains one or more electrons. In ionic bond formation, the atom that donates electron density is known as the **cation** (negatively charged ion), while the atom that gains electron density is known as the **anion** (positively charged ion). Examples of ions are alpha particle (He^{2+}), hydroxide (OH^-). Atoms can be converted to ions by radiation such as x rays or light of sufficient energy called ionising radiation.

ION ASSOCIATION: The electrostatic force of attraction which occurs in a solution of electrolytes, when ions of opposite electrical charge come together in solution to form a distinct chemical entity, based on the dielectric constant of the solvent.

ION CHANNEL: A protein present in the membranes that surround all biological cells, forming a water-filled pore through which ions can pass in or out of the cell or cell compartment. Ion channels are found in the plasma membrane and in certain internal cell membrane.

ION COLLECTOR: A device such as a Faraday cup collector (with a d.c. amplifier) or an electron multiplier, used in mass spectrometry for the capture of selected ions.

ION CYCLOTRON RESONANCE (ICR) MASS SPECTROMETER: A high-frequency mass spectrometer in which the ions to be detected, with a selected value of the quotient mass/charge, absorb maximum energy through the effect of a high-frequency electric field and a constant magnetic field perpendicular to the electric field. Maximum energy is gained by the ions which satisfy the cyclotron resonance condition and as a result they are separated from ions of different mass/charge.

ION ENGINE (ION THRUSTER): A type of electric propulsion engine used for propelling or controlling spacecraft, which creates thrust by accelerating ions. It consists of a unit producing a beam of ions which are accelerated by either electrostatic or electromagnetic force. Electrostatic ion thrusters use the Coulomb force to accelerate the ions in the direction of the electric field; and electromagnetic ion thrusters use the Lorentz force to accelerate the ions.

ION EXCHANGE: The exchange of ions of the same charge between two electrolytes or between an electrolyte solution, usually aqueous and a solid in contact with it. Ion exchangers are either cation exchangers that exchange positively charged ions (cations) or anion exchangers that exchange negatively charged ions (anions); and also amphoteric exchangers that are able to exchange both cations and anions simultaneously. The process occurs widely in nature especially in the absorption and retention of water-soluble fertilisers and is also used to remove certain ions from a solution or to make hard water soft and also to separate dangerous substances in nuclear waste.

ION EXCHANGER: A solid or liquid, inorganic or organic substance containing exchangeable ions with others of the same charge, present in a solution in which the ion exchanger is considered to be insoluble. A monofunctional ion exchanger contains only one type of ionogenic group; a bifunctional ion exchanger two types; and a polyfunctional

ion exchanger more than one type. In a macroporous ion exchanger the pores are large compared to atomic dimensions.

ION HYDROLYSIS: The reaction of water molecules with an ion such that O-H bonds are broken to give alkaline, neutral or acidic solution. There are two types: (i) cation hydrolysis, which always results in acidic solution; and (ii) anion hydrolysis, which results in basic solution. In hydrolyses of compounds with ionic bonds, the non water reactants are salts, acids or bases, participating in dissociation reactions.

ION IMPLANTATION: A process that utilises accelerated ions to penetrate a solid surface. It is used in implanting ions in the lattice of a semiconductor crystal in order to modify its electronic properties. The penetration depth can be controlled by adjusting the ion energy and the type of ions used. It is used as an alternative to diffusion or in conjunction with it, in the manufacture of integrated circuits and solid-state components and also to implant impurities on or near the surface of the substrate during the manufacture of semiconductors.

ION KINETIC ENERGY SPECTRUM: A spectrum obtained when a beam of ions is separated according to the translational energy-to-charge ratios of the ionic species contained within it. A radial electric field (E) achieves separation of the various ionic species in this way.

ION LASER: A gas laser in which the active species is an ion formed in the electric discharge. Most practically important lasers of this type have continuous wave (CW) output. Examples are the argon ion, kryptonion and helium-cadmium (HeCd) lasers.

ION MICROSCOPY: The use of the secondary ion mass spectrometry (SIMS) technique to obtain micrographs of the elemental or isotopic distribution at the surface of a sample with a spatial resolution of $2\mu\text{m}$ or better.

ION PAIR: A positive ion and an equal-charge negative ion, usually an electron formed by the transfer of an electron from one neutral atom or molecule to another by the action of radiation. For example, the ion pair Fe^{3+} and O^{2-} form the ionic compound Fe_2O_3 . An ion pair, whose constituent ions are in direct contact and not separated by an intervening solvent or other neutral molecule is called a **tight ion pair** or **intimate/contact ion pair**. A tight ion pair of X^+ and Y^- is symbolically represented as X^+Y^- . An ion pair whose constituent ions are separated by one or several solvent or other neutral molecules is described as a **loose ion pair**, symbolically represented as $\text{X}^+ || \text{Y}^-$. The members of a loose ion pair can readily interchange with other free or loosely paired ions in the solution.

ION PAIR RETURN: The recombination of a pair of ions R^+ and Z^- formed from ionisation of RZ . If the ions are paired as a tight ion pair and recombine without prior separation into a loose ion pair this is called **internal ion pair return**.

ION PLATING: A method of applying corrosion-resistant metal coating, utilising concurrent or periodic bombardment of the substrate and depositing film by atomic-sized energetic particles.

ION POLARISATION: The distortion or deformation of electron arrangements in an ion of opposite charge.

ION PROBE MICROANALYSIS (IPMA): Any technique in which the specimen is bombarded by a focused beam of (primary) ions (diameter less than $10\mu\text{m}$) and the (secondary) ions ejected from the specimen are detected after passage through a mass spectrometer.

ION PUMP: A type of vacuum pump that can reduce the pressure in a container to about 1 nano Pascal by passing a beam of electrons through the residual gas. The gas is ionised by electrons that have been generated by a high voltage and are spiralling in a high-intensity magnetic field. The positive ions formed are attracted to a cathode within the container where they are trapped. At very low pressures below 1 micropascal, the pump is useful. Ion pumps enable ions to flow through membranes in a thermodynamically uphill direction by the use of an energy source such as adenosine triphosphate (ATP) or light. They consist of sugar-containing hetero-peptide assemblies,

which open and close upon the binding and subsequent hydrolysis of ATP, usually transporting more than one ion towards the outside or the inside of the membrane.

ION SCATTERING SPECTROMETRY (ISS): Any technique using low energy (10 keV) ions in which the bombarding particles scattered by the sample are detected and recorded as a function of energy and/or angle. This technique is used mainly for determining the composition and structure of the first few atomic layers of a sample.

ION SOURCE: Generally, it is an assembly composed of the following: (i) an ionisation chamber in which a stream of electrons flows from a hot filament across a stream of gas to collector (the potential between filament and collector is usually between 50 and 70 V) and (ii) a device for the acceleration of these ions.

ION TRAP MASS SPECTROMETER: An arrangement in which ions with a desired range of quotients mass/charge are first made to describe stable paths under the effect of a high-frequency electric quadrupole field. They are then separated and presented to a detector by adjusting the field so as to selectively induce path instability according to their respective mass/charge ratios.

ION-BEAM ETCHING: A method of specimen preparation for scanning electron microscopy in which the specimen is placed under high vacuum and the surface is then eroded by a beam of ions to expose subsurface features. The specimen is then coated and examined. The technique is very successful with some materials but produces artefacts by uneven erosion of the surface of others.

ION-DIPOLE FORCE: The intermolecular attractive force that operates between an ion and a polar molecule (dipole).

ION-EXCHANGE CHROMATOGRAPHY: A chromatographic method in which the components of mixtures are separated by differences in their acid-base behaviour. The chromatogram column is filled with a charged resin that is easily able to exchange ions for any charged molecules that are passed through the column. Amino acid separation is often achieved by this method and the resin commonly used is sulfonated polystyrene in which the sulfonic acid groups are charged with sodium ions.

ION-EXCHANGE RESIN (ION-EXCHANGE POLYMER): A polymeric resinous substance, usually in bead form of 1-2 mm diameter, usually white or yellowish, fabricated from an organic polymer substrate that contains fixed groups with positive or negative charge. A cation exchange resin has negatively charged groups and is, therefore, useful in exchanging the cationic groups in a test sample. Ion-exchange resins are widely used in different separation, purification and decontamination processes. The most common examples are, zeolites, montmorillonite, clay and soil humus.

ION-FREE LAYER: An interphase in which only non-specifically adsorbed ions are present.

ION-MICROBE ANALYSIS: A technique for analysing the surface composition of solids.

ION-MOBILITY SPECTROMETRY (IMS): A technique for separating and detecting low concentration of specific compounds, based on the rate at which their ions migrate through an electric field. The instrument operates in the gas phase at atmospheric pressure.

ION-PAIR FORMATION: It involves an ionisation process in which a positive fragment ion and a negative fragment ion are among the products.

ION-SELECTIVE ELECTRODE (ISE; SPECIFIC ION ELECTRODE; SIE): A transducer or sensor that converts the activity of a specific ion dissolved in a solution into an electrical potential, which can be measured by a voltmeter or pH meter. The voltage is theoretically dependent on the logarithm of the ionic activity, according to the Nernst equation. The sensing part of the electrode is usually made as an ion-specific membrane, along with a reference electrode. Ion-selective electrodes are used in biochemical and biophysical research, where measurements of ionic concentration in an aqueous solution are required, usually on a real time basis.

IONIC BOND (ELECTROVALENT BOND): A chemical bond formed when atoms of different elements are transferred from one form to another, forming oppositely charged ions cations or anions being held together by electrostatic attraction or cation respectively; or a chemical bond formed by the attraction between two oppositely charged ions. These ions are almost spherical and attract one another due to their opposite charges. Ionic bonding takes place when the electronegativity difference between the two atoms is more than 1.7. This usually happens when a metal atom bonds with a non-metal atom. An example of ionic bonding is between magnesium (Mg) and oxygen (O₂) to form magnesium oxide (MgO). In this case, magnesium has two valence electrons and a lower electronegativity of 1.31, while oxygen has six valence electrons and an electronegativity of 3.44. Oxygen with its higher electronegativity, attracts the two valence electrons from the magnesium atom and these electrons are transferred from the magnesium atom to the oxygen atom. Magnesium loses two electrons to form Mg²⁺, and oxygen gains two electrons to form O²⁻. The attractive force between the oppositely charged ions is what holds the compound together. The balanced equation for the reaction is: 2Mg + O₂ → 2MgO. In ionic compounds, ions are held together in a crystal lattice structure by ionic bonds. Usually, the positively charged portion consists of metal cations and the negatively charged portion is an anion or polyatomic ion. Some properties of ionic compounds include the following: (i) ions are arranged in a lattice structure; (ii) ionic solids are crystalline at room temperature; (iii) the ionic bond is a strong electrostatic attraction and so ionic compounds are often hard and have high melting and boiling points; (iv) ionic compounds are brittle and bonds are broken along planes when the compound is put under pressure (stressed); (v) solid crystals do not conduct electricity, but do in a molten state or aqueous solution.

IONIC COMPOUND: A chemical compound such as sodium chloride (NaCl) in which ions are held together in a lattice structure by ionic bond. It has high electronegativity difference and it is brittle. Usually, the positively charged portion is an anion or polyatomic ion. They have high melting and boiling point and they conduct electricity in a molten state or aqueous solution. Metals and non-metals generally react to form ionic compounds and in that reaction metals generally lose electrons to complete their octet while non-metals gain electrons to complete their octet. Ionic compounds include hydroxides, salts, oxides, sulfides and most inorganic compounds.

IONIC CRYSTAL (ELECTROVALENT CRYSTAL; POLAR CRYSTAL): A crystal compound composed of ionic species bound together by their electrostatic attraction. In ionic crystals the ions and vacancies always arrange themselves such that there is no build up of one type of charge in any small volume of the crystal. Examples of such crystals are the alkali halides, including potassium fluoride, potassium chloride, potassium bromide, potassium iodide, sodium fluoride and other combinations of sodium, caesium, rubidium or lithium ions with fluoride, bromide, chloride or iodide ions.

IONIC DISSOCIATION: The decomposition of an ion into another ion of lower mass and one or more neutral species.

IONIC EQUATION: A chemical equation that includes only the ions involved as reactants in a chemical reaction. For example, in an equation $\text{NaOH}_{(\text{aq})} + \text{HCl}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$, the ionic equation is $\text{OH}^- + \text{H}^+ \rightarrow \text{H}_2\text{O}$.

IONIC HALF EQUATION: An equation that describes the reactions occurring at the electrodes of a chemical cell or in electrolysis. It may also be defined as an equation that shows explicitly, the total number of electrons lost or gained. It indicates which ion is losing electrons (oxidation) or gaining electrons (reduction). Examples are $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$.

IONIC POLYMER (ION-CONTAINING POLYMER): Polymer composed of macromolecules containing ionic or ionisable groups or both, irrespective of their nature, content and location.

IONIC POLYMERISATION: A chain polymerisation in which the kinetic-chain carriers are ions or ion pairs.

IONIC PRODUCT: Product of the concentration of ions in a liquid or solution, for example, in sodium chloride solution. The ionic product of sodium chloride is given by $[\text{Na}^+][\text{Cl}^-]$. In a pure liquid it results from the dissociation of molecules in the liquid.

IONIC PRODUCT OF WATER: The equilibrium constant K_w for the reaction in which water undergoes an acid-base reaction with itself. K_w is the product of hydrogen ion concentration and hydroxide ion concentration at a particular temperature. $K_w = [\text{H}^+][\text{OH}^-]$ at constant temperature.

IONIC RADIUS: A value assigned to the size or radius of an ion in a crystal lattice, based on the assumption that the ions are spherical with a definite size. It was developed independently by Victor Goldschmidt and Linus Pauling in the 1920s to summarise the data being generated in X-ray crystallography. Its value is in either picometres (pm) or Angstrom (Å), with 1 Å = 100 pm. Typical values range from 30 pm (0.3 Å) to over 200 pm (2 Å). Pauling's method of calculating ionic radius proved to be the more influential, but in a review of crystallographic data in 1976, a set of ionic radii had been published and these are preferred to Pauling's original values.

IONIC STRENGTH: A measure of the concentration of ions in a solution, which is a function, denoted by I , which expresses the effect of the charge of ions in a solution as the product of sum of the molality (concentration of solution) of each type of ion present (m) and the square of its charge (z). Mathematically, $I = \frac{1}{2} \sum m z^2$.

IONISATION (IONIZATION): The process by which ions are produced, typically occurring by collisions with atoms or electrons, or by interaction with electromagnetic radiation. It is the physical process in which an electrically neutral atom or molecule loses or gains electrons and acquires an electric charge. For example, neutral molecules of hydrogen chloride gas, HCl, react with similarly polar water molecules, H₂O, to produce positive hydronium ions, H₃O⁺ and negative chloride ions, Cl⁻; at the surface of a piece of metallic zinc in contact with an acidic solution. Zinc atoms, Zn, lose electrons to hydrogen ions and become colourless zinc ions, Zn²⁺. Electron transfer also causes ionisation in certain reactions and ions may also be formed when an atom or molecule loses one or more electrons.

IONISATION/IONIZATION BUFFER: A buffer added to increase the free-electron concentration by the flame gases, thus repressing and stabilising the degree of ionisation.

IONISATION/IONIZATION CHAMBER: A radiation detector for measuring or detecting ionising radiation, which employs an electric field for the collection at the electrodes of charges associated with the ions produced in the sensitive volume by ionising radiation, without charge multiplication.

IONISATION/IONIZATION CONSTANT: Equilibrium constant for the ionisation of a weak electrolyte.

IONISATION /IONIZATION CROSS-SECTION: A measure of the probability that a given ionisation process will occur when an atom or molecule interacts with an electron or a photon.

IONISATION/IONIZATION EFFICIENCY: The ratio of the number of ions formed to the number of electrons or photons used in an ionisation process. **Ionisation efficiency curve** is the number of ions produced as a function of the energy of the electrons or photons used to produce ionisation.

IONISATION/IONIZATION ENERGY: Minimum energy required to eject an electron out of a neutral atom or molecule in its ground state. The **adiabatic ionisation energy** refers to the formation of the molecular ion in its ground vibrational state; and the **vertical ionisation energy** applies to the transition to the molecular ion without change in geometry. This quantity was formerly called ionisation potential. The second ionisation energy of an atom is the energy required to eject the second electron from the atom.

IONISATION/IONIZATION EQUILIBRIUM: The condition in a hot gas when the rate of ionisation of a particular ion is balanced by the rate of recombination.

IONISATION/IONIZATION GAUGE: A vacuum gauge consisting of a three-electrode system inserted into the container in which the pressure is to be measured.

IONISATION/IONIZATION ISOMERS: Isomers that result from the interchange of ions inside and outside the compound.

IONISATION/IONIZATION POTENTIAL (IP; IONISATION ENERGY, E_i): The minimum energy required to remove an electron from a specified atom or molecule to such a distance that there is no electrostatic interaction between the ion and electron. In other words, it is the minimum amount of energy required to remove one mole of electron from an isolated gaseous atom to form gaseous cation. It can also be defined as the potential difference through which a bound electron must be raised to free it from the atom or molecule to which it is attached, which is a measure of its ability to enter into chemical reactions requiring ion formation or donation of electrons and is related to the nature of the chemical bonding in the compounds formed by elements.

IONISATION/IONIZATION TEMPERATURE: The temperature at which the average kinetic energy of gas molecules having a Maxwell distribution equals the ionisation energy.

IONISED/IONIZED LAYERS: Layers of increased ionisation within the ionosphere, produced by cosmic radiation. They are responsible for absorption and reflection of radio waves and are important in connection with communication and the tracking of satellites and other space vehicles.

IONISING/IONIZING COLLISION: An ion/neutral species interaction in which an electron or electrons are stripped from the ion and/or the neutral species in the collision. Generally, this term has come to be used to describe collisions of fast-moving ions with a neutral species in which the neutral species is ionised with no change in the number of charges carried by the ion. Care should be taken when this term is used to emphasise if charge stripping of the ion has taken place.

IONISING/IONIZING POWER: A term used for the tendency of a particular solvent to promote ionisation of an uncharged or, less often, charged solute. The term has been used both in a kinetic and in a thermodynamic context.

IONISING/IONIZING RADIATION: A radiation of sufficiently high energy as to cause ionisation in the medium through which it passes or subatomic particles or electromagnetic waves that are energetic enough to detach electrons from atoms or molecules, thus ionising them. This type of radiation can cause extensive damage to the molecular structure of a substance. Examples of ionising particles are energetic alpha particles, beta particles and neutrons.

IONISING/IONIZING VOLTAGE: The voltage difference through which electrons are accelerated before they are used to bring about electron impact ionisation. To obtain the true ionising voltage, corrections for any contact or surface potentials must be made.

IONOGENIC GROUPS: The fixed groupings in an ion exchanger which are either ionised or capable of dissociation into fixed ions and mobile counter-ions.

IONOMER: A thermoplastic resin with ionic bonds between the polymer chains, with physical properties such as electrical conductivity and increase in ionomer solution viscosity with increasing temperatures. Ionomers are used commercially for golf ball covers, semipermeable membranes and thermoplastic elastomers.

IONOMER CLUSTER: Ionic aggregate, in a polymer matrix of low polarity, formed through interactions of ionomer multiplets.

IONOMER MOLECULE: A macromolecule in which a small but significant proportion of the constitutional units have ionisable or ionic groups or both. Some protein molecules may be classified as ionomer molecules.

IONOPHORE: A relatively small hydrophobic molecule found in lipid membranes that facilitates the transport of ions across (lipid) membranes. Most ionophore channel formers combine to form a

channel in the membrane through which ions can flow and mobile ion carriers transport ions across a membrane by forming a complex with the ion. Examples of ionophores are valinomycin and nigericin.

IONOSPHERE: The part of the upper atmosphere between 50 and 1,000 km above the Earth's surface that contains electrically charged particles or ions, which affects the propagation of radio waves, for example, by reflecting them back to Earth.

IONOSPHERIC WAVE (SKY WAVE): A radio wave that is reflected or propagated back to Earth by the ionosphere or a communications satellite, permitting transmission around the curve of the Earth's surface. Skywave propagation permits broadcast signal from distant AM broadcasting stations at night or from a shortwave radio station or a low band TV station can sometimes be heard as clearly as local stations.

IONOTROPIC RECEPTOR: A recent protein that forms part of a ligand-gated ion channel, so that binding of ligand to the receptor causes opening of the channel, permitting ions to flow through it.

IP ADDRESS (IP NUMBER; "IP"): A code made up of numbers separated by three dots that identifies a particular computer on the Internet, for example, 1.160.10.240. It is a 32-bit numeric address written as four numbers separated by periods. Each number can be zero to 255. Every computer requires an IP address to connect to the Internet.

IP MULTICAST (INTERNET PROTOCOL MULTICAST): A method of sending data between a single sender and multiple receivers on a network that enables many recipients to share the same source. It is often employed for streaming media applications on the Internet and private networks.

IPAD: A new handheld tablet computing device from Apple Inc., designed for consumers who want a mobile device that is bigger than a smartphone but smaller than a laptop for entertainment multimedia. It was first launched in January 2010.

IPHONE: An innovative communications device announced by Apple on Jan. 9, 2007. iPhone combines into one small (11.6 mm thick), hand-held device three products: (i) a revolutionary mobile phone; (ii) a widescreen iPod with touch controls; and (iii) an Internet communications capability with desktop-class email, web browsing, maps and searching. The user interface is a touch display. Essentially, iPhone is a computer with a blank screen that users configure so they can operate the monitor with their fingers. The device comes with a built-in two megapixel camera.

IPOD: A brand of portable digital media player designed and marketed by Apple Computer. Devices in the iPod family provide a simple user interface designed around a central scroll wheel (with the exception of the iPod shuffle). The standard iPod model stores media on a built-in hard drive, while the smaller iPod shuffle and iPod nano use flash memory. Like most digital audio players, an iPod can serve as an external data storage device when connected to a computer. Since the release of the Apple iPod in 2001, under the iPod brand, Apple has released many variations of its product such as the iPod classic, iPod Touch, iPod Shuffle, iPod Mini, iPod Nano and several spin-off devices such as the iPod Photo.

IPSILATERAL: Of, relating to or affecting the same side of the body.

IPv4: The most widely used version of the Internet Protocol. It defines IP addresses in a 32-bit format, which looks like 123.123.123.123. Each three-digit section can include a number from 0 to 255. The total number of IPv4 addresses available is 4,294,967,296. Every computer requires an IP address to connect to the Internet. However, with the growth of computers connected to the Internet, the number of available IP addresses are predicted to run out in only a few years. IPv6 has been introduced to solve this problem.

IPv6 (IPNG; IP NEXT GENERATION): The upgrading of the IPv4 system, which uses 32-bit addresses and expected to run out due to the growth of computers connected to the Internet. IPv6 uses 128-bit

addresses, which increases the number of possible addresses by an exponential amount. IPv6 also supports auto-configuration to help correct most of the shortcomings in version 4 and it has integrated security and mobility features.

IrDA (INFRARED DATA ASSOCIATION): Founded in 1993 by around 50 companies, it is an industry-sponsored organisation that sets the standard by which wireless infrared communications may be used to beam data between two devices with built-in IrDA ports, for example, between two computers or a computer and a printer. For two IrDA devices to beam data between them, they must be about 1 metre from each other and must have a clear line of sight. Portable devices such as mobile telephones, laptops, cameras and printers, have IrDA ports.

IRIDESCENCE: The production of colours of varied hue by interference of light reflected from front and back of thin films (as in soap bubbles) or from faults and boundaries within crystalline solids such as mica or with opal. The colours of mother-of-pearl and some insects are due to iridescence.

IRIDIUM: Steel-grey metallic element in Group 9 (VIIIB) of the Periodic Table, with the chemical symbol Ir, atomic number 77, atomic weight 192.2, melting point 2,446 °C (2,719 K), boiling point 4,428 °C (4,701 K), specific gravity 22.562 at 293 K or 20 °C and valence 2, 3, 4, 6. It is a very hard and brittle, exceptionally corrosion-resistant, whitish-yellow metallic element occurring in platinum ores. It is used in high-temperature materials, electrical contacts and wear-resistant bearings, surgical tools and crucibles.

IRIDIUM ANOMALY: The occurrence of unusually high concentrations of the relatively scarce chemical element, iridium in a layer of rock strata. Two such layers have been discovered, one at the end of the Cretaceous, 65 million years ago and the second at the end of the Eocene 34 million years ago.

IRIS: A pigmented muscular tissue lying between the cornea and the lens, which regulates the amount of light reaching the retina in vertebrate eyes.

IRON: A hard, malleable and ductile silver-grey, metallic element, with the symbol Fe, atomic weight 55.847, atomic number 26, melting point 1540 °C (1813 K), boiling point 2800 °C (3073 K) and density 7.86 g/cm³. The two main ores from which it is obtained are haematite (hematite) Fe₂O₃ and limonite, Fe₂O₃·3H₂O. It is also obtained from siderite, a brownish carbonate. It is the fourth most abundant element in the Earth's crust and an essential component of haemoglobin in the red blood cells that transport oxygen to all parts of the human body. Iron is used in processed forms, such as wrought iron, cast iron and steel. Commercially, pure iron is used for the production of galvanised sheet metal and of electromagnets. Iron compounds are employed for medicinal purposes in the treatment of anaemia, when the amount of haemoglobin or the number of red blood corpuscles in the blood is lowered.

IRON PAN: A thin hard soil layer often found below the surface of sands and gravels in which iron oxides constituent the main cementing material. The various cementing materials include the following: iron-organic matter complexes; hydrous oxides of manganese and iron; and hydrous iron oxides.

IRON PYRITE (PYRITE; FOOL'S GOLD): Iron sulfide with the chemical formula FeS₂. It is a pale yellow-gold or brassy metallic lustre, resembling minerals such as arsenopyrite, chalcopyrite and pyrrhotite, in physical outlook and appearance. It is an ore of iron and a source of iron and sulfur. It occurs naturally in some types of crystals. One type, auriferous pyrite resembles gold so much that it is exploited commercially as gold.

IRON-SULFUR CLUSTER: A unit comprising two or more iron atoms and bridging sulfur ligands in an iron-sulfur protein. The recommended designation of a cluster consists of the iron and sulfur content, in square brackets, for example, [2Fe-2S], [3Fe-4S].

IRON-SULFUR PROTEINS: Proteins in which non-haem iron is coordinated with cysteine sulfur and usually also with inorganic sulfur. It is divided into three major categories: rubredoxins; 'simple iron-sulfur proteins', containing only iron-sulfur clusters; and 'complex iron-sulfur proteins', containing additional active redox centres such as flavin, molybdenum or haem. In most iron-sulfur proteins, the clusters function as electron-transfer groups, but in others they have other functions, such as catalysis of hydratase/dehydratase reactions, maintenance of protein structure or regulation of activity.

IRON(II)HYDROXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for ferrous hydroxide. Its chemical formula Fe(OH)₂, molar mass 89.86 g/mol and density 3.4 g/cm³. It is a white, water-insoluble, gelatinous solid that turns reddish-brown as it oxidises to ferric hydroxide. It is produced when iron(II) ions, from a compound such as iron(II) sulfate, react with hydroxide ions.

IRON(II)OXIDE (FERROUS OXIDE; IRON OXIDE; OXIDISED IRON; RUSTED IRON; OR BLACK SOLID): A greenish to black powder with the chemical formula FeO, consisting of the element iron in the oxidation state of 2 bonded to oxygen. Its density is 5.74 g/cm³, molar mass 71.844 g/mol, melting point 1377 °C (1650.15 K) and boiling point 3414 °C (3687.15 K). It occurs in nature as the mineral wuestite. It can be prepared by heating iron(II) oxalate in vacuo: FeC₂O₄ → FeO + CO + CO₂. The stoichiometric form can be prepared by heating Fe_{0.95}O with metallic iron at 770 °C and 36 kbar. It contains iron and oxygen and used to manufacture steel and enamels and used primarily as a pigment for glasses, in cosmetics and in some tattoo inks. It can also be used for filtering phosphates from home aquaria.

IRON(II)SULFATE (FERROUS SULFATE; GREEN VITRIOL; IRON VITRIOL): An off-white solid, which occurs in various states of hydration. Two forms are monoclinic with the chemical formula FeSO₄·H₂O and heptahydrate with the chemical formula FeSO₄·7H₂O. The heptahydrate is the best known iron(II) salt and is sometimes called green vitriol or copperas. It is obtained by the action of dilute sulfuric acid on iron in a reducing atmosphere. The anhydrous compound is very hygroscopic. It decomposes at red heat to give iron(III)oxide, sulfur trioxide and sulfur dioxide. A solution of iron(II) sulfate is gradually oxidised on exposure to air, a basic iron(III)sulfate being deposited. It is used medically to treat iron deficiency and also for industrial applications.

IRON(III) CHLORIDE (FERRIC CHLORIDE): A black-brown solid with the chemical formula FeCl₃, molar mass 162.2 g/mol (anhydrous) and 270.3 g/mol for the hexahydrate, density 2.898 g/cm³ (anhydrous), 1.82 g/cm³ (hexahydrate), melting point 306 °C (579 K) for the anhydrous and 37 °C (310 K) for the hexahydrate, boiling point 588 K (315 °C) for the anhydrous and decomposes and 553 K (280 °C) for the hexahydrate and decomposes and partial decomposition to FeCl₂ + Cl₂. When dissolved in water, iron(III) chloride undergoes hydrolysis and gives off heat in an exothermic reaction. Iron(III) chloride is prepared by passing dry chlorine over iron wire or steel wool. It is used in sewage treatment and drinking water production. It is also used as a leaching agent in chloride hydrometallurgy, in etching copper in two-step redox reaction to copper(I) chloride and then to copper(II) chloride in the production of printed circuit boards, etc.

IRON(III) OXIDE (FERRIC OXIDE): An odourless, red-brown solid, with the chemical formula is Fe₂O₃, molar mass 159.69 g/mol, density 5.242 g/cm³ (solid), melting point 1566 °C (1839 K) and decomposes. It is one of the three main oxides of iron, together with iron(II) oxide (FeO) and iron(II,III) oxide (Fe₃O₄). It contains iron and oxygen and is readily attacked by acids. Its sources are rust and hematite. It is a product of the oxidation of iron. It can be prepared in the laboratory by electrolyzing a solution of sodium bicarbonate, an inert electrolyte, with an iron anode: 4Fe + 3O₂ + 2H₂O → 4FeO(OH). It is the main source of iron for the steel industry. It is also used in pigment, in jeweller's rouge for polishing and on magnetic recording tape.

IRONSTONE: A fine-grained, heavy and compact sedimentary rock which contains iron, usually in the form of the mineral haematite, limonite or pyrite. Its main components are the carbonate or oxide of iron, clay and/or sand.

IRRADIANCE (RADIANT EMITTANCE; RADIANT EXITANCE):

The radiant flux per unit area of electromagnetic radiation reaching a surface. It is a radiometry term for the power per unit area at a surface. Its SI unit is watts per square metre (W/m^2). Its symbol is E .

IRRADIATION: An exposure to any form of radiation, especially ionising radiation and also to non-ionising radiation such as microwaves or to low frequency (50/60 Hz) power supply, high frequency such as cellular phones, radio and TV transmissions.

IRRADIATION ILLUSION: A distortion illusion in which a light area of the visual field looks larger than an otherwise identical dark area. It arises partly from scattering of light inside the eye. This has the effect of enlarging the image of a light area on the retina. It was discovered by Hermann von Helmholtz in the 19th century.

IRRATIONAL: 1. In mathematics, not expressible as a ratio of integers. 2. Of an equation, etc., containing one or more variables in irreducible radical form or raised to a fractional power, for example, $\sqrt{(x^2 + 1)} = x^{5/3}$.

IRRATIONAL NUMBER: A number that cannot be expressed as the ratio of two integers, usually defined as a limit of a sequence of rational numbers or as a partition of the rationals into those greater and those less than it. An irrational number may be a surd, such as $\sqrt{3}$, which can be expressed to any desired degree but cannot be assigned an exact value. Some other examples of irrational numbers are π , e and $\log_2 3$.

IRREDUCIBLE: 1. In mathematics, of a polynomial over a field, unable to be factorised into polynomials of lower degree over the same field; for example, $(x^2 + 1)$ is irreducible over the reals, but can be factorised as $(x + i)(x - i)$ over the complex numbers. Such a polynomial is also prime. 2. Of a non-zero element in an integral domain, not expressible as a product of two non-units: if $a = bc$ then either b or c is a unit. All other non-zero elements are either units or are reducible. The number 5 is irreducible in the integers but since $5 = (2 + i)(2 - i)$, it is reducible as a Gauss-Ian integer. By a theorem of Fermat, this is also true of any prime that is congruent to 1 modulo 4, being expressible as a sum of two integral squares. 3. Of a radical, unable to be expressed as a rational expression; for example, $f(x + 1)$ is irreducible.

IRREDUCIBLE REPRESENTATION: A representation of a symmetry operation of a group, which cannot be expressed in terms of a representation of lower dimension.

IRREFLEXIVE (ALIORELATIVE): A relation R on a set S is aliorelative or irreflexive provided that no element is related to itself or a relation that can never hold between itself and other relations. For example, the relation $<$ is an irreflexive relation on the set of natural numbers. An irreflexive relation is never reflexive on any subdomain whatever.

IRREGULAR (ZYGOMORPHIC): In botany, it refers to bilaterally symmetrical, such as a flower in which one or more members of a whorl or whorls are dissimilar in size and arrangement on the receptacle. Two mirror halves can be obtained by cutting the flower longitudinally through only on one particular vertical plane. An example is found in *Crotalaria* sp. (Rattle pod). On the other hand, **Regular** or **Actinomorphic** is a flower which has members of a whorl similar in shape and size. Examples are apple tree and lily. Two mirror halves can be obtained by cutting the flower longitudinally through many planes.

IRREGULAR GALAXY: A small galaxy that has no identifiable shape and that contains a great amount of dust and gas.

IRREGULAR MACROMOLECULE: A macromolecule with a structure which essentially comprises the repetition of more than one type of constitutional unit or a macromolecule with a structure which comprises constitutional units not all connected identically with respect to directional sense.

IRREGULAR SOLIDS: Solids that do not have clear defined dimensions that can be measured, therefore its length, width and

height cannot be measured to determine its volume. Volume can only be determined with the principle of liquid displacement. Examples of such solids include rocks, prisms, beans and screws.

IRREVERSIBLE PROCESS (IRREVERSIBILITY): A process that is not reversible. This concept arises most frequently in thermodynamics, as applied to processes. In thermodynamics, a change in the thermodynamic state of a system and all of its surroundings cannot be precisely restored to its initial state by infinitesimal changes in some property of the system without expenditure of energy. An irreversible process increases the entropy of the system, which is a measure of the microscopic disorder of the system whereas a reversible process does not. The second law of thermodynamics can be used to determine whether a process is reversible or not.

IRREVERSIBLE TRANSITION: A transition that changes the state of a system which cannot be readily reversed to restore the system to its original state. An example is the conversion of TiO_2 (anatase) to TiO_2 (rutile).

IRRIGATION: The artificial application of water to agricultural areas by means of dams and channels and pipelines; or adding water mechanically to the soil or the addition of water to plants to supplement that provided by rain for plant growth. There are several methods of irrigation, some of which include the following: (i) **Basin irrigation** is a method of irrigating land in which basins are created and filled with water and used to irrigate the field. Although this kind of irrigation is not suitable for crops such as potatoes, cassava, carrots and maize, it is especially suitable for water-loving crops such as paddy rice as its roots are submerged in water. (ii) **Border-strip irrigation**, also known as **border irrigation** is a type of surface irrigation in which water is applied at the upper end of a strip having dikes to confine the water to the strip. This type of irrigation is best suited to large mechanised farms for ease of machine operations. Crops best suited to this kind of irrigation are wheat, barley, fodder crops and legumes. Border slopes should be uniform, with a minimum slope of 0.05% to provide adequate drainage and a maximum slope of 2% to limit problems of soil erosion. (iii) **Furrow irrigation** is a type of surface irrigation in which water is allowed to flow along small, parallel ditches or channels (called furrows) dug between crop rows in a field. This type of irrigation is suitable for row crops such as maize, sunflower, sugarcane, soybean; it is also suitable for crops like tomatoes, vegetables, potatoes, beans as the water does not touch them directly to flood them. (iv) **Flood irrigation** is a method of surface irrigation using water brought down by a river in flood, which is applied and distributed over the soil surface by gravity. The floodwaters are led off into specially prepared basins or ditches and water is released from them to flood over the land. (v) **Sprinkler irrigation** is a system of watering fields using pipelines which carry water under pressure from a pump or elevated source with sprinklers spaced along a pipe, which sprays droplets of water in a continuous circle until the moisture reaches the root level of the crop.

IRRITABILITY (SENSITIVITY): The response to stimuli of an organism or its ability to detect, interpret and respond to changing stimuli of light, touch, chemicals, etc., in the environment.

IRRITANT: A non-corrosive mechanical, chemical or pathological agent that causes inflammation, pain, or tension to living tissue such as the skin, the eyes or airways.

IRROTATIONAL: 1. Of a vector field, having zero curl. 2. In physics, **irrotational flow**, also known as **acyclic motion** or **irrotational motion** is a fluid flow in which the curl of the velocity function is zero everywhere, so that the circulation of the velocity about any closed curve vanishes. The velocity is given by the gradient of a scalar field.

ISA BUS (INDUSTRY STANDARD ARCHITECTURE BUS): In computing, it is a 16-bit data bus introduced in 1984 with the IBM PC AT and still in common use in PCs, alongside the superior PCI bus. It is not compatible with plug and play and in 1993, Intel and Microsoft introduced a PnP ISA bus that allowed the computer to automatically detect and setup computer ISA peripherals, such as a modem or sound card.

ISCHAEMIA (ISCHEMIA): Reduction of blood supply to an organ or part of the body, usually due to a partially or completely blocked artery, causing a shortage of oxygen and nutrients to the affected part. When the cardiac arteries are partially or completely blocked and blood flow to the heart is reduced, preventing it from receiving enough oxygen and nutrients, it causes myocardial ischaemia (cardiac ischaemia) characterised by chest pain (angina pectoris) or serious abnormal heart rhythms (arrhythmia). A sudden, severe blockage of a coronary artery can lead to a heart attack (myocardial infarction). The reduced blood flow is usually the result of a partial or complete blockage of your heart's arteries (coronary arteries).

ISCHIUM: The rearmost and lowest of the three bones that make up each half of the pelvis, situated below the ilium and behind the pubis. Divided into three portions, they are body of ischium, containing a prominent spine which serves as the origin for gemellus superior; superior ramus of the ischium, which functions as a partial origin for obturator internus and obturator externus; and the inferior ramus of the ischium functioning as a partial origin for gracilis and adductor magnus.

ISDN (INTEGRATED SERVICES DIGITAL NETWORK): An international telecommunications standard for digital transfer of simultaneous voice and data signals over digital telephone lines or traditional telephone copper wires that can support data transfer rates of 64 Kbps (64,000 bits per second). Analogue signals such as from traditional telephones are converted into digital signals by an ISDN terminal adapter before being transmitted.

ISENTROPIC PROCESS: Any process that takes place without any increase or decrease of entropy.

ISIDIUM: A vegetative reproductive structure that can detach from the main body from the thallus of lichen. It is 0.01-0.03mm in diameter that includes both algal fungal cells. They are fragile structures and may break off and be distributed by wind, animals and splashing raindrops.

ISING MODEL: A model for magnetic systems in which atomic spins have to be aligned either parallel or antiparallel to a given direction. Spins may have one of two values: either the value +1 or the value -1 and only pairs of nearest neighbouring spins can interact. There are also random thermal fluctuations depending on the temperature of the system; at low temperatures there is a net magnetisation as a result of alignment of spins; and at high temperature there is no net magnetisation.

ISLAND: An area of land surrounded entirely by water that is not large enough to be called a continent. For example, Australia is classed as a continent rather than an island, because of its size. The largest island in the world is Greenland. The different types of islands include the following: (i) **Continental islands** were once connected to a continent that were formed by tectonic movement, erosion or changes in sea level; for example, Greenland and Madagascar broke away from the supercontinent Pangaea due to the slow movements of the Earth's crust that broke apart Pangaea into several pieces that began to drift apart. (ii) **Oceanic islands** such as Japan were formed by the explosion of underwater volcanoes. (iii) **Coral islands** are tropical islands formed mainly by organic material of skeletons of corals deposited over many years. (iv) An **atoll** is a circular coral reef surrounding a lagoon; formed when a coral reef grew up around a volcanic island that subsequently sank or was submerged by a rise in sea level. (v) **Barrier islands** are found in shallow-water coastal areas and are formed by the deposition of unconsolidated sediment, usually sand.

ISLETS OF LANGERHANS (PANCREATIC ISLETS): Small groups of cells in the pancreas that function as an endocrine gland or hormone-producing cells, with about one million islets in a healthy adult human's pancreas, distributed throughout the organ. The A cells secrete the hormones glucagon, the B cells secrete insulin and the D cells secrete somatostatin. It was discovered in 1869 by German pathological anatomist Paul Langerhans.

ISO: 1. Abbreviation for **Infrared Space Observatory**, an European Space Agency (ESA) orbiting infrared observatory for studying the Universe at infrared wavelengths of 2.5 to 240 micrometres. The ISO was launched on 17 November 1995 from the ELA-2 launch pad at the Guiana Space Centre near Kourou in French Guiana by an Ariane 44P rocket. 2. Abbreviation for **International Organization for Standardization**, also known as **International Organization For Standardization (IOS)** is an international organisation that sets standards in many businesses and technologies, including computing and data communications, but excluding electrical and electronic engineering. Some of its standards include ISO 9000 for Quality management, ISO 14000 for Environmental management, ISO 3166 for Country codes, ISO 26000 for Social responsibility, ISO 50001 for Energy management, ISO 31000 for Risk management, ISO 22000 for Food safety management, ISO 27001 for Information security management and ISO 45001 for Occupational health and safety.

ISO 9660: In computing, it is a standard file format for CD-ROM disks published by the International Organization for Standardization (ISO) that is compatible with most systems such as a PC, Mac or other major computer platform.

ISO-: A prefix denoting that a compound is an isomer, for example, isopentane.

ISOBAR: 1. Also known as **isobaric line**, they are lines on a weather map joining points of equal atmospheric pressure, used to define low-pressure regions or cyclones and anticyclones or high-pressure regions. Numbers written on isobars give the atmospheric pressure measured in millibars. The closer the isobars, the stronger the wind will be, the further away they are, the lighter the wind will be. 2. Nuclides of different chemical elements with the same nuclear mass number. For example, hydrogen 3 and helium 3 are isobars.

ISOBARIC PROCESS: A thermodynamic process that occurs at constant pressure ($\Delta P = 0$). This is usually obtained by allowing the volume to expand or contract in such a way to neutralise any pressure changes that would be caused by heat transfer.

ISOBARIC MASS-CHANGE DETERMINATION: A technique in which the equilibrium mass of a substance at constant partial pressure of the volatile product(s) is measured as a function of temperature while the substance is subjected to a controlled temperature program. The record is the isobaric mass-change curve. The mass should be plotted on the ordinate decreasing downwards and temperature on the abscissa increasing from left to right.

ISOC (INTERNET SOCIETY): Founded in 1992, it is an international non-profit organisation composed of companies, government agencies and foundations concerned with Internet standards, education and policy development.

ISOCHORIC PROCESS (ISOVOLUMETRIC PROCESS): A thermodynamic process that occurs at constant volume ($\Delta V = 0$). Since the volume is constant, the system does no work and $W = 0$.

ISOCHORIC MOTION: In physics, a motion in which volume is preserved and so the determinant of the deformation gradient is identically one.

ISOCHROMOSOME: A metacentric chromosome in which the two arms are genetically identical.

ISOCHRONE: A set of points with the property that a given process or trajectory will take the same length of time to complete starting from any point in the isochrone. The curve formed by such a set of points is called an **isochronous curve**.

ISOCLINE: 1. An isocline line on a map or chart is a line such as contour lines and isobar joining places or points of equal magnetic dip. 2. A geologic fold with rock beds that slope in the same direction.

ISOCLINIC POINT: A wavelength, wavenumber or frequency at which the first derivative of an absorption spectrum of a sample does not change upon a chemical reaction or physical change of the sample.

ISOCONJUGATE SYSTEMS: Molecular entities with similar conjugate networks which have the same number of π -electrons. Examples include cyclopentadienide anion and pyrrole or benzene and pyridine.

ISOCORTEX (NEOCORTEX; NEOPALLIUM): An area of the cerebral cortex of the vertebrate brain, forming the outer layer of the cerebral hemispheres. It is made up of six layers, labelled I to VI. It is most heavily developed in mammals in which it forms a surface layer covering most of the forebrain. It is also the major centre for the coordination of sensory and motor information.

ISOCRATIC ELUTION: Elution using a single solvent.

ISOCYANIDE (ISONITRILE; CARBYLAMINE): Organic compound such as methyl isocyanide (CH_3NC) of general formula RNC . It is a colourless liquid with an unpleasant odour and a salt of isocyanic acid. The prefix *iso* is used because R is an organic radical, which is the isomer of the related cyanide ($\text{R-C}\equiv\text{N}$). The organic fragment is connected to the isocyanide group through the nitrogen atom, but not through the carbon.

ISOCYANIDE TEST: A qualitative test for the identification of primary amines, by reaction with an alcoholic solution of potassium hydroxide and trichloromethane, forming the unpleasant odour of isocyanide.

ISODESMIC REACTION: A reaction (actual or hypothetical) in which the types of bonds that are made in forming the products are the same as those which are broken in the reactants.

ISODIAPHERE: Nuclides in which the difference between the number of neutrons and the number of protons is the same.

ISODIAMETRIC: 1. In biology, having a similar diameter in all planes, as in a cell. 2. In chemistry, having three equal axes, as in a crystal. 3. In mathematics, having all diameters of equal length.

ISODYNAMIC LINE: A line on a map or chart joining points or places at which the total magnetic field intensity are equal.

ISODYNE: Points of a cultivating implement having equal dynamometer pull; a line on a map of a cultivated field connecting points of equal dynamometer pull.

ISOELECTRIC: A macro-ion of a polyampholyte (in particular a protein) is said to be isoelectric if it exhibits no electrophoresis.

ISOELECTRIC pH: The pH at which a protein has no net charge.

ISOELECTRIC POINT: The pH of a medium at which a particular amino acid carries zero electric charge and therefore will not migrate in an electric field. Proteins precipitate most rapidly at their isoelectric points. This property can be utilised to separate mixtures of proteins or amino acids.

ISOELECTRONIC: A term describing species that have the same number of electrons, regardless of the nature of the elements involved. Examples are sodium ion and neon atom carbon monoxide.

ISOEMISSIVE POINT: Wavelength, wavenumber or frequency at which the total intensity of emission of light by a sample does not change during a chemical reaction or physical change. The terms isolampsic and isostilbic are sometimes used.

ISOENTROPIC: A reaction series is said to be isoentropic if the individual reactions of the series have the same standard entropy of activation.

ISOENZYME (ISOZYME): Any one of the multiple forms of a given enzyme, each with different kinetic characteristics. The different isoenzymes are usually formed by different combinations of the same two or more subunits.

ISOGAL LINE: A line on a map or chart joining points or places of equal magnetic declination.

ISOGAMY: A sexual reproduction in which the sexual cells or gametes, are similar in size and structure. It occurs in some prototists. Examples are certain protozoans and algae.

ISOGON: An equiangular polygon or a polygon whose angles are equal. **Isogonal** or **isogonic** is having, making or involving equal angles. An isogonal transformation in the plane preserves angles and

has the form $x' = a_1x + b_1y + c_1$, $y' = a_2x + b_2y + c_2$, where $a_1 = b_2$ and $a_2 = -b_1$ or else $a_2 = b_1$ and $a_1 = -b_2$.

ISOGONAL CONJUGATE: Lines that pass through the vertex of an angle and make equal angles with the bisector of the angle.

ISOIONIC POINT: The pH value at which the net electric charge of an elementary entity in pure water equals zero.

ISOKINETIC LINE (ISOTACH; ISOVEL): A line on a weather map or chart connecting points where winds of equal speeds have been recorded.

ISOKINETIC SAMPLING: A technique for collecting airborne particulate in which the sampling device has a collection efficiency of unity for all sizes of particles in sampled air, regardless of wind velocity and direction of the instrument. The air stream entering the collector has a velocity (speed and direction) equal to that of the air in the gas stream just ahead of the sampling port of the collector.

ISOKONT: Describing an organism having flagella similar in form and length, as have the motile species of algae in the Chlorophyta, hence the former name of the taxon, Isokontae.

ISOLATE: In mathematics, of a root of an equation, to determine an interval within which it is the only root.

ISOLATED DOUBLE BONDS: Those bonds which are neither conjugated nor cumulative.

ISOLATED ORDINAL: In mathematics, an ordinal with an immediate predecessor and so not a limit ordinal.

ISOLATED POINT (ACNODE): A point not lying on a curve, but whose coordinates satisfy its equation.

ISOLATED SINGULARITY: Of a complex function, a point in a punctured neighbourhood of which the function is analytic and at which the function is not continuous. The singularity is either removable, if $\lim_{z \rightarrow a} (z - a)f(z) = 0$, or is essential or is a pole.

ISOLATED SYSTEM: A system that does not interact with its surroundings, hence does not permit the transfer of either mass or energy to or from its surroundings.

ISOLATING MECHANISM: Any of the biological properties such as behaviour, morphology or genetics of organisms that prevent interbreeding between members of different species that inhabit the same geographical area. The mechanism includes seasonal isolation, in which the breeding season of the different population do not overlap and behaviour ensuring that mating takes place only between members of the same species. Isolating mechanisms can be divided into two groups, prezygotic isolating mechanisms and postzygotic isolating mechanisms.

ISOLECITHAL (HOMOLECITHAL): Of an egg, having small amount of yolk fairly uniformly distributed throughout the cytoplasm.

ISOLEPTIC COMPLEX: A metal complex in which all the ligands are the same.

ISOLEUCINE: An amino acid with the chemical formula $\text{C}_6\text{H}_{13}\text{NO}_2$. Isoleucine is synthesised from threonine. It is an essential amino acid and found in most proteins.

ISOLINE (ISOGRAM; CONTOUR LINES): A line that connects equal values such as places of equal altitude on a map, graph or chart.

ISOMER: One of several species or molecular entities that have the same atomic composition (molecular formula) but different line formulae or different stereochemical formulae and hence different physical and/or chemical properties. Isomerism arises because the comparatively large number of atoms which make up many organic compounds can be arranged in more than one way. The two chief types of isomers are **stereoisomers**, which have the same structural formula; and **structural isomers**, which have different structural formulas. **Structural isomers** may be subdivided into **positional isomers**, which have the same functional groups occupying the different positions on the carbon skeleton; and **functional isomers**, which have different functional groups.

ISOMERASE: 1. Any of the class of enzymes that catalyse the structural rearrangement of the atoms within the molecule, thereby converting one isomer into another. 2. Catalyse reactions of the form $A \rightarrow B$, where B is an isomer of A.

ISOMERIC STATE: A nuclear state having a mean life long enough to be observed.

ISOMERIC TRANSITION: A nuclear process in which a nucleus with excess energy following the emission of an alpha particle or a beta particle emits energy without changing its number of protons or neutrons. Isomeric transitions can occur through the emission of a gamma ray or through the process known as **internal conversion**.

ISOMERISM: 1. The existence of two or more nuclides that have the same atomic number and the same mass number but different energy states. 2. The existence of two or more substances that have the same molecular formula but different structural formulae or different arrangement of atoms in space. The different types of isomers include structural isomers (ethanol and methoxymethane), cis-trans isomers (trans-but-2-ene and cis-but-2-ene), etc. Other isomers include meri-isomers and fac-isomer.

ISOMERISATION: A chemical reaction, whose principal product is isomeric with the principal reactant. An intramolecular isomerisation that involves the breaking or making of bonds is a special case of a molecular rearrangement. Isomerisation does not necessarily imply molecular rearrangement such as in the case of the interconversion of conformational isomers.

ISOMEROUS: In botany, having an equal number of parts, as in successive floral whorls. This includes equal number of petals, sepals and stamen. An example is flowers of tulip.

ISOMETRIC: 1. A line or graph which interrelates temperature and pressure at constant volume. 2. **Isometric projection** is a method for visually representing three-dimensional objects in two dimensions in technical drawing or a projection in which the three axes are equally inclined to the surface of the drawing and lines are drawn to scale. The angle between any two of the axes is 120 degrees.

ISOMETRIC TRANSFORMATION: The process of transformation of shape, size, length and angle of a figure in which they remain the same during the transformation. Translation, reflection and rotation are isometric transformations.

ISOMETRY: A geometric transformation such as the rotation of a plane in which the distance between any two points is preserved.

ISOMORPHISM: 1. In biology, similarity in form or appearance between organisms of different ancestry or between different stages in the life cycle of the same organism. 2. In chemistry, similarity of crystal structure between two or more distinct substances so that they are able to form solid solutions or the existence of substances of different chemical compositions but with similar crystalline form. Isomorphic substances usually have similar chemical formulas. Examples are sodium nitrate and calcium sulfate. 3. In mathematics, properties which have a one-to-one correspondence such that the result of an operation such as addition or multiplication on elements of one set corresponds to the result of the analogous operation on their images in the other set. For example, in three sided figures, an equilateral triangle and an isosceles triangle are isomorphic.

ISOMORPHOUS SUBSTITUTION: The replacement of one atom by another of similar size in a crystal lattice without disrupting or changing its crystal structure. When a substituting cation is of a smaller valence than the cation it is replacing, there is a net negative charge on the structure.

ISONITRILE (ISOCYANIDE; CARBYLAMINE): An organic compound containing the group $-NC$, in which the bonding is to the nitrogen atom. The prefix *iso* is used because it is the isomer of the related cyanide ($R-C \equiv N$). The organic fragment is connected to the isocyanide group through the nitrogen atom, but not through the carbon.

ISONOME METHOD: A technique used to study the distribution of plants in a particular area, where there are obvious variations in species distribution.

ISOOPTOACOUSTIC POINT: Wavelength, wavenumber or frequency at which the total energy emitted by a sample as heat does not change upon a chemical reaction or a physical change of the sample. Its position depends on the experimental conditions.

ISOPENTENYL ADENOSINE (IPA): A natural cytokinin, similar in structure to zeatin. In addition to being a very active cytokinin, it has been found as a minor base in certain transfer RNA molecules. In the tRNA specific for the amino acid serine, IPA is found following the anticodon and thus may serve to mark the end of the anticodon.

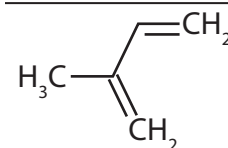
ISOPERIMETRIC PROBLEMS: Extensions of the classical variational problem of finding which of all plane figures with equal perimeter has greatest area.

ISOPLETH (ISOLINE): A vertical line in liquid-vapour phase consisting of a line of constant composition of the whole system as the pressure is changed or a line on a map connecting points with the same value for variables such as temperature or air pressure.

ISOPOD: Any member of the order Isopoda. Isopods are segmented crustaceans, the best known of which are woodlice, pill bugs, sow bugs and other land-dwelling species. Most isopods are small; however, there are about nine species of so-called giant isopods in the genus *Bathynomus*.

ISOPOLYMORPHISM: The property of a substance of more than one crystalline structure that is isomorphous with the crystalline structures of another substance.

ISOPRENE (2-METHYL-1,3-BUTADIENE; TERPENE): A colourless flammable liquid hydrocarbon with the molecular formula C_5H_8 , density 0.681 g/cm^3 , melting point -144°C (129 K) and boiling point 34°C (307 K). It is insoluble in water but soluble in many organic solvents. It is the monomer of natural rubber and also a common structure motif to an immense variety of other naturally occurring compounds, collectively termed the isoprenoids. It is produced and emitted by many species of trees into the atmosphere. Major producers are oaks, poplars, eucalyptus and some legumes. It is used in the manufacture of synthetic rubber.



Chemical structure of Isoprene

ISOPRENOIDS: Compounds formally derived from isoprene (2-methylbuta-1,3-diene), the skeleton of which can generally be discerned in repeated occurrence in the molecule. The skeleton of isoprenoids may differ from strict additivity of isoprene units by loss or shift of a fragment, commonly a methyl group. The class includes both hydrocarbons and oxygenated derivatives.

ISOPROPANOL (ISOPROPYL ALCOHOL; 2-PROPANOL; ISOPROPYL ALCOHOL): One of the two isomers of propanol with the chemical formula C_3H_8O . It is a colourless liquid that is miscible with water with melting point -89°C (184 K) and boiling point 82°C (355 K). It is one of the cheapest alcohols and has many uses because of its similar solvent properties. Isopropanol is made commercially by dissolving propylene gas in sulfuric acid forming sulfate ester, which is hydrolysed. When isopropanol is oxidised, it becomes acetone.

ISOPTERA: An order of a social exopterygote that comprises the termites. They generally have gradual metamorphosis, lack of true larval and pupal stages, biting and prognathous mouthparts, two pairs of subequal wings and the abdomen joined broadly to the thorax. They are mainly tropical insects and have a complex system of castes, including wingless workers and soldiers and primary and secondary winged reproductive members.

ISOPYCNIC: An adjective describing components of a multicomponent system with equal partial specific volumes.

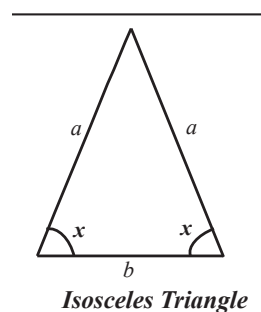
ISOQUINOLINE ALKALOIDS: A group of alkaloids, the structures of which are based on the isoquinoline nucleus. They are derived from the amino acids tyrosine and phenylalanine. Examples are morphine, from the opium poppy (*Papaver somniferum*) and curare, from the bark of Strychnos species.

ISOREFRACTIVE: A term used to describe components of a multicomponent system having zero refractive index increments with respect to each other.

ISOSCELES TETRAHEDRON: A tetrahedron in which each pair of opposite sides have the same length. This is usually a triangular pyramid whose base is an equilateral triangle.

ISOSCELES TRAPEZIUM: A quadrilateral with 1 pair of parallel sides, 1 pair of equal, non-parallel sides, equal diagonals and 2 pairs of equal angles.

ISOSCELES TRIANGLE: A triangle with two or at least two equal sides (as in the case of the equilateral triangle which has all three sides and angles equal). The angles opposite the equal sides are also equal. It has one line of symmetry.



ISOSPIN (ISOTOPIC SPIN; ISOBARIC SPIN; I-SPIN): A quantum characteristic of families of related subatomic particles, such as baryons and mesons that relates to the number of different values of electric charge they can have. Isospin symmetry remains an important concept in particle physics and a close examination of this symmetry historically led directly to the discovery and understanding of quarks and of the development of Yang-Mills theory. Its symbol is I .

ISOSTASY: The theoretical equilibrium that tends to exist in the Earth's crust between forces such as accumulated ice pushing down on a section of the Earth's surface and those pushing up from below. Isostasy can cause rising sea level, called **isotatic sea level change**.

ISOSTATIC PRESSING: Application of a hydrostatic pressure through a liquid to achieve densification prior to the production of a uniform compact monolith through ceramisation of the densified liquid.

ISOSTATIC LAND MOVEMENTS: The vertical movements of the earth's crust or land mass. When the loading of the lithosphere is changed by alterations in land ice mass, ocean mass, sedimentation, erosion, or mountain building, vertical isostatic adjustment results, in order to balance the new load.

ISOSURFACE: A three-dimensional diagram showing surfaces connecting points such as pressure, temperature, velocity, density of equal values.

ISOTACTIC MACROMOLECULE: A tactic macromolecule, essentially comprising only one species of configurational base unit, which has chiral or prochiral atoms in the main chain in a unique arrangement with respect to its adjacent constitutional units.

ISOTHERM: 1. A line on a map of a particular region of the Earth's surface linking all places having the same temperature at a given time or a curve on a graph representing readings taken at constant temperature. The relative spacing of the isothermal lines indicates a temperature gradient, which is the amount of temperature change over a given distance. 2. A line on a graph showing the relationship between variables, especially pressure and volume, at a constant temperature.

ISOTHERMAL LAYER: 1. Vertical layer in the atmosphere where temperature remains unchanged. In the Earth's atmosphere, three isothermal layers are found in the lower regions of the stratosphere, mesosphere and the thermosphere. 2. A layer of the ocean characterised by an essentially constant temperature.

ISOTHERMAL PROCESS: Any change that takes place at a constant temperature ($\Delta T = 0$); the change occurs slowly enough to allow the system to continually adjust to the temperature through heat exchange to maintain a constant temperature.

ISOTOMIC CONJUGATE: Two points on the side of a triangle are isotomic if they are equidistant from the midpoint of that side. Two points inside a triangle are isotomic conjugates if the corresponding cevians through these points meet the opposite sides in isotomic points.

ISOTONE: 1. In mathematics, of a function between ordered spaces order-preserving; monotone increasing. 2. In physics, an atom whose nucleus has the same number of neutrons, but a different number of protons (i.e. a different atomic number).

ISOTONIC: A term describing a solution whose osmotic pressure is equal to that of a body fluid, such as blood plasma or tears, to which the solution is compared. No water movement by osmosis occurs across the membrane.

ISOTOPE: A form of the same chemical element having the same number of protons but different number of neutrons in the atomic nucleus. For example, carbon 12 and carbon 14 are isotopes of carbon. They both have 6 protons but different number of neutrons. Carbon-12 has 6 neutrons and carbon-14 has 8 neutrons. Most chemical elements have several stable isotopes. For example, lithium has two stable isotopes, one with three neutrons in the nucleus (${}^6\text{Li}$) and one with four (${}^7\text{Li}$). Isotopes of a single element possess almost identical chemical properties but have very different nuclear properties. There are 275 isotopes of the 81 stable elements (non-radioactive), in addition to over 800 radioactive isotopes and every element has known isotopic forms. Radioisotopes or radionuclides are radioactive isotopes.

ISOTOPE DILUTION: Mixing of a given nuclide with one or more of its isotopes.

ISOTOPE DILUTION ANALYSIS: A kind of quantitative analysis based on the measurement of the isotopic abundance of a nuclide after isotope dilution with the test portion.

ISOTOPE EFFECT: The effect on the rate or equilibrium constant of two reactions that differ only in the isotopic composition of one or more of their otherwise chemically identical components is referred to as a kinetic isotope effect or a thermodynamic (or equilibrium) isotope effect, respectively.

ISOTOPE EXCHANGE: A chemical reaction in which the reactant and product chemical species are chemically identical but have different isotopic composition. In such a reaction the isotope distribution tends towards equilibrium (as expressed by fractionation factors) as a result of transfers of isotopically different atoms or groups.

ISOTOPE EXCHANGE ANALYSIS: A kind of quantitative analysis based on the isotope exchange between isotopes of the element to be determined and other isotopes of this element in different valency states or in different molecules.

ISOTOPE PATTERN: A set of peaks related to ions with the same chemical formula but containing different isotopes, for example, the 16 and 17 mass/charge peaks in a CH_4^+ sample arising from ${}^{12}\text{CH}_4^+$ and ${}^{13}\text{CH}_4^+$ ions.

ISOTOPE SEPARATION: The physical separation of the isotopes of an element from each other on the basis of slight differences in their physical properties. Uranium isotope separation has the greatest industrial importance, because uranium is used as a fuel for nuclear power reactors. An example of isotope separation is production of heavy water, which is obtained by isotope separation of light hydrogen (${}^1\text{H}$) and heavy hydrogen (${}^2\text{H}$) in natural water.

ISOTOPIC ABUNDANCE: The relative number of atoms of a particular isotope in a mixture of the isotopes of an element, expressed as a fraction of all the atoms of the element.

ISOTOPIC CARRIER: A carrier which differs only in isotopic composition from the trace it has to carry.

ISOTOPIC DISCRIMINATION: The uptake or assimilation by living organisms of a particular isotope in preference to another isotope of the same element. For example, the preferential fixation by photosynthetic organisms of the lighter isotope carbon-12 compared with the heavier isotope-13.

ISOTOPIC ENRICHMENT: Any process by which the isotopic abundance of a specified isotope in a mixture of isotopes of an element is increased.

ISOTOPIC ENRICHMENT FACTOR: For a material enriched in a specified isotope, the ratio between the isotopic abundance and the natural abundance of that isotope.

ISOTOPIC ION: Any ion containing one or more of the less abundant naturally occurring isotopes of the elements that make up its structure.

ISOTOPIC LABELLING: Labelling in which the resulting product is only different from the initial one by its isotopic composition.

ISOTOPIC NUMBER (NEUTRON EXCESS): The difference between the number of neutrons in an isotope and the number of protons in a nucleus.

ISOTOPIC SIGNATURE (ISOTOPIC FINGERPRINT): The ratio of stable or unstable isotopes of particular elements found in a sample of tissue, water, air or other material. In biology, the isotopic signature of organic matter, especially carbon, nitrogen and oxygen, can provide relevant information on the growing conditions, lifestyle and metabolism of the organism from which the material was derived.

ISOTOPIC SPIN (ISOSPIN; ISOBARIC SPIN): A quantum number applied to hadrons strong interaction to distinguish between members of a set of particles that differ in their electromagnetic properties but are otherwise apparently identical. For example, if electromagnetic interactions and weak interactions are ignored, the proton cannot be distinguished from the neutron in their strong interactions. Isotopic spin is introduced to make a distinction between them.

ISOTOPIC TRACER: A stable or radioactive isotope that can be used to label a metabolite and consequently follow its fate in an intact organism. Many elements found in living organisms have rare isotopes that are useful as tracers.

ISOTOPIC WEIGHT (ISOTOPIC MASS): Atomic mass of an isotope. **Atomic mass** is the mass of an atom or molecule on a scale where the mass of a carbon-12 (^{12}C) atom is exactly 12.0. The mass of any atom is approximately equal to the total number of its protons and neutrons multiplied by the atomic mass unit, $u = 1.6605397 \times 10^{-24}\text{gram}$.

ISOTOPICALLY MODIFIED: A term used to describe a compound that has a macroscopic composition such that the isotopic ratio of nuclides for at least one element deviates measurably from that occurring in nature. It is either an isotopically substituted compound or an isotopically labelled compound.

ISOTOPICALLY SUBSTITUTED: A term used to describe a compound that has a composition such that essentially all the molecules of the compound have only the indicated nuclide(s) at each designated position. For all other positions, the absence of nuclide indication means that the nuclide composition is the natural one.

ISOTOPOLOGUES: Substances that differ only in isotopic composition, for example, CH_3OH , CH_2DOH , CHD_2OH , CD_3OH , CD_3OD .

ISOTOPOMERS (ISOTOPIC ISOMERS): Molecules or isomers that have the same numbers of each type of isotopic atom but different arrangements in the molecule, for example, CH_2DOH and CH_3OD .

ISOTROPIC: Having the same physical properties independent of direction.

ISOTROPIC BAND (I BAND; I-BAND): The region of striated muscle fibre that contains only thin filaments. The thin filament is

an assembly of three proteins: actin (the main one), tropomyosin and troponin. I bands are visible as light bands at either end of a sarcomere. Unlike the A-band, the I-band can vary in width depending upon the state of contraction of the muscle when fixed. **Striated muscle** also called **skeletal muscle**, **striped muscle** or **voluntary muscle** is attached to bone and produces all the movements of body parts in relation to each other. Unlike smooth and cardiac muscles, striated muscle is under voluntary control.

ISOTYPE: A biological specimen that is a duplicate of or very similar to the type of specimen. It can be used as a reference specimen if the type specimen is lost.

ISOTYPE SWITCHING (IMMUNOGLOBULIN CLASS SWITCHING; ISOTYPIC COMMUTATION; CLASS SWITCH RECOMBINATION, CSR): The biological mechanism by which B cells can change the type of antibody such as isotype that is produced during the course of an immune response from one class to another.

ISOZYMES (ISOENZYME): One of several forms of an enzyme in an individual or population that differ in amino acid sequence such as substrate affinity and maximum rate of enzyme-substrate reaction but catalyse the same chemical reaction.

ISTHMUS: 1. A narrow passage or connection between parts of the body. 2. A narrow strip of land that joins two larger areas of land usually with water on either side. For example, the Isthmus of Suez in Egypt connects Africa and Asia.

IT INFRASTRUCTURE: The hardware, software, network resources and services that makes communication possible.

ITACONIC ACID (2-METHYLIDENEBUTANEDIOIC ACID; METHYLENESUCCINIC ACID): A production of the fermentation of the filamentous fungus, *Aspergillus niger*. It is a colourless crystalline compound, with the chemical formula $\text{C}_5\text{H}_6\text{O}_4$, density 1.63 g/cm^3 , melting point $162\text{--}164^\circ\text{C}$ ($435\text{--}437 \text{ K}$) and decomposes at 438 K (165°C). It is used commercially in the production of adhesive, plasticisers and paint.

ITERATED INTEGRAL: A repeated integral of the form that arises when evaluating a multiple integral by Fubini's theorem, $\int_a^b \text{d}x \int_c^d \text{d}y \int_e^f f(x, y, z) \text{d}z$, in which one first integrates with respect to z , then y then x , treating the remaining variables as parameters.

ITERATED SERIES: A double or multiple series of the form $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{n,m}$.

ITERATION: 1. The process of successive approximations, used as a technique for solving a mathematical problem. 2. The repetition of a sequence of instructions in a computer program until a result is achieved. Iterations of functions are common in computer programming, since they allow multiple blocks of data to be processed in sequence.

ITEROPARITY: Reproduction repeatedly or more than once over a reproductive span that usually covers a major part of the total life span. Iteroparous vertebrates include all birds, most reptiles, virtually all mammals and most fish. Among invertebrates, most mollusca and many insects (for example, mosquitoes and cockroaches) are iteroparous. Most perennial plants are iteroparous. In botany, the term **polycarpic** is also sometimes used.

IUPAC: The abbreviation for International Union of Pure and Applied Chemistry. It is an international organisation of national chemistry associations established in 1919 which recommends rules and conventions in the practice of chemistry. These rules bring uniformity and order into chemical writings/texts so that all chemists can

understand and communicate with one another without confusion. The recommendations cover system of naming, quantities, units, symbols, atomic and relative atomic masses, etc, in chemistry.

IUPAC NOMENCLATURE: A system of naming chemical compounds according to rules of the International Union of Pure and Applied Chemistry (IUPAC).

IVORY: 1. The hard, white substance of which the teeth and tusks of certain animals such as the elephant, hippopotamus, walrus, mammoth and narwhal are made. Ivory has been used for ornamental objects and also for billiard balls, piano keys. Ivory is mostly obtained

from elephants, resulting in their indiscriminate killing, leading to the banning of trade in ivory worldwide. 2. A creamy white colour like that of an elephant's tusk.

IXORA: A genus of flowering plants in the Rubiaceae family, with about 500 species of tropical evergreen trees and shrubs. The plants are adapted to tropical and subtropical areas worldwide. The leaves of the plant are leathery with opposite leaf arrangement. The flowers are in tiny clusters of red and flower all year round. The plant prefers acidic soils. They are planted ornamentally in warm greenhouses, gardens and landscapes.



j: 1. A unit vector, usually one in the direction of the y -axis. It is conventionally printed in bold type. 2. A unit quaternion.

j: In physics, it is the symbol used for the imaginary number, $\sqrt{-1}$.

J: The notation or symbol for joule, the derived unit of energy, work or amount of heat in the International System of Units.

J-J COUPLING: A type of coupling in many-fermion systems, such as electrons in atoms and nucleons in nuclei, in which the energies associated with the spin-orbit interactions are much higher than the energies associated with electrostatic repulsion.

J-VALUE: An effective first order rate constant for a photochemical reaction (photodecomposition, etc.) of light absorbing impurities in the atmosphere.

J# (J SHARP): A programming language from Microsoft based on Java that includes most of the features of Visual J++. It is specifically designed to allow Java-language developers easy transition to the .NET Framework and to create .NET applications.

J2ME (JAVA 2 PLATFORM, MICRO EDITION): A technology that allows programmers to use the Java programming language and related tools to create applications and programs for wireless and mobile devices such as cellular phones, TV set-top boxes, and personal digital assistants (PDAs). J2ME consists of programming specifications and a special virtual machine, the K Virtual Machine that allows a J2ME-encoded program to run in the mobile device.

JABLONSKI DIAGRAM: A diagram that represents the electronic energy levels of molecules and the transitions between them, named after the Polish physicist Aleksander Jabłoński. It enables radiative transitions between energy levels in molecules as well as such phenomena as internal conversions to be shown.

JACCARD COEFFICIENT: 1. In biology, it is an association coefficient of numerical taxonomy, which is the proportion of characters that match, excluding those that both organisms lack. 2. In mathematics, it measures similarity between finite sample sets, and is defined as the size of the intersection divided by the size of the union of the sample sets.

JACCARD INDEX (INTERSECTION OVER UNION): A statistic used for comparing the similarity and diversity of sample sets.

JACINTH (HYACINTH): 1. Red or yellowish-red variety of zircon, used as a gemstone. 2. Any genus of liliaceous Mediterranean plant, especially *Hyacinthus orientalis* having narrow leaves and a terminal of fragrant flowers with shades of blue, purple, red, pink, white and yellow.

JACK: 1. A portable hand-operated mechanical, pneumatic or hydraulic device for lifting, hoisting or moving heavy objects for short heights. An example is the automobile jack, used to lift a car to change a tire. 2. A male donkey or mule.

JACKSON-BERNSTEIN THEOREMS: A set of theorems concerning Chebyshev approximation that describe the best possible convergence

rates for classes of functions that have given smoothness properties and, conversely, deduce smoothness properties from the rates. For example, if f is a Lipschitz function satisfying the Lipschitz condition of order α the error in best approximation by a trigonometric polynomial of degree n is at worst $O(n^{-\alpha})$ and if this rate holds for $0 < \alpha < 1$, the function has Lipschitz order α . Correspondingly, a function is entire if and only if the error tends to zero better than geometrically.

JACOB-MONOD MODEL (OPERON MODEL): A theory to explain how genetic regulation of protein synthesis in prokaryotes in which the structural genes that determine the organisation of the proteins are controlled by other regions of DNA upstream from the structural genes. It was proposed by F. Jacob and J. Monod in 1960. The theory is well substantiated for prokaryotes but there is little evidence that it is directly applicable to eukaryotic cells. The operon model shows how messenger RNA synthesis and thus the quantity of enzymes present, may be regulated.

JACOBI ITERATION: The iterative method of solving a linear matrix system, $Ax = b$, by repeatedly solving $x_{\text{NEW}(i)} = \frac{1}{a_{ij}} \left[b_i - \sum_{l \neq j} a_{il} x_{\text{OLD}(l)} \right]$.

This assumes $a_{jj} \neq 0$ and will not always converge.

JACOBI, KARL GUSTAV JACOB (1804-51): German mathematician, who made notable contributions to elliptic functions, dynamics, differential equations and number theory.

JACOBI'S METHOD: A method for solving a first-order partial differential equation of the form $F\left(x_1, x_2, \dots, x_n, \frac{\partial z}{\partial x_1}, \frac{\partial z}{\partial x_2}, \dots, \frac{\partial z}{\partial x_n}\right) = 0$,

in which the dependent variable z does not occur explicitly. This is an elaboration of Charpit's method for equations with more than two independent variables.

JACOBIAN ELLIPTIC FUNCTIONS: A class of elliptic functions which arise by inversion of elliptic integrals of the first kind. The three basic functions are denoted sn , cn and dn . For modulus k , the function $y = sn(z) = sn(z, k)$ is obtained by inversion of $z = \int_0^y (1-t^2)^{-\frac{1}{2}} (1-k^2 t^2)^{-\frac{1}{2}} dt$, and the other two functions are defined by $sn^2 z + cn^2 z = 1$, $k^2 sn^2 z + dn^2 z = 1$ and $cn(0) = dn(0) = 1$, then sn is doubly periodic with periods $4K(k)$ and $2iK(k')$, where K is the complete elliptic integral of the first kind and k' is the complementary modulus.

JACOBIAN MATRIX: A matrix whose i^{th} row is the vector of the partial derivatives of the i^{th} function, evaluated at the point in question. Consider two functions, f_1 and f_2 , their Jacobian is given by:

$$J(f_1, f_2) = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{vmatrix}, \text{ which can be generalised as}$$

$$J(f_1, \dots, f_n) = \begin{vmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{vmatrix}.$$

JADE: A hard dark green to almost white semiprecious mineral consisting mainly of jadeite or nephrite used as gemstones. Jadeite contains sodium aluminium silicate and used as ornaments and jewelry. Nephrite contains calcium, magnesium and iron.

JAGGIES: An informal term for various kinds of anomalies in computer graphics outputs or display imaging. The result displays shape edges composed of small squares or "steps" instead of a smoothly contoured line.